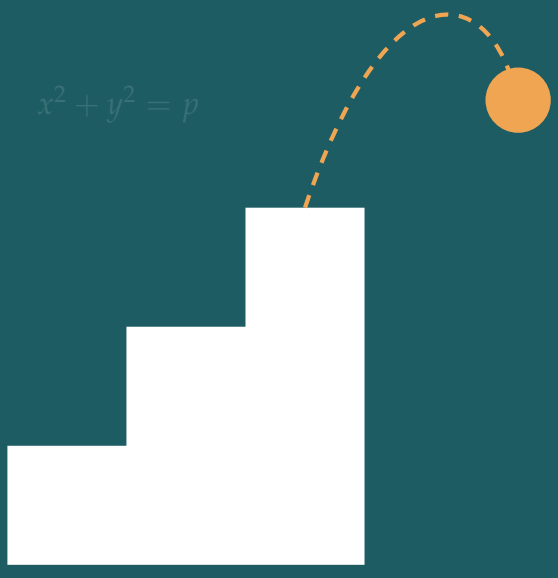
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2008 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2008 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2008 Worked Solutions

Worked solutions to all 39 questions of the 2008 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2008, Question 1

Pure • Proof by Contradiction

> **STEP I 2008, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/08-s1-q1

The key idea

Rational numbers are closed under addition, subtraction, multiplication and division by a non-zero number. Almost everything here follows from that one fact: if some numbers are rational, anything you build from them with these operations is rational too. In the final part, assume two of the four numbers are rational, combine them cleverly, and produce one of the numbers we were told is irrational.

Solution

Definition

A real number x is *rational* if it can be written as $x = \frac{m}{n}$ with m and n integers and $n \neq 0$. A real number is *irrational* if it is not rational, that is, if there are no integers m and $n \neq 0$ with $x = \frac{m}{n}$.

Statement A

Suppose, for a contradiction, that pq is irrational but neither p nor q is irrational. Then both are rational, so we can write $p = \frac{a}{b}$ and $q = \frac{c}{d}$ with a, b, c, d integers and $b, d \neq 0$. Then

$$pq = \frac{ac}{bd},$$

where ac and bd are integers and $bd \neq 0$ because neither b nor d is zero. So pq is rational, which contradicts our assumption. Hence at least one of p and q is irrational.

Statement B

In the same way, suppose $p + q$ is irrational but $p = \frac{a}{b}$ and $q = \frac{c}{d}$ are both rational, with $b, d \neq 0$. Then

$$p + q = \frac{ad + bc}{bd},$$

a ratio of integers with non-zero denominator, so $p+q$ is rational. This contradiction shows that at least one of p and q is irrational.

Statement C

Take $p=\sqrt{2}$ and $q=-\sqrt{2}$. Both are irrational, but $p+q=0=\frac{0}{1}$ is rational, so C is false.

For completeness, here is why $\sqrt{2}$ is irrational. If $\sqrt{2}=\frac{m}{n}$ in lowest terms, then $m^2=2n^2$, so m^2 is even and therefore m is even (the square of an odd number is odd). Writing $m=2k$ gives $n^2=2k^2$, so n is even too, which contradicts lowest terms. And $-\sqrt{2}$ is irrational because if $-\sqrt{2}=\frac{m}{n}$ then $\sqrt{2}=\frac{-m}{n}$.

The final part

We are told that π, π^2 and $\mathrm{e}\pi$ are irrational (these three facts are all we need). Two simple observations will be used repeatedly:

- if r and s are rational, then so are $r+s$, $r-s$, rs, and r/s when $s\neq 0$ (the argument is exactly the one used for A and B);
- consequently, if r is rational, then so is $\frac{1}{2}r$, since $\frac{1}{2}$ is rational.

Note also that $\pi+\mathrm{e}>0$ and $\pi-\mathrm{e}>0$, since $\pi>3>\mathrm{e}>0$; in particular neither is zero.

Suppose, for a contradiction, that two of the four numbers

$$\pi+\mathrm{e},\quad \pi-\mathrm{e},\quad \pi^2-\mathrm{e}^2,\quad \pi^2+\mathrm{e}^2$$

are rational. There are six possible pairs, and we rule out each one.

1. $\pi+\mathrm{e}$ *and* $\pi-\mathrm{e}$ *rational.* Adding, 2π is rational, so π is rational. Contradiction.
2. $\pi^2-\mathrm{e}^2$ *and* $\pi^2+\mathrm{e}^2$ *rational.* Adding, $2\pi^2$ is rational, so π^2 is rational. Contradiction.
3. $\pi+\mathrm{e}$ *and* $\pi^2+\mathrm{e}^2$ *rational.* Squaring the first, $(\pi+\mathrm{e})^2$ is rational, and
$$2\mathrm{e}\pi=(\pi+\mathrm{e})^2-(\pi^2+\mathrm{e}^2),$$
so $\mathrm{e}\pi$ is rational. Contradiction.
4. $\pi-\mathrm{e}$ *and* $\pi^2+\mathrm{e}^2$ *rational.* Similarly,
$$2\mathrm{e}\pi=(\pi^2+\mathrm{e}^2)-(\pi-\mathrm{e})^2$$
is rational, so $\mathrm{e}\pi$ is rational. Contradiction.
5. $\pi+\mathrm{e}$ *and* $\pi^2-\mathrm{e}^2$ *rational.* Since $\pi+\mathrm{e}\neq 0$,
$$\pi-\mathrm{e}=\frac{\pi^2-\mathrm{e}^2}{\pi+\mathrm{e}}$$

is a quotient of rationals with non-zero denominator, so it is rational. Now $\pi + \mathrm{e}$ and $\pi - \mathrm{e}$ are both rational, which is impossible by case 1.

6. $\pi - \mathrm{e}$ *and* $\pi^2 - \mathrm{e}^2$ *rational.* Since $\pi - \mathrm{e} \neq 0$,

$$\pi + \mathrm{e} = \frac{\pi^2 - \mathrm{e}^2}{\pi - \mathrm{e}}$$

is rational, and again case 1 gives a contradiction.

Ask yourself Why do we need $\pi - \mathrm{e} \neq 0$ in case 6?

Because dividing by zero is not allowed: the step "rational divided by rational is rational" only works when the denominator is non-zero. If we had $\pi = \mathrm{e}$, then $\pi^2 - \mathrm{e}^2 = 0$ would be rational without telling us anything about $\pi + \mathrm{e}$. Here $\pi - \mathrm{e} \approx 0.42$, so all is well.

Every pair leads to a contradiction, so no two of the four numbers can both be rational. Hence at most one of them is rational, as required.

Checks

The identities used are easy to confirm numerically: with $\pi \approx 3.14159$ and $\mathrm{e} \approx 2.71828$, we get $(\pi + \mathrm{e})(\pi - \mathrm{e}) \approx 5.85987 \times 0.42331 \approx 2.48055 \approx \pi^2 - \mathrm{e}^2$, and $(\pi + \mathrm{e})^2 - (\pi^2 + \mathrm{e}^2) \approx 17.0795 \approx 2\mathrm{e}\pi$. Count the pairs as well: four numbers give $\binom{4}{2} = 6$ pairs, and all six are covered. Notice that the irrationality of e and e^2 was never used; by symmetry we could have used them in place of π and π^2 in cases 1 and 2.

Common slips

- Proving the converse of A or B by mistake. A says "product irrational $\Rightarrow$ a factor irrational"; its converse is false ($\sqrt{2} \cdot \sqrt{2} = 2$).
- Dividing by $\pi \pm \mathrm{e}$ without remarking that it is non-zero.
- Using statement C as if it were true, for example claiming "π and e irrational, so $\pi + \mathrm{e}$ is irrational". Whether $\pi + \mathrm{e}$ is irrational is in fact an open problem.

Beyond STEP

The closure facts used here say that the rationals form a *field*: a set where you can add, subtract, multiply and divide (except by zero) without leaving it. Fields are a central object in a first-year algebra course, and questions such as "is $\pi + \mathrm{e}$ irrational?" lead into transcendence theory, where remarkably little is known. For the algebra, try Allenby, *Rings, Fields and Groups*; for the number theory of rationals and irrationals, Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 2008, Question 2

Pure • Integration by Substitution

STEP I 2008, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q2

The key idea

The substitution is chosen so that the square root becomes $t - x$, which is simple. Rather than solving for x and differentiating a messy quotient, we square to remove the root and then differentiate the resulting relation implicitly. The derivative contains exactly the factor $t - x$ that cancels the square root in the integrand, so the integral collapses to $\int \frac{1}{t+b}\,\mathrm{d}t$. The final part is a test of when that cancellation and division are legitimate.

Solution

The derivative

Write $R = \sqrt{x^2 + 2bx + c}$, so that $t - x = R$. Squaring,

$$t^2 - 2tx + x^2 = x^2 + 2bx + c \quad \Longrightarrow \quad t^2 - 2tx - 2bx - c = 0.$$

Differentiating this with respect to t (with x a function of t),

$$2t - 2x - 2t\frac{\mathrm{d}x}{\mathrm{d}t} - 2b\frac{\mathrm{d}x}{\mathrm{d}t} = 0, \qquad \text{so} \qquad (t+b)\frac{\mathrm{d}x}{\mathrm{d}t} = t - x.$$

To divide by $t + b$ we need it to be non-zero. Since $b^2 < c$,

$$x^2 + 2bx + c = (x+b)^2 + (c - b^2) > (x+b)^2,$$

so $R > |x + b|$ and therefore

$$t + b = (x + b) + R > (x + b) + |x + b| \geq 0.$$

Hence $t + b > 0$ for every x, and

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{t - x}{t + b},$$

as required. (The squared relation can also be solved directly: $x = \dfrac{t^2 - c}{2(t+b)}$, and differentiating this gives the same answer after some algebra.)

The integral

The quadratic $x^2 + 2bx + c$ is positive for all x, so the integrand is defined everywhere. Substituting, and using $R = t - x$,

$$\int \frac{1}{\sqrt{x^2+2bx+c}}\,\mathrm{d}x = \int \frac{1}{t-x}\,\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \int \frac{1}{t-x}\cdot\frac{t-x}{t+b}\,\mathrm{d}t$$
$$= \int \frac{1}{t+b}\,\mathrm{d}t = \ln(t+b) + C,$$

where no modulus is needed because $t + b > 0$. The cancellation of $t - x = R$ is fine because $R > 0$. Returning to x,

$$\int \frac{1}{\sqrt{x^2+2bx+c}}\,\mathrm{d}x = \ln\left(x + b + \sqrt{x^2+2bx+c}\right) + C.$$

You can confirm this by differentiating: the derivative of $x + b + R$ is $1 + \dfrac{x+b}{R} = \dfrac{R+x+b}{R}$, and dividing by $x + b + R$ leaves $\dfrac{1}{R}$.

The case $b^2 = c$

Now $x^2 + 2bx + c = (x+b)^2$, so $\sqrt{x^2+2bx+c} = |x+b|$ and the integrand is $\dfrac{1}{|x+b|}$ (defined for $x \neq -b$).

Ask yourself Why is $\sqrt{(x+b)^2}$ not simply $x + b$?

The square root symbol means the non-negative root, so $\sqrt{(x+b)^2} = |x+b|$, which equals $-(x+b)$ when $x + b < 0$. This sign is exactly what decides whether the formula survives.

If $x + b > 0$: directly,

$$\int \frac{1}{x+b}\,\mathrm{d}x = \ln(x+b) + C.$$

The formula gives

$$\ln\left(x + b + (x+b)\right) = \ln 2 + \ln(x+b),$$

which differs from the direct answer only by the constant $\ln 2$. So the result still holds.

If $x + b < 0$: directly,

$$\int \frac{1}{-(x+b)}\,\mathrm{d}x = -\ln\left(-(x+b)\right) + C = -\ln|x+b| + C.$$

But the formula gives $\ln\left(x + b + |x+b|\right) = \ln\left((x+b) - (x+b)\right) = \ln 0$, which is undefined. So the result does not hold here.

It is worth seeing where the derivation breaks. With $x + b < 0$ we have $t = x - (x+b) = -b$ for every x: t is constant, so it cannot be used as a new variable at all, and $t + b = 0$

is exactly the quantity we divided by. The strict inequality $b^2 < c$ was what guaranteed $t + b > 0$. When $b^2 = c$ only the weak inequality $t + b \geq 0$ survives, with equality precisely when $x + b \leq 0$.

Checks

- With $b = 0$, $c = 1$ the result reads $\int \frac{1}{\sqrt{x^2+1}}\,\mathrm{d}x = \ln\left(x + \sqrt{x^2+1}\right) + C = \operatorname{arsinh} x + C$, the standard result.
- Numerically, with random b and $c > b^2$, Simpson's rule for the integral between random limits agrees with the difference of the log formula to better than 10^{-7}. For $b^2 = c$ with $b = 1.3$, $\int_{0.5}^{4} \frac{\mathrm{d}x}{|x+b|} \approx 1.07992$, matching $\ln \frac{5.3}{1.8}$ from the formula.

Common slips

- Writing $\ln|t + b|$ without noticing that $t + b > 0$, and so missing what goes wrong when $b^2 = c$.
- Taking $\sqrt{(x+b)^2} = x + b$ in the final part, which hides the failure for $x + b < 0$ entirely.

Beyond STEP

The substitution $t = x + \sqrt{x^2 + 2bx + c}$ is one of *Euler's substitutions*, which turn any integral of a rational function of x and $\sqrt{ax^2 + bx + c}$ into the integral of a rational function of t. The final part illustrates why a change of variable needs the new variable to depend smoothly and one-to-one on the old one: when t is constant the method collapses. Bartle and Sherbert, *Introduction to Real Analysis*, states the change-of-variables theorem with these hypotheses made precise.

WORKED SOLUTION

STEP I 2008, Question 3

Pure • Inequalities by Rearrangement

> **STEP I 2008, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/08-s1-q3

The key idea

The inequality $(*)$ is just a product of two non-negative factors in disguise. Once we have it, every later inequality is a matter of choosing a, b, c, d so that the two sides of $(*)$ become the two sides we want, while making sure the conditions $c \geq a$ and $d \geq b$ really hold. Because the target inequalities are symmetric, we are free to put the variables in order first, and that ordering is what supplies those conditions.

Solution

Proving $(*)$

Bring everything to one side and factorise:

$$ab + cd - bc - ad = a(b-d) - c(b-d) = (c-a)(d-b).$$

If $c \geq a$ and $d \geq b$, both factors are non-negative, so their product is non-negative. Hence $ab + cd \geq bc + ad$, with equality exactly when $c = a$ or $d = b$.

Part (i)

Two variables. Suppose $x \geq y$. In $(*)$ take

$$a = y, \quad b = y, \quad c = x, \quad d = x.$$

The conditions $c \geq a$ and $d \geq b$ both say $x \geq y$, which is given. Then $ab + cd = y^2 + x^2$ and $bc + ad = xy + xy$, so $(*)$ gives

$$x^2 + y^2 \geq 2xy.$$

Three variables. Now suppose also that $x \geq z$ and $y \geq z$. Take

$$a = z, \quad b = z, \quad c = x, \quad d = y.$$

Here $c \geq a$ is $x \geq z$ and $d \geq b$ is $y \geq z$, both given. Now $ab + cd = z^2 + xy$ and $bc + ad = zx + zy$, so

$$z^2 + xy \geq xz + yz.$$

Adding this to $x^2 + y^2 \geq 2xy$ (which we may use, since $x \geq y$):

$$x^2 + y^2 + z^2 + xy \geq 2xy + xz + yz,$$

and subtracting xy from both sides gives

$$x^2 + y^2 + z^2 \geq xy + yz + zx.$$

So far this is proved only when $x \geq y \geq z$.

Ask yourself How can a result proved only for $x \geq y \geq z$ give one for all x, y, z?

Both sides of $x^2 + y^2 + z^2 \geq xy + yz + zx$ are symmetric: swapping any two of the variables leaves each side unchanged. Given any real x, y, z, list them in decreasing order as $X \geq Y \geq Z$. By what we have just shown, $X^2 + Y^2 + Z^2 \geq XY + YZ + ZX$. But this is the same inequality as the one for x, y, z, only with the terms in a different order. So the inequality holds for all real x, y and z.

(Equality needs equality in both inequalities we added, which forces $x = y = z$.)

Part (ii)

The expression is the sum of three pairs, each a number plus its reciprocal:

$$\left(\frac{s}{t} + \frac{t}{s}\right) + \left(\frac{t}{r} + \frac{r}{t}\right) + \left(\frac{r}{s} + \frac{s}{r}\right).$$

So it is enough to show that $\dfrac{p}{q} + \dfrac{q}{p} \geq 2$ for any positive p, q, the analogue of $x^2 + y^2 \geq 2xy$.

The expression $\frac{p}{q} + \frac{q}{p}$ is symmetric in p and q, so we may label them so that $p \geq q > 0$. In $(*)$ take

$$a = q, \quad b = \frac{1}{p}, \quad c = p, \quad d = \frac{1}{q}.$$

Then $c \geq a$ is $p \geq q$, and $d \geq b$ is $\frac{1}{q} \geq \frac{1}{p}$, which also follows from $p \geq q$ because p and q are positive. Now

$$ab + cd = \frac{q}{p} + \frac{p}{q}, \qquad bc + ad = \frac{p}{p} + \frac{q}{q} = 2,$$

so $(*)$ gives $\dfrac{p}{q} + \dfrac{q}{p} \geq 2$.

Applying this to the pairs (s,t), (t,r) and (r,s) and adding the three results,

$$\frac{s}{t} + \frac{t}{r} + \frac{r}{s} + \frac{t}{s} + \frac{r}{t} + \frac{s}{r} \geq 2 + 2 + 2 = 6$$

for all positive r, s and t, as required. Equality holds exactly when $r = s = t$, since each pair needs its two numbers equal.

Checks

- Try $x = 3$, $y = 1$, $z = -2$ in part (i): the left side is 14 and the right side is $3 - 2 - 6 = -5$. Both sides are unchanged if we permute the three values, as the symmetry argument needs.
- An independent proof of part (i): $2(x^2 + y^2 + z^2 - xy - yz - zx) = (x - y)^2 + (y - z)^2 + (z - x)^2 \geq 0$.
- With $r = 2$, $s = 1$, $t = 1$ the expression in part (ii) is $1 + \frac{1}{2} + 2 + 1 + 2 + \frac{1}{2} = 7 \geq 6$. A computer checking every step on 20,000 random exact rational cases found no failures, and values such as $r = 1.001$, $s = 1$, $t = 0.999$ give a sum just above 6.

Common slips

- Using $(*)$ without checking $c \geq a$ and $d \geq b$. With the wrong assignment the inequality can point the other way.
- In part (ii), forgetting that $\frac{1}{q} \geq \frac{1}{p}$ uses positivity: for negative numbers taking reciprocals does not reverse the order in the same way.
- Proving part (i) only for $x \geq y \geq z$ and not explaining why symmetry gives the general case.

Beyond STEP

The inequality $(*)$ is the two-term case of the *rearrangement inequality*: if two lists are sorted, the sum of products is largest when they are paired in the same order and smallest when paired in opposite orders. Both results here, and AM–GM for two terms, drop out of it, and it is closely related to Chebyshev's sum inequality. Siklos, *Advanced Problems in Mathematics*, has more STEP-style inequality practice, and Bartle and Sherbert, *Introduction to Real Analysis*, shows how such order arguments underpin analysis.

WORKED SOLUTION

STEP I 2008, Question 4

Pure • Convex Functions

STEP I 2008, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q4

The key idea

Both functions are exponentials, $e^{h(x)}$, and differentiating twice gives $\left(h'' + (h')^2\right)e^h$. The exponential factor is always positive, so convexity comes down to the sign of $h'' + (h')^2$, a purely trigonometric inequality. In (i), that inequality turns out to compare exactly the two curves we're asked to sketch. In (ii), it simplifies to a comparison of $\sin 2x$ with $\sin 2\alpha$.

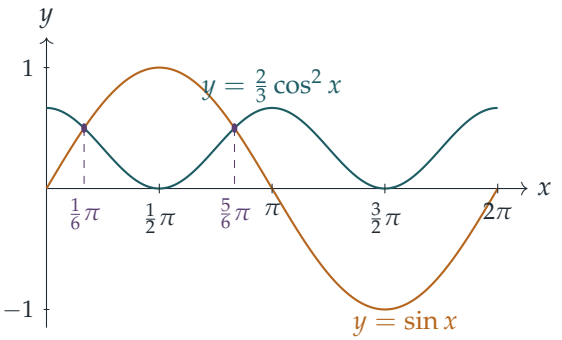
Solution

Part (i)

The sketch. The curve $y = \frac{2}{3}\cos^2 x$ lies between 0 and $\frac{2}{3}$. It equals $\frac{2}{3}$ at $x = 0, \pi, 2\pi$ and touches the axis at $x = \frac{1}{2}\pi, \frac{3}{2}\pi$, with period π. The curve $y = \sin x$ is the usual sine wave. To find where they cross, put $s = \sin x$ and $\cos^2 x = 1 - s^2$:

$$\tfrac{2}{3}(1 - s^2) = s \iff 2s^2 + 3s - 2 = 0 \iff (2s - 1)(s + 2) = 0.$$

Since $s = -2$ is impossible, $\sin x = \frac{1}{2}$, so the curves cross at $x = \frac{1}{6}\pi$ and $x = \frac{5}{6}\pi$ only. On $(\pi, 2\pi)$ the sine curve is negative while the other curve is non-negative, so there are no crossings there.



Convexity of f. With $f(x) = \mathrm{e}^{\frac{2}{3}\sin x}$, the chain rule gives $f'(x) = \frac{2}{3}\cos x\, f(x)$, and then by the product rule

$$\begin{aligned} f''(x) &= -\tfrac{2}{3}\sin x\, f(x) + \tfrac{2}{3}\cos x\, f'(x) \\ &= f(x)\left(\tfrac{4}{9}\cos^2 x - \tfrac{2}{3}\sin x\right) \\ &= \tfrac{2}{3}f(x)\left(\tfrac{2}{3}\cos^2 x - \sin x\right). \end{aligned}$$

Ask yourself Why does the bracket settle everything?

Since $f(x)$ is an exponential, it's strictly positive, so $f''(x) \geq 0$ exactly when $\frac{2}{3}\cos^2 x \geq \sin x$. In other words, f is convex exactly where the teal curve lies on or above the amber one. That's why the question asks for the sketch first.

From the sketch, and the crossing points found above, $\frac{2}{3}\cos^2 x \geq \sin x$ for $0 < x \leq \frac{1}{6}\pi$ and for $\frac{5}{6}\pi \leq x < 2\pi$. Between $\frac{1}{6}\pi$ and $\frac{5}{6}\pi$ the sine curve is above (for instance at $x = \frac{1}{2}\pi$, where the two values are 0 and 1). So f is convex on

$$0 < x \leq \tfrac{1}{6}\pi \qquad \text{and} \qquad \tfrac{5}{6}\pi \leq x < 2\pi,$$

and on no larger interval. (Whether you include the endpoints $\frac{1}{6}\pi$ and $\frac{5}{6}\pi$, where $f'' = 0$, is a matter of taste.)

Part (ii)

With $g(x) = \mathrm{e}^{-k\tan x}$ we have $g'(x) = -k\sec^2 x\, g(x)$. Using $\frac{\mathrm{d}}{\mathrm{d}x}\sec^2 x = 2\sec^2 x\tan x$ and the product rule,

$$\begin{aligned} g''(x) &= -2k\sec^2 x\tan x\, g(x) - k\sec^2 x\, g'(x) \\ &= g(x)\left(k^2\sec^4 x - 2k\sec^2 x\tan x\right) \\ &= k\sec^2 x\, g(x)\left(k\sec^2 x - 2\tan x\right). \end{aligned}$$

Here $k = \sin 2\alpha > 0$ because $0 < 2\alpha < \frac{1}{2}\pi$, and $\sec^2 x > 0$, $g(x) > 0$. So $g''(x) \geq 0$ exactly when $k\sec^2 x \geq 2\tan x$. Multiplying by $\cos^2 x > 0$, this becomes

$$k \geq 2\tan x\cos^2 x = 2\sin x\cos x = \sin 2x.$$

So g is convex exactly where

$$\sin 2x \leq \sin 2\alpha.$$

Ask yourself Where is $\sin 2x \leq \sin 2\alpha$ when $0 < x < \frac{1}{2}\pi$?

As x runs over $(0, \frac{1}{2}\pi)$, $2x$ runs over $(0, \pi)$. On that interval sin increases up to $\frac{1}{2}\pi$ and then decreases, symmetrically about $\frac{1}{2}\pi$. The value $\sin 2\alpha$ (with $0 < 2\alpha < \frac{1}{2}\pi$) is

reached at $2x = 2\alpha$ and at $2x = \pi - 2\alpha$. So $\sin 2x \leq \sin 2\alpha$ exactly when $2x \leq 2\alpha$ or $2x \geq \pi - 2\alpha$. That is,

$$0 < x \leq \alpha \qquad \text{or} \qquad \tfrac{1}{2}\pi - \alpha \leq x < \tfrac{1}{2}\pi.$$

In particular, for $0 < x < \alpha$ we have $0 < 2x < 2\alpha < \frac{1}{2}\pi$, where sin is increasing, so $\sin 2x < \sin 2\alpha = k$ and $g''(x) > 0$. Hence g is convex in $0 < x < \alpha$, as required.

The other interval is

$$\tfrac{1}{2}\pi - \alpha < x < \tfrac{1}{2}\pi.$$

This is genuinely different from the first, because $\alpha < \frac{1}{4}\pi$ gives $\frac{1}{2}\pi - \alpha > \frac{1}{4}\pi > \alpha$. On the middle stretch $\alpha < x < \frac{1}{2}\pi - \alpha$, g is not convex.

Checks

- At $x = \frac{1}{6}\pi$: $\frac{2}{3}\cos^2 x = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2} = \sin x$, so $f'' = 0$ there, as it should be at a boundary.
- In (ii), the problem is symmetric under $x \mapsto \frac{1}{2}\pi - x$, since $\sin 2x$ is unchanged. That explains why the second interval mirrors the first.
- Numerically, with $\alpha = 0.5$ ($k \approx 0.841$), a finite-difference second derivative of g is positive at $x = 0.3$ and at $x = 1.3$, and negative at $x = 0.8$. Since $\frac{1}{2}\pi - \alpha \approx 1.071$, this matches the intervals found. A computer testing many random α and x agreed with both parts.

Common slips

- Forgetting the $(h')^2$ term when differentiating $\mathrm{e}^{h(x)}$ twice. The second derivative is not just $h''\mathrm{e}^h$.
- Missing the whole of $(\pi, 2\pi)$ in (i). There $\sin x < 0 \leq \frac{2}{3}\cos^2 x$, so f is convex all the way from $\frac{5}{6}\pi$ to 2π.
- Stopping at $\sin 2x \leq \sin 2\alpha \Rightarrow x \leq \alpha$. Because sin isn't monotonic on $(0, \pi)$, this loses the second interval.

Beyond STEP

The second-derivative test is only one face of convexity. The real definition is geometric: every chord lies on or above the graph. For differentiable functions this is equivalent to f' being increasing, and it leads to Jensen's inequality, which underlies AM–GM and much of probability and optimisation. Convex functions are treated carefully in Bartle and Sherbert, *Introduction to Real Analysis*. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

WORKED SOLUTION

STEP I 2008, Question 5

Pure • Polynomial Bounds

STEP I 2008, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q5

The key idea

For $n = 1$ the polynomial is a straight line, and its values at the two ends of the interval are tied together: they differ by exactly 2, so they can't both be small. In part (ii) the trick is to divide by the leading coefficient so that Chebyshev's theorem applies to the curve itself. The theorem then forces $|y|$ to be large somewhere in $[-1, 1]$, while the values at the endpoints are small. A curve that climbs higher than both of its endpoint values (or dips lower than both) must turn round somewhere in between.

Solution

Part (i)

The case $n = 1$. Here $\mathrm{p}(x) = x + a_0$, and we need $M \geq 2^0 = 1$. Since M is at least the value of $|\mathrm{p}|$ at any point of $[-1, 1]$, it's at least the values at the endpoints:

$$M \geq |\mathrm{p}(1)| = |1 + a_0| \quad \text{and} \quad M \geq |\mathrm{p}(-1)| = |a_0 - 1|.$$

Adding, and using the triangle inequality $|u| + |v| \geq |u - v|$,

$$2M \geq |1 + a_0| + |a_0 - 1| \geq \big|(1 + a_0) - (a_0 - 1)\big| = 2,$$

so $M \geq 1 = 2^{1-n}$, which proves the theorem for $n = 1$. (Equality holds for $\mathrm{p}(x) = x$, so the bound can't be improved.)

(a) For $\mathrm{p}(x) = x^2 - \frac{1}{2}$ and $|x| \leq 1$ we have $0 \leq x^2 \leq 1$, so $-\frac{1}{2} \leq \mathrm{p}(x) \leq \frac{1}{2}$. The value $\frac{1}{2}$ is attained at $x = \pm 1$ (and $-\frac{1}{2}$ at $x = 0$), so

$$M = \tfrac{1}{2} = 2^{1-2}.$$

The theorem holds, with equality.

(b) For $\mathrm{p}(x) = x^3 - x$ we need $M \geq 2^{1-3} = \frac{1}{4}$. It's enough to find a single point where $|\mathrm{p}| \geq \frac{1}{4}$. The stationary points are where $3x^2 - 1 = 0$, that is $x = \pm\frac{1}{\sqrt{3}}$, both in $[-1, 1]$,

and

$$\mathrm{p}\left(-\tfrac{1}{\sqrt{3}}\right) = -\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{2}{3\sqrt{3}}.$$

Since $\mathrm{p}(\pm 1) = 0$, this is in fact the greatest value: $M = \frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9} \approx 0.385$. To compare with $\frac{1}{4}$ without a calculator, square both sides:

$$\left(\frac{2}{3\sqrt{3}}\right)^2 = \frac{4}{27} > \frac{4}{64} = \frac{1}{16} = \left(\frac{1}{4}\right)^2,$$

so $M > \frac{1}{4}$ and the theorem holds (this time with room to spare).

Part (ii)

Write $f(x) = 64x^5 + 25x^4 - 66x^3 - 24x^2 + 3x + 1$. Chebyshev's theorem is about polynomials with leading coefficient 1, so we apply it to

$$\mathrm{p}(x) = \tfrac{1}{64} f(x) = x^5 + \tfrac{25}{64}x^4 - \tfrac{66}{64}x^3 - \tfrac{24}{64}x^2 + \tfrac{3}{64}x + \tfrac{1}{64},$$

which has the required form with $n = 5$. The theorem gives $\max_{|x|\le 1} |\mathrm{p}(x)| \ge 2^{-4} = \frac{1}{16}$, and multiplying by 64,

$$\max_{|x|\le 1} |f(x)| \ge 4.$$

So there is some $x_0 \in [-1, 1]$ with $|f(x_0)| \ge 4$.

Now look at the endpoints:

$$f(1) = 64 + 25 - 66 - 24 + 3 + 1 = 3,$$
$$f(-1) = -64 + 25 + 66 - 24 - 3 + 1 = 1.$$

Both have modulus less than 4, so x_0 is not an endpoint: $-1 < x_0 < 1$.

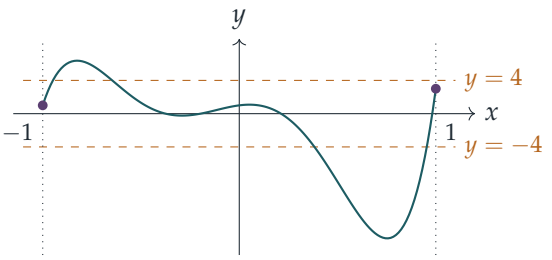
Ask yourself We know $|f|$ gets big inside the interval. Why does that force a turning point?

There are two cases.

- If $f(x_0) \ge 4$, then $f(x_0)$ is bigger than both $f(-1) = 1$ and $f(1) = 3$. The continuous function f attains a greatest value on the closed interval $[-1, 1]$; this value is at least $f(x_0) \ge 4$, so it isn't attained at either endpoint. It is therefore attained at an interior point, which is a local maximum of f.
- If $f(x_0) \le -4$, then $f(x_0)$ is below both endpoint values, and in the same way the least value of f on $[-1, 1]$ is attained at an interior point, which is a local minimum.

Either way f has a local maximum or minimum strictly inside $(-1, 1)$. There $f' = 0$, and since f is a non-constant polynomial, f' has only finitely many zeros. So f' keeps one sign just to the left of the point and one sign just to the right. Because the point is a local maximum or minimum, those signs must be opposite, so it is a genuine turning point. Hence the curve has at least one turning point with $-1 \le x \le 1$, as required.

Notice that we never had to solve the quartic $f'(x) = 0$. That's the point of the question.



The sketch shows the curve on $[-1, 1]$. Both endpoint values lie between the dashed lines $y = \pm 4$, but the curve goes outside that band. In fact it has four turning points in the interval, the deepest near $x \approx 0.75$ with $y \approx -15$.

Checks

- For $n = 1$, trying a few values of a_0 shows that $\max(|1 + a_0|, |1 - a_0|)$ is never below 1, and equals 1 only when $a_0 = 0$.
- In (b), $\frac{2}{3\sqrt{3}} \approx 0.3849$, comfortably above 0.25. Sampling $x^3 - x$ on a fine grid confirms this is the maximum of $|x^3 - x|$.
- In (ii), a numerical scan of f on $[-1, 1]$ gives $\max |f| \approx 15.0$ (at $x \approx 0.753$, where $f \approx -15.0$), well above the guaranteed 4. It also finds f' changing sign at $x \approx -0.825$, -0.292, 0.052, 0.753.

Common slips

- Applying the theorem to f directly. Its leading coefficient is 64, not 1, so you must divide by 64 first. The bound becomes $64 \cdot 2^{-4} = 4$.
- Stopping at "$|f| \geq 4$ somewhere". That alone gives no turning point: you need the endpoint values 1 and 3 to show that the extreme value is reached inside the interval.
- In (b), comparing $\frac{2}{3\sqrt{3}}$ with $\frac{1}{4}$ carelessly. Squaring both positive numbers makes the comparison exact.

Beyond STEP

Equality in Chebyshev's theorem holds exactly for $2^{1-n}T_n(x)$, where T_n is the Chebyshev polynomial defined by $T_n(\cos\theta) = \cos n\theta$. Part (a) is $\frac{1}{2}T_2$, and the $n = 1$ case is $T_1(x) = x$. These polynomials are the starting point of approximation theory: they explain why interpolation works best at Chebyshev nodes rather than at equally spaced points. The argument in part (ii) uses the extreme value theorem (a continuous function on a closed interval attains its bounds), which is proved carefully in a first analysis course. Abbott's *Understanding Analysis* is a good place to read it.

WORKED SOLUTION

STEP I 2008, Question 6

Pure • Inverse Functions and Areas

STEP I 2008, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q6

The key idea

The graph of an inverse function is the reflection of the original graph in the line $y = x$, so an integral of g with respect to y is an area measured *sideways*, between the curve $y = \mathrm{f}(x)$ and the y-axis. The upper limit k is no accident: it is exactly $\mathrm{f}(\frac{1}{2})$. Once you spot that, the two areas fit together to fill a rectangle.

Solution

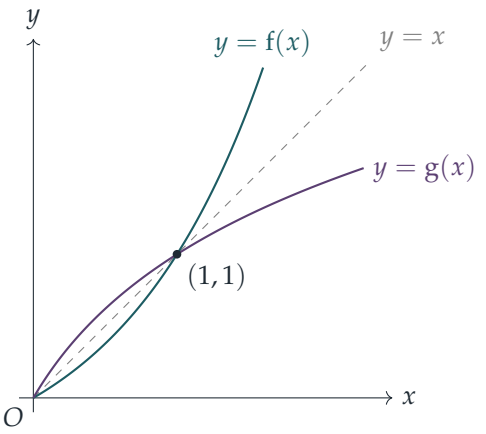
The sketch

The function f is increasing, with $\mathrm{f}(0) = 0$ and $\mathrm{f}(1) = 1$. It is convex, since $\mathrm{f}''(x) = \mathrm{e}^x/(\mathrm{e}-1) > 0$, and its gradient at the origin is $\mathrm{f}'(0) = 1/(\mathrm{e}-1) \approx 0.58 < 1$. So on $0 < x < 1$ the curve lies below the line $y = x$, it crosses that line at $(1,1)$, and then grows exponentially.

To find g, solve $y = \dfrac{\mathrm{e}^x - 1}{\mathrm{e} - 1}$ for x:

$$\mathrm{g}(x) = \ln\bigl(1 + (\mathrm{e}-1)x\bigr), \qquad x \geq 0,$$

since the range of f is $[0, \infty)$. Its graph is the reflection of $y = \mathrm{f}(x)$ in $y = x$: it passes through $(0,0)$ with gradient $\mathrm{e} - 1 \approx 1.72$, is concave, crosses $y = x$ at $(1,1)$ and then grows only logarithmically.



The first integral

$$\int_0^{1/2} \mathrm{f}(x)\,\mathrm{d}x = \frac{1}{\mathrm{e}-1}\Big[\mathrm{e}^x - x\Big]_0^{1/2} = \frac{1}{\mathrm{e}-1}\Big(\sqrt{\mathrm{e}} - \tfrac{1}{2} - 1\Big) = \frac{\sqrt{\mathrm{e}} - \frac{3}{2}}{\mathrm{e}-1}.$$

The second integral

First notice that

$$\mathrm{f}(\tfrac{1}{2}) = \frac{\sqrt{\mathrm{e}}-1}{\mathrm{e}-1} = \frac{\sqrt{\mathrm{e}}-1}{(\sqrt{\mathrm{e}}-1)(\sqrt{\mathrm{e}}+1)} = \frac{1}{\sqrt{\mathrm{e}}+1} = k,$$

using the difference of two squares $\mathrm{e}-1 = (\sqrt{\mathrm{e}})^2 - 1$. Equivalently $\mathrm{g}(k) = \frac{1}{2}$, and $1 + (\mathrm{e}-1)k = 1 + (\sqrt{\mathrm{e}}-1) = \sqrt{\mathrm{e}}$.

? **Ask yourself** How do we integrate $\ln\big(1 + (\mathrm{e}-1)x\big)$?

Substitute $u = 1 + (\mathrm{e}-1)x$, so $\mathrm{d}x = \mathrm{d}u/(\mathrm{e}-1)$, with limits $u = 1$ and $u = \sqrt{\mathrm{e}}$. Then use $\int \ln u\,\mathrm{d}u = u\ln u - u$ (integration by parts with $\ln u$ times 1):

$$\begin{aligned}\int_0^k \mathrm{g}(x)\,\mathrm{d}x &= \frac{1}{\mathrm{e}-1}\int_1^{\sqrt{\mathrm{e}}} \ln u\,\mathrm{d}u = \frac{1}{\mathrm{e}-1}\Big[u\ln u - u\Big]_1^{\sqrt{\mathrm{e}}} \\ &= \frac{1}{\mathrm{e}-1}\Big(\tfrac{1}{2}\sqrt{\mathrm{e}} - \sqrt{\mathrm{e}} - (0-1)\Big) = \frac{1 - \frac{1}{2}\sqrt{\mathrm{e}}}{\mathrm{e}-1},\end{aligned}$$

using $\ln\sqrt{\mathrm{e}} = \frac{1}{2}$.

Adding

$$\int_0^{1/2} f(x)\,\mathrm{d}x + \int_0^k \mathrm{g}(x)\,\mathrm{d}x = \frac{\sqrt{\mathrm{e}} - \frac{3}{2} + 1 - \frac{1}{2}\sqrt{\mathrm{e}}}{\mathrm{e}-1} = \frac{\frac{1}{2}(\sqrt{\mathrm{e}}-1)}{(\sqrt{\mathrm{e}}-1)(\sqrt{\mathrm{e}}+1)} = \frac{1}{2(\sqrt{\mathrm{e}}+1)},$$

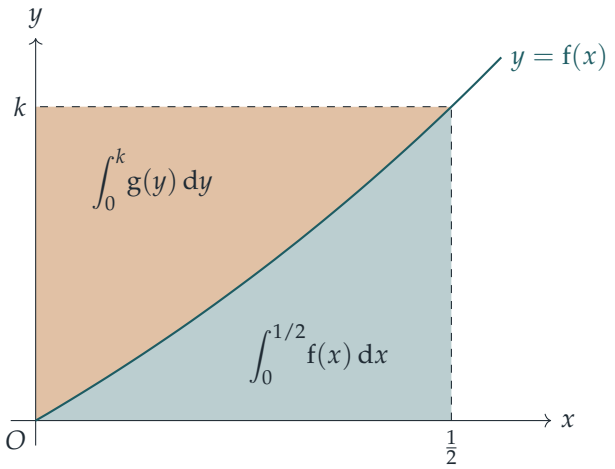
as required.

Explaining the result

The answer is $\frac{1}{2} \times k$, which is the area of the rectangle with corners $(0,0)$ and $(\frac{1}{2},k)$. Here is why.

> **Ask yourself** Where is $\int_0^k \mathrm{g}(x)\,\mathrm{d}x$ on the graph of f?

Reflecting in $y = x$ preserves area. The region under $y = \mathrm{g}(x)$ for $0 \le x \le k$ reflects to the region between the curve $y = \mathrm{f}(x)$ and the y-axis, for $0 \le y \le k$. In other words, it is $\int_0^k x\,\mathrm{d}y$ along the curve $y = \mathrm{f}(x)$: the area to the *left* of the curve. Since $\mathrm{f}(\frac{1}{2}) = k$, the curve runs from the corner $(0,0)$ to the corner $(\frac{1}{2},k)$ of the rectangle and splits it into two pieces: the area below the curve (the first integral, teal) and the area to its left (the second integral, amber).



So the two integrals together make up the whole rectangle, of area $\frac{1}{2}k = \dfrac{1}{2(\sqrt{\mathrm{e}}+1)}$. The same picture gives the general fact that, for an increasing function with $\mathrm{f}(0) = 0$,

$$\int_0^a \mathrm{f}(x)\,\mathrm{d}x + \int_0^{\mathrm{f}(a)} \mathrm{f}^{-1}(y)\,\mathrm{d}y = a\,\mathrm{f}(a).$$

Checks

Numerically, the first integral is about 0.08655 and the second about 0.10222; Simpson's rule on each agrees with the exact forms to twelve decimal places, and their sum 0.18877 equals $1/\big(2(\sqrt{\mathrm{e}}+1)\big)$. As a sanity check on the sketch, both pieces are positive and the teal piece is smaller, as it should be: the convex curve sags below the diagonal of the rectangle.

Common slips

- Not noticing that $k = \mathrm{f}(\frac{1}{2})$, and so missing the simplification $1 + (\mathrm{e} - 1)k = \sqrt{\mathrm{e}}$.
- Drawing f above $y = x$ near the origin: its gradient there is $1/(\mathrm{e} - 1) < 1$.
- In the diagram, shading the area under $y = \mathrm{g}(x)$ on a separate picture without saying how it relates to the region beside $y = \mathrm{f}(x)$; the point is that reflection carries one to the other.

Beyond STEP

The identity $\int_0^a \mathrm{f} + \int_0^{\mathrm{f}(a)} \mathrm{f}^{-1} = a\,\mathrm{f}(a)$ is the equality case of *Young's inequality*: for any $b \geq 0$, $\int_0^a \mathrm{f} + \int_0^b \mathrm{f}^{-1} \geq ab$. Taking $\mathrm{f}(x) = x^{p-1}$ gives $ab \leq a^p/p + b^q/q$ with $1/p + 1/q = 1$, the starting point for Hölder's inequality in analysis. Abbott's *Understanding Analysis* is a good place to see how integrals like these are made rigorous.

WORKED SOLUTION

STEP I 2008, Question 7

Pure • Rotations and Distances

STEP I 2008, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q7

The key idea

A perpendicular distance to a line through the origin becomes easy once we rotate the plane so that the line lies along the x-axis: the distance is then just the size of the new y-coordinate. The rotation formula from the first part does exactly this for two of the three sides. For the final part, the trick is to drop the modulus signs: the three *signed* distances always add up to the same constant, and the moduli can only make the total bigger.

Solution

The rotation formulae

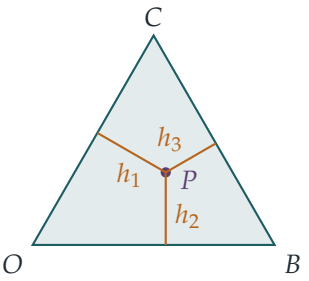
Write $x = r\cos\theta$, $y = r\sin\theta$, where $r = OP$ and θ is the angle from the positive x-axis to OP. Rotating OP clockwise by 60° keeps r and changes the angle to $\theta - 60^\circ$, so the new y-coordinate is

$$\begin{aligned} r\sin(\theta - 60^\circ) &= r\sin\theta\cos 60^\circ - r\cos\theta\sin 60^\circ \\ &= \tfrac{1}{2}y - \tfrac{\sqrt{3}}{2}x = \tfrac{1}{2}\left(y - \sqrt{3}\,x\right), \end{aligned}$$

as required. For an anticlockwise rotation the angle becomes $\theta + 60^\circ$, and the same expansion with the sign of $\sin 60^\circ$ reversed gives the new y-coordinate

$$\tfrac{1}{2}\left(y + \sqrt{3}\,x\right).$$

The three distances



h_1. The line OC makes an angle of $60°$ with the x-axis. Rotating the whole picture clockwise by $60°$ about O carries OC onto the x-axis and preserves distances. In the rotated picture the distance from P to the x-axis is the modulus of its new y-coordinate, so

$$h_1 = \tfrac{1}{2}\left| y - \sqrt{3}\,x \right|.$$

h_2. The line OB is the x-axis, so $h_2 = |y|$.

h_3. Now rotate anticlockwise by $60°$ about O. This sends $B = (1,0)$ to $(\tfrac{1}{2}, \tfrac{1}{2}\sqrt{3})$ and C (at angle $60°$) to the point at angle $120°$, namely $(-\tfrac{1}{2}, \tfrac{1}{2}\sqrt{3})$. So the line BC becomes the horizontal line $y = \tfrac{1}{2}\sqrt{3}$, while P acquires the y-coordinate $\tfrac{1}{2}(y + \sqrt{3}\,x)$. Hence

$$h_3 = \left| \tfrac{1}{2}(y + \sqrt{3}\,x) - \tfrac{1}{2}\sqrt{3} \right| = \tfrac{1}{2}\left| \sqrt{3}\,x + y - \sqrt{3} \right|.$$

(Equivalently: BC has equation $\sqrt{3}\,x + y = \sqrt{3}$, and the usual distance formula gives the same answer, since $\sqrt{3+1} = 2$.)

The final part

Define the signed quantities

$$s_1 = \tfrac{1}{2}(\sqrt{3}\,x - y), \qquad s_2 = y, \qquad s_3 = \tfrac{1}{2}(\sqrt{3} - \sqrt{3}\,x - y),$$

so that $h_i = |s_i|$ for $i = 1, 2, 3$. Adding,

$$s_1 + s_2 + s_3 = \tfrac{1}{2}\sqrt{3}\,x - \tfrac{1}{2}y + y + \tfrac{1}{2}\sqrt{3} - \tfrac{1}{2}\sqrt{3}\,x - \tfrac{1}{2}y = \tfrac{1}{2}\sqrt{3}$$

for *every* point (x, y).

Ask yourself How do the signs of s_1, s_2, s_3 relate to the triangle?

Each s_i is a linear expression that vanishes on one side of the triangle. Testing the centroid $(\tfrac{1}{2}, \tfrac{1}{6}\sqrt{3})$, which is inside, gives $s_1 = s_2 = s_3 = \tfrac{1}{6}\sqrt{3} > 0$. So $s_2 \geq 0$ is the side of OB containing C, $s_1 \geq 0$ is the side of OC containing B, and $s_3 \geq 0$ is the side of BC containing O. The closed triangle OBC is exactly the intersection of these three half-planes, so

$$P \text{ lies on or in } OBC \iff s_1 \geq 0,\ s_2 \geq 0,\ s_3 \geq 0.$$

Now, for any real number s we have $|s| \geq s$, with equality if and only if $s \geq 0$. Therefore

$$h_1 + h_2 + h_3 = |s_1| + |s_2| + |s_3| \;\geq\; s_1 + s_2 + s_3 = \tfrac{1}{2}\sqrt{3},$$

with equality if and only if $|s_i| = s_i$ for each i, that is, if and only if all three s_i are non-negative. By the previous paragraph, this happens exactly when P lies on or in the triangle OBC. This proves both directions at once:

$$h_1 + h_2 + h_3 = \tfrac{1}{2}\sqrt{3} \iff P \text{ lies on or in triangle } OBC.$$

Note also that outside the triangle the sum is strictly greater than $\frac{1}{2}\sqrt{3}$.

Checks

- At a vertex, say $P = B = (1, 0)$: $h_1 = \frac{1}{2}\sqrt{3}$, $h_2 = 0$, $h_3 = 0$, sum $\frac{1}{2}\sqrt{3}$. Good.
- At the centroid each distance is $\frac{1}{6}\sqrt{3}$, and three of them make $\frac{1}{2}\sqrt{3} \approx 0.8660$, the height of the triangle.
- Outside, at $P = (2, 0)$: $h_1 = \sqrt{3}$, $h_2 = 0$, $h_3 = \frac{1}{2}\sqrt{3}$, sum $\frac{3}{2}\sqrt{3} > \frac{1}{2}\sqrt{3}$.
- We compared all three formulae with directly computed point-to-line distances at 10^5 random points (agreement to 10^{-15}), and tested the "if and only if" at 2×10^5 points, including many on the edges, with no exceptions.

Common slips

- Getting the sign of the rotation wrong: clockwise means the angle *decreases*, giving $\theta - 60^\circ$.
- Proving only one direction. Checking that the sum is $\frac{1}{2}\sqrt{3}$ inside the triangle is not enough; you must also show it is *not* $\frac{1}{2}\sqrt{3}$ outside. The inequality $|s| \geq s$ handles both at once.
- Forgetting the modulus signs in h_1, h_2, h_3: distances are never negative.

Beyond STEP

This is Viviani's theorem: in an equilateral triangle the sum of the distances from an interior point to the sides equals the height. The signed distances s_1, s_2, s_3 are (up to scaling) the *barycentric* or *trilinear coordinates* of P, which always sum to a constant and are all non-negative exactly inside the triangle; they are a standard tool in geometry, computer graphics and convex analysis. The rotation formula is the second row of a 2×2 rotation matrix, and Strang, *Introduction to Linear Algebra*, is a good place to see rotations as linear maps.

WORKED SOLUTION

STEP I 2008, Question 8

Pure • Clairaut Differential Equations

STEP I 2008, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q8

The key idea

Both equations involve y' but not y'', and they are not the kind we can separate or integrate directly. Differentiating once more is the trick: the terms in y' alone cancel, and what is left factorises with y'' as one factor. Each factor gives a case, and in each case we substitute back into the *original* equation rather than integrating, which pins down the arbitrary constant for free.

Solution

Throughout we write $p = y'$, so $p' = y''$.

Part (i)

Differentiate $p^2 - xp + y = 0$ with respect to x, using the product rule on xp:

$$2p\,y'' - (p + x\,y'') + p = 0 \quad \Longrightarrow \quad y''\,(2p - x) = 0.$$

The two p terms cancel, so at each point either $y'' = 0$ or $2y' = x$, as required.

Case $y'' = 0$. Then y' is a constant m and $y = mx + c$. Substituting into $(*)$:

$$m^2 - xm + (mx + c) = 0 \quad \Longrightarrow \quad c = -m^2.$$

So the curve is the straight line $y = mx - m^2$.

Case $2y' = x$. Put $p = \frac{1}{2}x$ into $(*)$:

$$\tfrac{1}{4}x^2 - \tfrac{1}{2}x^2 + y = 0 \quad \Longrightarrow \quad 4y = x^2.$$

Ask yourself Why not just integrate $y' = \frac{1}{2}x$ to get $y = \frac{1}{4}x^2 + k$?

We could, but differentiating $(*)$ threw information away: any $y = \frac{1}{4}x^2 + k$ satisfies $2y' = x$, yet only $k = 0$ satisfies $(*)$ itself (for example, $k = 1$ at $x = 0$ gives $0 - 0 + 1 \neq 0$).

Substituting straight back into $(*)$ uses all the information at once and gives the constant directly. The same remark explains why $c = -m^2$ in the first case.

So the curve is either a line $y = mx + c$ with $c = -m^2$, or the parabola $4y = x^2$.

The two answers are linked: the line $y = mx - m^2$ meets $4y = x^2$ where $x^2 - 4mx + 4m^2 = 0$, that is $(x - 2m)^2 = 0$, so each line is the tangent to the parabola at $(2m, m^2)$. The parabola is the *envelope* of the family of lines.

Part (ii)

Differentiate $(x^2 - 1)p^2 - 2xyp + y^2 - 1 = 0$ with respect to x:

$$2xp^2 + 2(x^2 - 1)p\,y'' - 2yp - 2xp^2 - 2xy\,y'' + 2yp = 0.$$

The terms without y'' cancel in pairs, leaving

$$2y''\big[(x^2 - 1)p - xy\big] = 0.$$

Case $y'' = 0$. Then $y = mx + c$. It helps to notice that the equation can be rearranged as

$$(y - xp)^2 = 1 + p^2,$$

since $(y - xp)^2 - p^2 - 1 = y^2 - 2xyp + (x^2 - 1)p^2 - 1$. For the line, $y - xp = c$ and $p = m$, so

$$c^2 = 1 + m^2, \qquad \text{that is,} \qquad y = mx \pm \sqrt{1 + m^2}.$$

Case $(x^2 - 1)p = xy$. Where $x^2 \neq 1$ we have $p = \dfrac{xy}{x^2 - 1}$, so

$$y - xp = y - \frac{x^2 y}{x^2 - 1} = \frac{-y}{x^2 - 1}.$$

Substituting into $(y - xp)^2 = 1 + p^2$ and multiplying by $(x^2 - 1)^2$:

$$\begin{aligned} y^2 &= (x^2 - 1)^2 + x^2 y^2 \\ y^2(1 - x^2) &= (1 - x^2)^2 \\ y^2 &= 1 - x^2, \end{aligned}$$

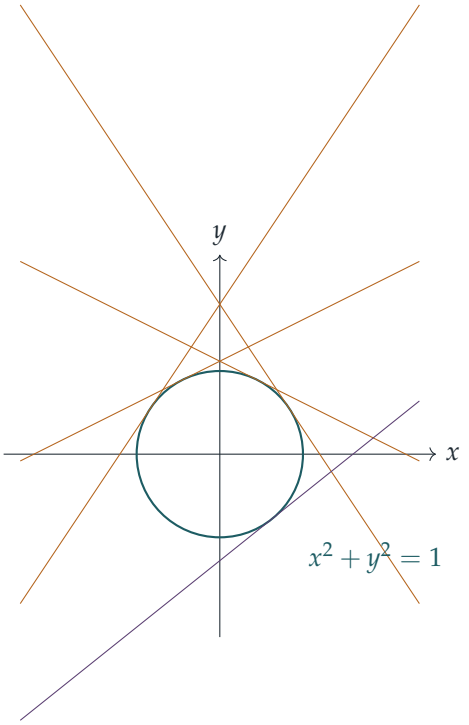
dividing by $1 - x^2 \neq 0$. So the curve is the circle $x^2 + y^2 = 1$. (At $x = \pm 1$ the condition $(x^2 - 1)p = xy$ forces $y = 0$, giving the points $(\pm 1, 0)$, which lie on the same circle.)

Conversely the circle does work: on it $p = -x/y$, so $y - xp = (y^2 + x^2)/y = 1/y$ and $1 + p^2 = (x^2 + y^2)/y^2 = 1/y^2$, which agree.

So the curve is either a straight line $y = mx + c$ with $c^2 = 1 + m^2$, or the unit circle $x^2 + y^2 = 1$.

Ask yourself What do the lines $y = mx \pm \sqrt{1 + m^2}$ have to do with the circle?

The distance from the origin to $y = mx + c$ is $|c|/\sqrt{1 + m^2}$, which here is exactly 1. So the lines are precisely the (non-vertical) tangents to the unit circle, just as the lines in (i) were the tangents to the parabola.



Checks

- Take $m = 1$ in (i): $y = x - 1$ has $p = 1$, and $1 - x + x - 1 = 0$. It touches $4y = x^2$ at $(2, 1)$.
- In (ii), the point $(\frac{3}{5}, \frac{4}{5})$ on the circle has $p = -\frac{3}{4}$; then $(x^2 - 1)p^2 - 2xyp + y^2 - 1 = -\frac{16}{25} \cdot \frac{9}{16} + \frac{18}{25} + \frac{16}{25} - 1 = 0$.
- Numerically, we checked both derivative factorisations along random cubic curves, and confirmed the lines, parabola and circle give residual 0 (to 10^{-13}), while $y = \frac{1}{4}x^2 + 1$ and $x^2 + y^2 = 4$ do not.

Common slips

- Integrating $y' = \frac{1}{2}x$ or $y'' = 0$ and leaving an arbitrary constant: the constant must be fixed by the original equation.
- Losing the $\pm$ in $c = \pm\sqrt{1 + m^2}$: there are two parallel tangents of each gradient.

Beyond STEP

Part (i) is a *Clairaut equation*, $y = xp - f(p)$: its general solution is the family of lines $y = mx - f(m)$, and it also has a *singular solution*, the envelope of that family, which

no choice of constant produces. Part (ii) is the same phenomenon, $y = xp \pm \sqrt{1+p^2}$, with the circle as envelope. Singular solutions show that a first-order equation need not have a unique solution through each point; when uniqueness holds is the subject of the existence and uniqueness theorems met in a first ODE course. Boyce and DiPrima, *Elementary Differential Equations*, is a good place to read about this.

WORKED SOLUTION

STEP I 2008, Question 9

Mechanics • Rolling Hoop Momentum

STEP I 2008, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q9

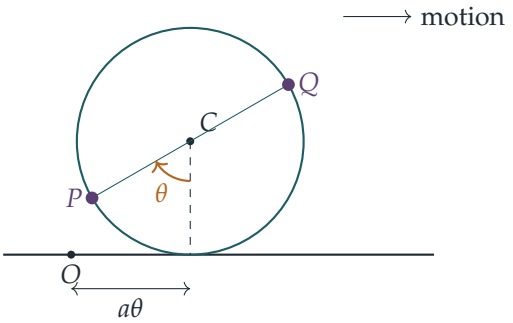
The key idea

Rolling without slipping ties the translation to the rotation: when the hoop has turned through θ, its centre has moved a distance $a\theta$, the length of rim that has touched the table. So each particle's position is "centre plus a rotating radius", and differentiating gives the velocities. For the last part, notice that the external forces have no horizontal component, so the total horizontal momentum can't change, and that momentum turns out to be a multiple of $\dot{\theta}$.

Solution

Position and speed of P

Model P and Q as particles on a light hoop. Take the origin at P's starting point, the x-axis along the line of rolling (in the direction the hoop moves) and the y-axis vertically up. Let the hoop roll in the positive x-direction, so that it turns clockwise in the picture below.



Since the hoop doesn't slip, the arc of rim that has rolled onto the table equals the distance moved by the centre, so the centre C is at $(a\theta, a)$. Initially CP points straight

down; after a clockwise turn through θ the vector $\overrightarrow{CP}$ is $(-a\sin\theta,\ -a\cos\theta)$. Hence

$$\boxed{P = \big(a\theta - a\sin\theta,\ a - a\cos\theta\big)}$$

(a cycloid). Differentiating with respect to t, using the chain rule,

$$\dot{x}_P = a\dot{\theta}(1-\cos\theta), \qquad \dot{y}_P = a\dot{\theta}\sin\theta,$$

so

$$v_P^2 = a^2\dot{\theta}^2\big[(1-\cos\theta)^2 + \sin^2\theta\big] = 2a^2\dot{\theta}^2(1-\cos\theta) = 4a^2\dot{\theta}^2\sin^2\tfrac{1}{2}\theta,$$

using $1-\cos\theta = 2\sin^2\frac{1}{2}\theta$. The speed of P is therefore

$$v_P = 2a\left|\dot{\theta}\sin\tfrac{1}{2}\theta\right| \qquad \big(= 2a\dot{\theta}\sin\tfrac{1}{2}\theta \text{ for } 0 \le \theta \le 2\pi,\ \dot{\theta} \ge 0\big).$$

This vanishes when θ is a multiple of 2π, exactly when P is the point of contact, as rolling requires.

Kinetic energy

Q is at the other end of the diameter, so $\overrightarrow{CQ} = -\overrightarrow{CP}$ and

$$Q = \big(a\theta + a\sin\theta,\ a + a\cos\theta\big), \qquad \dot{x}_Q = a\dot{\theta}(1+\cos\theta), \quad \dot{y}_Q = -a\dot{\theta}\sin\theta.$$

In the same way,

$$v_Q^2 = a^2\dot{\theta}^2\big[(1+\cos\theta)^2 + \sin^2\theta\big] = 2a^2\dot{\theta}^2(1+\cos\theta) = 4a^2\dot{\theta}^2\cos^2\tfrac{1}{2}\theta.$$

The hoop is light, so the kinetic energy is that of the particles alone:

$$T = \tfrac{1}{2}mv_P^2 + \tfrac{1}{2}mv_Q^2 = 2ma^2\dot{\theta}^2\big(\sin^2\tfrac{1}{2}\theta + \cos^2\tfrac{1}{2}\theta\big) = 2ma^2\dot{\theta}^2,$$

as required.

Constant speed

The external forces are the weights mg of the particles and the reaction of the table, and all of them are vertical. So the total horizontal force on the system is zero, and Newton's second law for the system says its total horizontal momentum is constant. (The hoop is light, so it carries no momentum.)

> **Ask yourself** Why is the horizontal momentum enough, when the particles also move vertically?

Because the horizontal momentum happens to depend only on $\dot{\theta}$. Adding the horizontal velocities,

$$m\dot{x}_P + m\dot{x}_Q = ma\dot{\theta}(1-\cos\theta) + ma\dot{\theta}(1+\cos\theta) = 2ma\dot{\theta}.$$

The $\cos\theta$ terms cancel: the system's centre of mass is C, which always moves horizontally with velocity $a\dot{\theta}$. Since $2ma\dot{\theta}$ is constant, $\dot{\theta}$ is constant, and the hoop's speed (the speed $a\dot{\theta}$ of its centre) is constant. □

Or by energy. The heights of P and Q add to $(a - a\cos\theta) + (a + a\cos\theta) = 2a$, so the potential energy $2mga$ never changes. The reaction acts at the point of contact, which is instantaneously at rest when rolling, so it does no work. Hence $T = 2ma^2\dot{\theta}^2$ is constant, so $|\dot{\theta}|$ is constant; and since $\dot{\theta}$ varies continuously and can't jump from one sign to the other without passing through 0, $\dot{\theta}$ itself is constant. The momentum argument avoids this last step, which is why we used it first.

Checks

- At $\theta = \pi$, P is at the top: the formula gives $P = (a\pi,\, 2a)$ and speed $2a\dot{\theta}$, twice the speed of the centre, the familiar result for the top of a rolling wheel.
- The kinetic energy equals $\frac{1}{2}(2m)(a\dot{\theta})^2 + \frac{1}{2}(2ma^2)\dot{\theta}^2$: translation of the centre of mass plus rotation about it, with moment of inertia $2ma^2$. Both terms give $ma^2\dot{\theta}^2$.
- Numerically, with $a = 1.3$, $\dot{\theta} = 2.1$ and $\theta = 1$, differentiating the positions by finite differences gives $v_P^2 + v_Q^2 = 4a^2\dot{\theta}^2 \approx 29.8$, and the horizontal accelerations of P and Q cancel when $\dot{\theta}$ is constant.

Common slips

- Putting the centre at $(a,\, a)$ or forgetting it moves at all: the $a\theta$ term is what rolling contributes.
- Getting the sign of $\sin\theta$ wrong in P's position; check that P starts with zero velocity.
- Concluding constant speed from constant kinetic energy without saying why $\dot{\theta}$ can't change sign.

Beyond STEP

This is a two-particle example of the general fact that the total momentum of a system changes only through external forces: internal forces (here carried by the hoop) cancel in pairs. The kinetic-energy split in the Checks is König's theorem, kinetic energy equals that of the centre of mass plus that of the motion relative to it, which is the starting point for rigid-body dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rolling motion and these ideas carefully.

WORKED SOLUTION

STEP I 2008, Question 10

Mechanics • Projectile Motion

STEP I 2008, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q10

The key idea

Gravity acts only vertically, so the horizontal velocity $V\cos\alpha$ never changes and the range is simply $V\cos\alpha$ times the total time of flight. The time of flight comes from the vertical motion alone, which splits into three stages: up through the bottom layer, the excursion above height h, and back down through the bottom layer. Each stage is a constant-acceleration problem, and symmetry does most of the work.

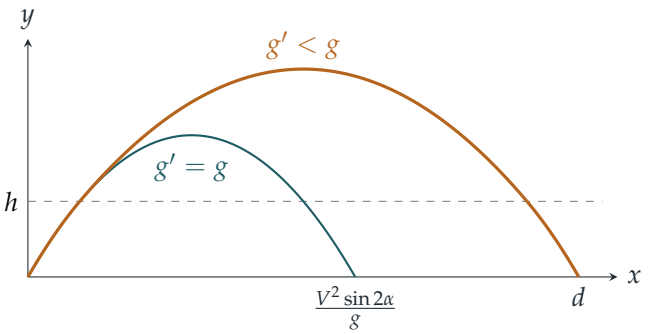
Solution

We model the particle as a point with no air resistance, moving in a vertical plane over flat ground. Take x horizontal and y vertically up, with the origin at the point of projection.

The sketch

Below height h both trajectories obey exactly the same equations, so they coincide there. The condition $V^2\sin^2\alpha > 2gh$ says the initial vertical kinetic energy is enough to rise through height h under acceleration g, so the particle does pass above $y = h$.

Above h, the case $g' < g$ has weaker gravity, so the vertical speed falls off more slowly: the particle climbs higher and spends longer up there, tracing a flatter, wider parabola. The velocity is continuous at $y = h$ (only the acceleration jumps), so the two arcs join smoothly with no corner, although the curvature changes abruptly. On the way down the particle re-enters the bottom layer with the same speed and angle at which it left (by the symmetry of the top arc), and its final arc is a mirror image of the first. So the $g' < g$ path lands further away.



The range

The vertical component of the initial velocity is $V\sin\alpha$, with acceleration $-g$ while $y < h$.

Stage 1: up to height h. Let w be the (upward) vertical speed on reaching $y = h$. By $v^2 = u^2 + 2as$,

$$w^2 = V^2\sin^2\alpha - 2gh = \sin^2\alpha\left(V^2 - 2gh\,\mathrm{cosec}^2\alpha\right) = V'^2\sin^2\alpha,$$

so $w = V'\sin\alpha$, taking the positive root since $\sin\alpha > 0$ and the particle is still rising. (The given condition is exactly what makes V' real and positive.) The time taken is, by $v = u + at$,

$$t_1 = \frac{V\sin\alpha - V'\sin\alpha}{g} = \frac{(V - V')\sin\alpha}{g}.$$

Stage 2: above height h. Now the particle starts at $y = h$ moving upwards at w with acceleration $-g'$. It returns to $y = h$ when $wt - \frac{1}{2}g't^2 = 0$, that is, after

$$t_2 = \frac{2w}{g'} = \frac{2V'\sin\alpha}{g'},$$

and it is then moving downwards with vertical speed w.

Ask yourself Why does the last stage take the same time as the first?

In stage 3 the particle starts at $y = h$ moving downwards at w, with acceleration g downwards, and falls to $y = 0$. That is stage 1 run backwards in time: constant-acceleration motion is reversible, and stage 1 went from $y = 0$ at speed $V\sin\alpha$ to $y = h$ at speed w under the same acceleration. So it reaches the ground with vertical speed $V\sin\alpha$ after time $t_3 = t_1$. (Directly: the vertical speed goes from w to $V\sin\alpha$ at rate g, which takes $(V\sin\alpha - w)/g$.)

The total time of flight is therefore

$$T = 2t_1 + t_2 = 2\sin\alpha\left(\frac{V - V'}{g} + \frac{V'}{g'}\right).$$

There is no horizontal force, so the horizontal velocity stays $V\cos\alpha$ throughout, and

$$\begin{aligned} d = V\cos\alpha \cdot T &= \left(\frac{V-V'}{g} + \frac{V'}{g'}\right) 2V\sin\alpha\cos\alpha \\ &= \left(\frac{V-V'}{g} + \frac{V'}{g'}\right) V\sin 2\alpha, \end{aligned}$$

using $2\sin\alpha\cos\alpha = \sin 2\alpha$. This is the required result.

Checks

- If $g' = g$ the bracket becomes V/g, and $d = V^2\sin 2\alpha/g$, the familiar range on Earth.
- If $g' < g$ then $V'/g' > V'/g$, so d exceeds the uniform-gravity range, matching the sketch.
- If $V^2\sin^2\alpha = 2gh$ exactly then $V' = 0$: the particle just touches $y = h$ and never feels g', and again $d = V^2\sin 2\alpha/g$.
- Numerically, with $V = 10$, $\alpha = 60^\circ$, $g = 10$, $h = 2$ and $g' = 5$ we get $V' = \sqrt{140/3} \approx 6.83$ and $d \approx 14.576$; stepping the motion numerically gives the same landing point to within 10^{-5} relative error. (These are the values used in the sketch.)

Common slips

- Forgetting the descent through the bottom layer, giving $t_1 + t_2$ instead of $2t_1 + t_2$.
- Taking the vertical speed at height h to be V' rather than $V'\sin\alpha$: V' is a speed scaled by $\operatorname{cosec}\alpha$, not the vertical component itself.
- Drawing a corner where the arcs meet: the velocity is continuous, so the join is smooth.

Beyond STEP

Gravity that depends on position is the norm rather than the exception: real gravity weakens as $1/r^2$ with distance from the centre of a planet, and projectile paths become arcs of ellipses rather than parabolas. Splitting a motion into regions with simple force laws and matching position and velocity at the boundaries is a standard technique, used again for refraction of particles and in quantum mechanics with step potentials. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both the constant-acceleration toolkit and orbital motion under inverse-square gravity.

WORKED SOLUTION

STEP I 2008, Question 11

Mechanics • Limiting Friction Equilibrium

STEP I 2008, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q11

The key idea

Because $\angle P_1OP_2$ is a right angle, the radius OP_2 is parallel to the tangent at P_1, and the radius OP_1 is parallel to the tangent at P_2. So if we resolve along the two radii, each equation involves just one normal reaction and one friction force. Add a moment equation about O, where both normal reactions have no moment, and limiting friction $F = \mu R$ at each end gives three equations from which α drops out. The final part is the double-angle formulae in disguise.

Solution

The model and the forces

Treat the rod as a uniform rigid rod with centre of mass G at its midpoint, the rings as light, and let the wire have radius a. Since the rings are light, the wire's force on each ring is passed straight to the rod, so we can treat the rod as touching the wire at P_1 and P_2. At each end there is a normal reaction along the radius and a friction force along the tangent.

Take axes with origin O, x to the right and y upwards, with P_1 to the left of the lowest point of the wire. Since $\angle P_1OP_2 = 90^\circ$ and P_1 makes angle α with the downward vertical, OP_2 makes angle $90^\circ - \alpha$ with it, on the other side:

$$P_1 = a(-\sin\alpha, -\cos\alpha), \qquad P_2 = a(\cos\alpha, -\sin\alpha).$$

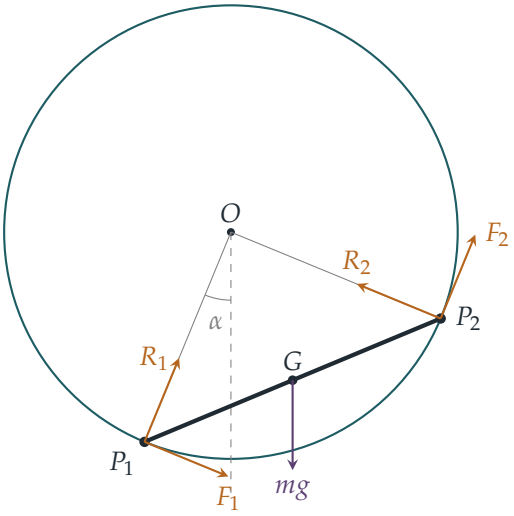
As $0 < \alpha < 45^\circ$, $\cos\alpha > \sin\alpha$, so P_1 is indeed lower than P_2, and both lie below O.

The midpoint is $G = \frac{a}{2}(\cos\alpha - \sin\alpha, -\cos\alpha - \sin\alpha)$, which lies to the *right* of the vertical through O. So the weight has a clockwise moment about O, and the rod is about to slip clockwise round the wire (P_2 moving down, P_1 moving up). Friction opposes this, so it acts *anticlockwise* at both ends: down the wire at P_1 and up the wire at P_2.

Let $\mathbf{u}_1$ and $\mathbf{u}_2$ be the unit vectors from P_1 and P_2 towards O:

$$\mathbf{u}_1 = (\sin\alpha, \cos\alpha), \qquad \mathbf{u}_2 = (-\cos\alpha, \sin\alpha), \qquad \mathbf{u}_1 \cdot \mathbf{u}_2 = 0.$$

The anticlockwise tangent at P_1 is $-\mathbf{u}_2$ and at P_2 it is $\mathbf{u}_1$. Let the normal reactions (towards O) be R_1, R_2 and the friction forces be F_1, F_2.



The equations of equilibrium

The weight mg acts vertically downwards, so its components along $\mathbf{u}_1$ and $\mathbf{u}_2$ are $-mg\cos\alpha$ and $-mg\sin\alpha$.

Resolving along $\mathbf{u}_1$ (that is, along P_1O): R_1 and F_2 contribute, while R_2 and F_1 are perpendicular to it. So

$$R_1 + F_2 = mg\cos\alpha. \tag{1}$$

Resolving along $\mathbf{u}_2$ (along P_2O): R_2 contributes $+R_2$ and F_1, which points along $-\mathbf{u}_2$, contributes $-F_1$. So

$$R_2 - F_1 = mg\sin\alpha. \tag{2}$$

Taking moments about O: the normal reactions pass through O. Each friction force is tangential at distance a from O, giving anticlockwise moments aF_1 and aF_2. The weight acts at horizontal distance $\frac{a}{2}(\cos\alpha - \sin\alpha)$ from O, clockwise. Hence

$$F_1 + F_2 = \tfrac{1}{2}mg(\cos\alpha - \sin\alpha). \tag{3}$$

Limiting friction and $\tan\alpha$

The rod is on the point of slipping at both ends, so $F_1 = \mu R_1$ and $F_2 = \mu R_2$. Writing $c = \cos\alpha$, $s = \sin\alpha$, equations (1) and (2) become

$$R_1 + \mu R_2 = mgc, \qquad -\mu R_1 + R_2 = mgs,$$

with solution

$$R_1 = \frac{mg(c - \mu s)}{1 + \mu^2}, \qquad R_2 = \frac{mg(s + \mu c)}{1 + \mu^2}.$$

Both are positive when μ is small (as it must be here; see below), so the reactions do act towards O as drawn. Now (3) says $\mu(R_1 + R_2) = \frac{1}{2}mg(c - s)$, that is,

$$\begin{aligned} 2\mu\big(c - \mu s + s + \mu c\big) &= (1+\mu^2)(c-s) \\ c\,(2\mu + 2\mu^2 - 1 - \mu^2) + s\,(2\mu - 2\mu^2 + 1 + \mu^2) &= 0 \\ s\,(1 + 2\mu - \mu^2) &= c\,(1 - 2\mu - \mu^2). \end{aligned}$$

Dividing by $c\,(1 + 2\mu - \mu^2)$ (nonzero, since $c > 0$ and $\tan\alpha > 0$ forces μ to be small, as we see next),

$$\tan\alpha = \frac{1 - 2\mu - \mu^2}{1 + 2\mu - \mu^2},$$

as required.

Ask yourself How do we know the friction acts anticlockwise rather than clockwise?

The weight's moment about O is clockwise because G is to the right of O, and the normal reactions have no moment about O. So friction alone must supply an anticlockwise moment, and the rod would slip clockwise without it. If you had drawn friction the other way, you would get the same working with μ replaced by $-\mu$, giving $\tan\alpha > 1$, which contradicts $\alpha < 45^\circ$.

The angle of the rod

The vector from P_1 to P_2 is $a(\cos\alpha + \sin\alpha,\ \cos\alpha - \sin\alpha)$, whose components are both positive. So the acute angle θ that the rod makes with the horizontal satisfies

$$\tan\theta = \frac{\cos\alpha - \sin\alpha}{\cos\alpha + \sin\alpha} = \frac{1 - \tan\alpha}{1 + \tan\alpha} = \tan(45^\circ - \alpha),$$

and since $0 < 45^\circ - \alpha < 45^\circ$, we get $\theta = 45^\circ - \alpha$.

Now put $\mu = \tan\lambda$ in the formula for $\tan\alpha$ and multiply top and bottom by $\cos^2\lambda$:

$$\tan\alpha = \frac{(\cos^2\lambda - \sin^2\lambda) - 2\sin\lambda\cos\lambda}{(\cos^2\lambda - \sin^2\lambda) + 2\sin\lambda\cos\lambda} = \frac{\cos 2\lambda - \sin 2\lambda}{\cos 2\lambda + \sin 2\lambda} = \frac{1 - \tan 2\lambda}{1 + \tan 2\lambda} = \tan(45^\circ - 2\lambda).$$

Since $0 < \lambda < 22.5^\circ$, the angle $45^\circ - 2\lambda$ lies strictly between 0° and 45°, as does α. The tangent function is one-to-one on this interval, so $\alpha = 45^\circ - 2\lambda$. Therefore

$$\theta = 45^\circ - \alpha = 2\lambda,$$

which is what we had to show. (This also explains the condition $\lambda < 22.5^\circ$, i.e. $\mu < \sqrt{2} - 1$: it is exactly what makes $\tan\alpha > 0$.)

Checks

- With $\mu = 0$ (smooth wire) we get $\tan\alpha = 1$, so $\alpha = 45^\circ$ and the rod is horizontal, which is the only position in which a rod on a smooth circle can rest.

- With $\mu = 0.2$: $\tan\alpha = 0.56/1.36$, so $\alpha \approx 22.38^\circ$ and $\theta = 45^\circ - \alpha \approx 22.62^\circ$, which matches $2\arctan 0.2 \approx 22.62^\circ$.
- Numerically, for 10^4 random values of $\mu \in (0, \sqrt{2}-1)$, a and m, substituting these R_1, R_2, F_1, F_2 into the full vector equations gave zero resultant force and zero moment (to about 10^{-14}), with $R_1, R_2 > 0$ and $\theta = 2\lambda$.

Common slips

- Resolving horizontally and vertically. It works, but every equation then contains all four forces; resolving along the two perpendicular radii is much cleaner.
- Getting the direction of friction wrong at one end. Both rings slip the same way round the circle, so friction is anticlockwise at *both* ends.
- Forgetting that $\theta = 45^\circ - \alpha$ needs a word of justification: the chord is perpendicular to the bisector of $\angle P_1 O P_2$.

Beyond STEP

There is a slicker geometric view: at limiting friction the total reaction at each end makes angle λ with the radius, and a body in equilibrium under three forces needs them to be concurrent, which leads straight to $\theta = 2\lambda$. This "angle of friction" idea, and the fact that friction is an inequality $|F| \le \mu R$ rather than an equation, are the starting points for statics problems with several contacts, where you must ask which contacts slip first. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body equilibrium with this kind of care.

WORKED SOLUTION

STEP I 2008, Question 12

Probability and Statistics • Maximum of Discrete Uniforms

STEP I 2008, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q12

The key idea

X is just the larger of the two scores. Asking for "the larger equals r" directly means counting several cases, but "the larger is *at most* r" is easy: it happens exactly when *both* scores are at most r. So we work with the cumulative distribution $\mathrm{P}(X \leq r)$ first and take differences. The same cumulative distribution then makes the median almost immediate.

Solution

Let X_1 and X_2 be the two independent scores, and $X = \max(X_1, X_2)$ (the two cases in the definition agree when $X_1 = X_2$, so X is well defined).

Part (i)

For $r = 0, 1, \ldots, N$, the event $\{X \leq r\}$ is the event $\{X_1 \leq r \text{ and } X_2 \leq r\}$. Each of these has probability r/N, and they are independent because X_1 and X_2 are, so

$$\mathrm{P}(X \leq r) = \frac{r}{N} \cdot \frac{r}{N} = \frac{r^2}{N^2}.$$

This holds for $r = 0$ too, giving 0. Hence, for $r = 1, 2, \ldots, N$,

$$\mathrm{P}(X = r) = \mathrm{P}(X \leq r) - \mathrm{P}(X \leq r-1) = \frac{r^2 - (r-1)^2}{N^2} = \frac{2r-1}{N^2},$$

as required. (As a sanity check, $\sum_{r=1}^{N}(2r-1) = N^2$, so these add to 1.)

Part (ii)

Using the two given sums,

$$\begin{aligned}\mu = \sum_{r=1}^{N} r\,\frac{2r-1}{N^2} &= \frac{1}{N^2}\left(2\sum_{r=1}^{N} r^2 - \sum_{r=1}^{N} r\right)\\ &= \frac{1}{N^2}\left(\frac{N(N+1)(2N+1)}{3} - \frac{N(N+1)}{2}\right)\\ &= \frac{N+1}{N}\cdot\frac{2(2N+1)-3}{6} = \frac{(N+1)(4N-1)}{6N}.\end{aligned}$$

When $N = 100$,

$$\mu = \frac{101\times 399}{600} = \frac{40\,299}{600} = 67.165,$$

as required.

Part (iii)

From part (i), $\mathrm{P}(X \le m) = m^2/N^2$ and $\mathrm{P}(X \ge m) = 1 - \mathrm{P}(X \le m-1) = 1 - (m-1)^2/N^2$. The two conditions on the median become

$$\mathrm{P}(X \le m) \ge \tfrac{1}{2} \iff m^2 \ge \tfrac{1}{2}N^2 \iff m \ge \frac{N}{\sqrt{2}},$$

$$\mathrm{P}(X \ge m) \ge \tfrac{1}{2} \iff (m-1)^2 \le \tfrac{1}{2}N^2 \iff m-1 \le \frac{N}{\sqrt{2}},$$

taking positive square roots, which is valid since $m \ge 1$ and so m and $m-1$ are non-negative. So m must be an integer with

$$\frac{N}{\sqrt{2}} \le m \le \frac{N}{\sqrt{2}} + 1.$$

Ask yourself An interval of length 1 can contain two integers. Could there be two medians?

Only if both ends were integers, which would need $N/\sqrt{2}$ to be an integer. But $N/\sqrt{2} = k$ would give $\sqrt{2} = N/k$, a rational number, which is impossible. So both inequalities are strict, the open interval of length 1 contains exactly one integer, and

$$m = \left\lceil \frac{N}{\sqrt{2}} \right\rceil,$$

the smallest integer greater than $N/\sqrt{2}$.

When $N = 100$, $N/\sqrt{2} = 50\sqrt{2} \approx 70.71$, so $m = 71$. Directly: $71^2 = 5041 \ge 5000 = \frac{1}{2}\times 100^2$ and $70^2 = 4900 \le 5000$, so both conditions hold.

Part (iv)

From part (iii), $m = N/\sqrt{2} + \theta$ for some $0 < \theta < 1$, so

$$\frac{m}{N} = \frac{1}{\sqrt{2}} + \frac{\theta}{N} \to \frac{1}{\sqrt{2}} \quad \text{as } N \to \infty.$$

From part (ii),

$$\frac{\mu}{N} = \frac{(N+1)(4N-1)}{6N^2} = \frac{1}{6}\left(1 + \frac{1}{N}\right)\left(4 - \frac{1}{N}\right) \to \frac{4}{6} = \frac{2}{3}.$$

Therefore, for large N,

$$\frac{\mu}{m} = \frac{\mu/N}{m/N} \approx \frac{2/3}{1/\sqrt{2}} = \frac{2\sqrt{2}}{3}.$$

Checks

- With $N = 2$: the four equally likely pairs give $X = 1$ once and $X = 2$ three times, matching $\frac{1}{4}$ and $\frac{3}{4}$ from part (i), and $\mu = \frac{7}{4}$, matching $\frac{3\times 7}{12}$.
- Listing all N^2 pairs by computer for $N \leq 40$ confirms parts (i) and (ii) exactly; testing every integer against the median definition for $N \leq 2000$ always finds exactly one median, equal to $\lceil N/\sqrt{2} \rceil$.
- For $N = 10^5$, $\mu/m \approx 0.942812$, against $2\sqrt{2}/3 \approx 0.942809$.

Common slips

- Counting $\mathrm{P}(X = r)$ as $2 \times \frac{r}{N} \cdot \frac{1}{N}$, forgetting that the case $X_1 = X_2 = r$ is then counted twice; the correct count is $2(r-1) + 1 = 2r - 1$.
- Writing $\mathrm{P}(X \geq m) = 1 - m^2/N^2$: that is $\mathrm{P}(X > m)$, not $\mathrm{P}(X \geq m)$.
- Giving $m = 70$ for $N = 100$ by rounding 70.71 to the nearest integer rather than up.

Beyond STEP

X is the simplest example of an *order statistic*: the maximum of a sample. As $N \to \infty$, X/N approaches a continuous random variable on $[0, 1]$ with density $2x$, whose mean is $\frac{2}{3}$ and whose median is $1/\sqrt{2}$, which is exactly why the ratio in part (iv) appears. Order statistics of larger samples lead to the beta distributions. Blitzstein and Hwang, *Introduction to Probability*, treats order statistics in an accessible way.

WORKED SOLUTION

STEP I 2008, Question 13

Probability and Statistics • Circular Arrangements

STEP I 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q13

The key idea

At a round table only the people's positions relative to each other matter, so we fix one person's chair and arrange the other five around them. That gives $5! = 120$ equally likely arrangements, and every probability is a count out of 120. A couple who must sit together can be glued into a single "block" that has two internal orders. For "exactly" and "none" we either count directly or use inclusion–exclusion over the events "couple k sits together".

Solution

Call the couples A, a; B, b; C, c. Rotating everyone one place round the table gives the same seating relative to each other. So we fix A's chair and seat the other five people in the remaining chairs in order going clockwise. This gives $5! = 120$ arrangements, all equally likely.

If we labelled the chairs instead, there would be $6! = 720$ arrangements. Each relative arrangement then appears 6 times, once for each chair A could occupy, so the probabilities are the same.

A useful general count: n objects round a circle can be arranged in $(n-1)!$ ways (fix one object, then order the rest).

Part (i)

If every couple sits together, glue each couple into a block. Then we have 3 blocks round the table, which can be arranged in $(3-1)! = 2$ ways. Inside each block the husband can sit on either side of his wife, which gives $2^3 = 8$ choices. So

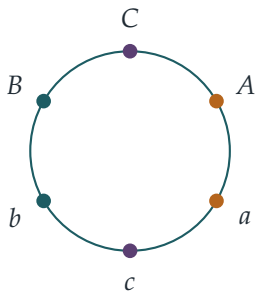
$$\mathrm{P}(\text{all three couples together}) = \frac{2 \times 8}{120} = \frac{16}{120} = \frac{2}{15},$$

as required.

Part (ii)

First choose which couple is *not* together. There are 3 choices. Say it is C, c, so A, a and B, b are together but C and c are not.

Glue Aa and Bb into blocks. Round the table we now have 4 units, Aa, Bb, C, c, which can be arranged in $3! = 6$ ways. Fix C and look at where c goes among the other three places of this 4-unit circle. In two of them c is next to C, and in one c is directly opposite. So C and c are apart in exactly $6/3 = 2$ of the 6 arrangements. In those, one block fills each of the two gaps between C and c, as in the diagram:



Ask yourself Why is this simpler than it looks?

With C and c opposite, the only freedom left is which block sits on which side (2 ways) and the order within each block ($2 \times 2 = 4$ ways). That gives $2 \times 4 = 8$ arrangements. Taking all three choices of the separated couple,

$$\mathrm{P}(\text{exactly two couples together}) = \frac{3 \times 8}{120} = \frac{24}{120} = \frac{1}{5}.$$

Part (iii)

Let E_k be the event that couple k sits together, for $k = 1, 2, 3$. We count each type of arrangement out of 120.

- $|E_1|$: glue couple 1 into a block. That leaves 5 units round the table, so there are $4! = 24$ arrangements, each with 2 internal orders, giving 48.
- $|E_1 \cap E_2|$: two blocks plus two people make 4 units. There are $3! = 6$ arrangements, with $2^2 = 4$ internal orders, giving 24.
- $|E_1 \cap E_2 \cap E_3| = 16$, from part (i).

By symmetry the same counts hold for any choice of couples. By inclusion–exclusion, the number of arrangements with at least one couple together is

$$|E_1 \cup E_2 \cup E_3| = 3(48) - 3(24) + 16 = 144 - 72 + 16 = 88.$$

So $120 - 88 = 32$ arrangements have no couple together, and

$$\mathrm{P}(\text{no husband next to his wife}) = \frac{32}{120} = \frac{4}{15}.$$

Ask yourself Can part (ii) be read off the same counts?

Yes. Each $|E_i \cap E_j|$ counts the arrangements with exactly the pair i, j together, plus those with all three together. So the number with exactly two couples together is $3(24) - 3(16) = 24$, matching part (ii).

Checks

- The four cases must account for all 120 arrangements. The count with exactly one couple together is $3(48) - 2 \cdot 3(24) + 3(16) = 48$, and $32 + 48 + 24 + 16 = 120$.
- A computer enumeration of all 720 labelled seatings gives probabilities $\frac{4}{15}, \frac{2}{5}, \frac{1}{5}, \frac{2}{15}$ for $0, 1, 2, 3$ couples together. These match the answers above.
- Sanity check: a given wife has two neighbours chosen from 5 people, so her husband is next to her with probability $\frac{2}{5}$. This agrees with $|E_1|/120 = 48/120$.

Common slips

- Using 6! arrangements but counting the favourable seatings as circular arrangements, or the other way round. The numerator and denominator must use the same convention.
- Forgetting the factor of 2 for the order of the two people within each block.
- In part (ii), forgetting to exclude the arrangements where the third couple also sits together. Without that, $\frac{3\times 24}{120} = \frac{3}{5}$ is far too big.

Beyond STEP

This is the *ménage problem* in miniature. The classical version also requires men and women to alternate, and its general answer (Touchard's formula) comes from exactly this kind of inclusion–exclusion. In our unrestricted version with n couples, the expected number of couples sitting together is $n \cdot \frac{2}{2n-1} \to 1$. The probability that no couple sits together tends to e^{-1}, just like the famous derangement limit. In the alternating ménage version it tends to e^{-2}. Inclusion–exclusion and such limits are treated well in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP II 2008, Question 1

Pure • Complex Iteration

STEP II 2008, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q1

The key idea

The expressions $x^2 - y^2$ and $2xy$ are the real and imaginary parts of $(x + \mathrm{i}y)^2$. So if we write each point as a complex number $z_n = x_n + \mathrm{i}y_n$, the two-component rule becomes the single equation $z_{n+1} = z_n^2 + c$ with one complex constant c. Then "constant" and "period 2" become quadratic equations in one complex unknown, which we can solve with the quadratic formula.

Solution

Put $z_n = x_n + \mathrm{i}y_n$ and $c = a + (b+2)\mathrm{i}$. Since

$$z_n^2 + c = (x_n^2 - y_n^2 + a) + \mathrm{i}\,(2x_n y_n + b + 2),$$

and two complex numbers are equal exactly when their real and imaginary parts agree, the rule is equivalent to

$$z_{n+1} = z_n^2 + c.$$

Because a, b, x_n and y_n are real, every point (x, y) corresponds to exactly one complex number and vice versa. So nothing is lost by working with z.

Part (i)

With $a = 1$, $b = -1$ we have $c = 1 + \mathrm{i}$. The sequence is constant exactly when $z_2 = z_1$, since then every later step repeats the same calculation. So we need

$$z^2 - z + (1 + \mathrm{i}) = 0, \qquad z = \frac{1 \pm \sqrt{1 - 4(1+\mathrm{i})}}{2} = \frac{1 \pm \sqrt{-3 - 4\mathrm{i}}}{2}.$$

To find a square root of $-3 - 4\mathrm{i}$, look for $(p + q\mathrm{i})^2 = -3 - 4\mathrm{i}$ with p, q real: $p^2 - q^2 = -3$ and $pq = -2$. Then $p = 1$, $q = -2$ works, since $(1 - 2\mathrm{i})^2 = 1 - 4\mathrm{i} - 4 = -3 - 4\mathrm{i}$. Hence

$$z = \frac{1 \pm (1 - 2\mathrm{i})}{2} = 1 - \mathrm{i} \quad \text{or} \quad \mathrm{i}.$$

So the sequence is constant exactly when

$$(x_1, y_1) = (1, -1) \quad \text{or} \quad (x_1, y_1) = (0, 1).$$

Directly: $(1,-1) \mapsto (1-1+1,\ -2+1) = (1,-1)$ and $(0,1) \mapsto (0-1+1,\ 0+1) = (0,1)$.

Part (ii)

Now $z_1 = -1 + \mathrm{i}$, so $z_1^2 = 1 - 2\mathrm{i} - 1 = -2\mathrm{i}$ and

$$z_2 = c - 2\mathrm{i}.$$

Write $w = z_2$. The sequence has period 2 when $z_3 = z_1$ but $z_2 \neq z_1$. Once $z_3 = z_1$, the rule makes the sequence repeat as $z_1, z_2, z_1, z_2, \ldots$. Using $c = w + 2\mathrm{i}$,

$$z_3 = w^2 + c = w^2 + w + 2\mathrm{i},$$

so $z_3 = z_1$ becomes

$$w^2 + w + 2\mathrm{i} = -1 + \mathrm{i} \quad \Longleftrightarrow \quad w^2 + w + (1 + \mathrm{i}) = 0.$$

? **Ask yourself** Why choose $w = z_2$ as the unknown, and not c?

The condition $z_3 = z_1$ is a quadratic in c either way, but in terms of w the constant sequence $w = z_1$ must be one root. So we already know one root of the quadratic, and the other is the one we want. Indeed $w = -1 + \mathrm{i}$ satisfies it: $(-1+\mathrm{i})^2 + (-1+\mathrm{i}) + 1 + \mathrm{i} = -2\mathrm{i} + 2\mathrm{i} = 0$. The roots sum to -1, so the other root is $w = -\mathrm{i}$. (The quadratic formula gives the same: $w = \frac{1}{2}\left(-1 \pm \sqrt{-3-4\mathrm{i}}\right) = \frac{1}{2}\left(-1 \pm (1-2\mathrm{i})\right)$, the same square root as in (i).)

- $w = -1 + \mathrm{i} = z_1$ gives $c = -1 + 3\mathrm{i}$, that is $a = -1$, $b = 1$. Here $z_2 = z_1$ and the sequence is constant, so its period is 1, not 2.
- $w = -\mathrm{i}$ gives $c = \mathrm{i}$, that is $a = 0$ and $b + 2 = 1$.

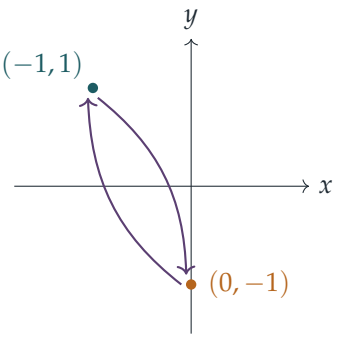
So the sequence has period 2 exactly when

$$a = 0, \qquad b = -1.$$

Check in coordinates: with $a = 0$, $b = -1$,

$$(-1, 1) \mapsto (1 - 1 + 0,\ -2 + 1) = (0, -1) \mapsto (0 - 1 + 0,\ 0 + 1) = (-1, 1),$$

and the two points are different, so the period is exactly 2.



Checks

- Iterating the original coordinate rule shows that $(1,-1)$ and $(0,1)$ are fixed when $a=1$, $b=-1$, and that $(-1,1)\to(0,-1)\to(-1,1)$ when $a=0$, $b=-1$.
- The square root $1-2\mathrm{i}$ can be checked by squaring it, and so can the factorisation $w^2+w+1+\mathrm{i}=(w+1-\mathrm{i})(w+\mathrm{i})$: the product of the roots is $(-1+\mathrm{i})(-\mathrm{i})=1+\mathrm{i}$.
- A numerical root search over random starting values found no other fixed points in (i) and no other (a,b) with $z_3=z_1$ in (ii) apart from $(-1,1)$, which gives the constant sequence.

Common slips

- Forgetting the "+2" in the second component, so taking $c=a+b\mathrm{i}$ in place of $a+(b+2)\mathrm{i}$.
- Giving $a=-1$, $b=1$ as a second answer to (ii). That sequence is constant, so its period is 1.
- Working only in real coordinates and getting lost in simultaneous quadratics. This works, but the complex form is much shorter.

Beyond STEP

The map $z\mapsto z^2+c$ is the most famous example in complex dynamics. The Mandelbrot set is the set of c for which the orbit of 0 stays bounded, and its two largest pieces, the main cardioid and the disc to its left, consist of the c for which the map has an attracting fixed point or an attracting 2-cycle. Here, too, the equation for 2-cycles had a factor coming from the fixed points: in general $(z^2+c)^2+c-z=(z^2-z+c)(z^2+z+c+1)$. For a readable introduction to the geometry behind this, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP II 2008, Question 2

Pure • Partial Fractions and Series

STEP II 2008, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q2

The key idea

Partial fractions turn one awkward rational function into three pieces whose expansions we know by heart: a geometric series, its derivative, and a geometric series in $-x^2$. The first two give a coefficient growing like n; the third only ever contributes $\pm\frac{1}{2}$, repeating with period 4, and that is where the "either/or" comes from. For the last part, look for a value of x that turns the function into the given fraction.

Solution

The partial fractions

Since $(1-x)^2$ is a repeated linear factor and $1+x^2$ is an irreducible quadratic, we write

$$\frac{1+x}{(1-x)^2(1+x^2)} = \frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{Cx+D}{1+x^2},$$

so that, for all x,

$$1+x = A(1-x)(1+x^2) + B(1+x^2) + (Cx+D)(1-x)^2.$$

- Putting $x=1$: $2 = 2B$, so $B=1$.
- Putting $x=0$: $1 = A+B+D$, so $A=-D$.
- Putting $x=\mathrm{i}$ (allowed, since this is a polynomial identity): $1+\mathrm{i} = (C\mathrm{i}+D)(1-\mathrm{i})^2 = (C\mathrm{i}+D)(-2\mathrm{i}) = 2C-2D\mathrm{i}$. Comparing real and imaginary parts, $C=\frac{1}{2}$ and $D=-\frac{1}{2}$, hence $A=\frac{1}{2}$.

As a check, the coefficient of x^3 on the right is $-A+C=0$, matching the left. So

$$\frac{1+x}{(1-x)^2(1+x^2)} = \frac{1}{2(1-x)} + \frac{1}{(1-x)^2} - \frac{1-x}{2(1+x^2)}.$$

The coefficients

For $|x| < 1$ all three expansions converge, and

$$\frac{1}{2(1-x)} = \tfrac{1}{2}\sum_{n\geq 0} x^n, \qquad \frac{1}{(1-x)^2} = \sum_{n\geq 0}(n+1)x^n,$$

$$\begin{aligned} -\frac{1-x}{2(1+x^2)} &= -\tfrac{1}{2}(1-x)\left(1-x^2+x^4-x^6+\cdots\right) \\ &= -\tfrac{1}{2}\left(1-x-x^2+x^3+x^4-x^5-x^6+x^7+\cdots\right). \end{aligned}$$

The bracket in the last line has coefficients $1, -1, -1, 1$ repeating with period 4: for $n = 2k$ the coefficient of x^n is $(-1)^k$, and for $n = 2k+1$ it is $-(-1)^k$. So the third piece contributes $-\frac{1}{2}$ to x^n when $n \equiv 0$ or $3 \pmod 4$, and $+\frac{1}{2}$ when $n \equiv 1$ or $2 \pmod 4$.

Ask yourself Why does the period-4 pattern produce exactly two cases rather than four?

Because only the sign matters: adding the three contributions,

$$a_n = \tfrac{1}{2} + (n+1) \mp \tfrac{1}{2} = \begin{cases} n+1, & n \equiv 0 \text{ or } 3 \pmod 4, \\ n+2, & n \equiv 1 \text{ or } 2 \pmod 4, \end{cases}$$

which is the required result. The sequence begins $1, 3, 4, 4, 5, 7, 8, 8, 9, 11, 12, 12, \ldots$

The decimal approximation

Ask yourself Which value of x makes the function equal to $\frac{11\,000}{8181}$?

The denominators suggest $x = \frac{1}{10}$: then $(1-x)^2 = 0.81$ and $1 + x^2 = 1.01$, so $(1-x)^2(1+x^2) = 0.8181$, and

$$\frac{1+\frac{1}{10}}{0.8181} = \frac{1.1}{0.8181} = \frac{11\,000}{8181}.$$

Since $\frac{1}{10} < 1$, the series is valid, and

$$\frac{11\,000}{8181} = \sum_{n\geq 0}\frac{a_n}{10^n} = 1 + 0.3 + 0.04 + 0.004 + 0.0005 + 0.00007 + \cdots.$$

While $a_n \leq 9$ (that is, up to $n = 8$) the coefficients are simply the decimal digits: 1.34457889. After that the coefficients have two digits, so we must carry:

$$\begin{aligned} \sum_{n=0}^{8}\frac{a_n}{10^n} &= 1.344\,578\,89 \\ +\,11\times 10^{-9} &\to 1.344\,578\,901 \\ +\,12\times 10^{-10} &\to 1.344\,578\,902\,2 \\ +\,12\times 10^{-11} &\to 1.344\,578\,902\,32 \\ +\,13\times 10^{-12} &\to 1.344\,578\,902\,333. \end{aligned}$$

The later terms only change the eleventh significant figure onwards, so to nine significant figures

$$\frac{11\,000}{8181} \approx 1.344\,578\,90.$$

Checks

- Setting $x = 0$ in the function gives $1 = a_0$, and $n = 0$ falls in the $n + 1$ case. Multiplying out $(1 + x)(1 + 2x + \cdots)(1 - x^2 + \cdots)$ gives $1 + 3x + \cdots$, so $a_1 = 3 = 1 + 2$.
- Long division gives $11\,000/8181 = 1.344\,578\,902\,33\ldots$, agreeing with the series. We also checked a_n for $n < 60$ by exact power-series division, and the partial fractions identity at many random rational points.

Common slips

- Forgetting the carries: reading the digits straight off the a_n gives $1.34457889\ldots$, which is wrong in the ninth significant figure.
- Writing $\frac{D}{1+x^2}$ instead of $\frac{Cx+D}{1+x^2}$; the irreducible quadratic needs a linear numerator.
- Rounding $1.344578902\ldots$ to 1.34457890 and then dropping the final zero; it is a significant figure here.

Beyond STEP

A rational function always has a power series whose coefficients are a polynomial in n plus terms like r^n and periodic patterns, one for each root of the denominator; this is the theory of *linear recurrences* and *generating functions*. Here the roots 1 (twice) and $\pm$i give the linear growth and the period-4 wobble. It is also why every fraction has an eventually repeating decimal. Wilf's *generatingfunctionology* is a lively and freely available introduction, and Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

WORKED SOLUTION

STEP II 2008, Question 3

Pure • Inequalities and Contradiction

STEP II 2008, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q3

The key idea

The sketch shows that the cubic never dips below the x-axis once $x \geq 0$, and rearranging "$y \geq 0$" gives the inequality $x^2(1-x) \leq 4/27$. For the contradiction, notice that the three numbers $bc(1-a)$, $ca(1-b)$, $ab(1-c)$ look awkward one at a time, but their *product* regroups as $a^2(1-a)\cdot b^2(1-b)\cdot c^2(1-c)$, which the first result controls. Part (ii) is the same trick with a quadratic in place of the cubic.

Solution

Part (i): the curve

Differentiating $y = 27x^3 - 27x^2 + 4$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 81x^2 - 54x = 27x(3x-2),$$

which is zero at $x = 0$ and $x = \frac{2}{3}$. At $x = 0$, $y = 4$; at $x = \frac{2}{3}$,

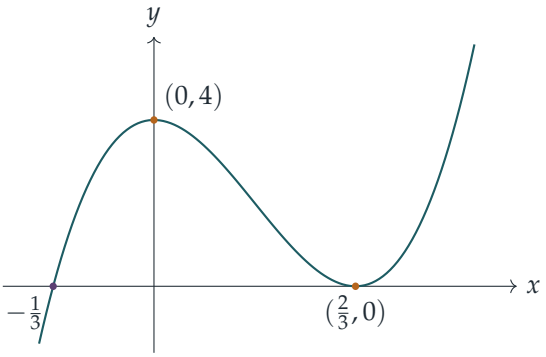
$$y = 27\cdot\tfrac{8}{27} - 27\cdot\tfrac{4}{9} + 4 = 8 - 12 + 4 = 0.$$

Since $\mathrm{d}^2y/\mathrm{d}x^2 = 162x - 54$ is $-54 < 0$ at $x = 0$ and $54 > 0$ at $x = \frac{2}{3}$, the turning points are a maximum at $(0,4)$ and a minimum at $(\frac{2}{3},0)$.

The minimum lies on the x-axis, so the curve touches the axis there and $(3x-2)^2$ is a factor. Comparing the coefficient of x^3 and the constant term gives

$$27x^3 - 27x^2 + 4 = (3x-2)^2(3x+1),$$

so the only other place the curve meets the axis is $x = -\frac{1}{3}$. The curve rises from $-\infty$, crosses the axis at $x = -\frac{1}{3}$, peaks at $(0,4)$, comes down to touch the axis at $(\frac{2}{3},0)$ and then rises again.



Part (i): the inequality

For $x \geq 0$ we have $3x + 1 > 0$ and $(3x - 2)^2 \geq 0$, so the factorisation gives $27x^3 - 27x^2 + 4 \geq 0$. (Equivalently, from the sketch: on $x \geq 0$ the lowest point of the curve is the minimum $(\frac{2}{3}, 0)$.) Rearranging,

$$4 \geq 27x^2 - 27x^3 = 27x^2(1 - x), \qquad \text{so} \qquad x^2(1 - x) \leq \frac{4}{27} \quad \text{for all } x \geq 0,$$

with equality only at $x = \frac{2}{3}$. This is the required result.

Part (i): the contradiction

Suppose, for contradiction, that all three numbers exceed $4/27$:

$$bc(1 - a) > \tfrac{4}{27}, \qquad ca(1 - b) > \tfrac{4}{27}, \qquad ab(1 - c) > \tfrac{4}{27}.$$

All six quantities here are positive, so we may multiply the three inequalities together:

$$bc(1 - a) \cdot ca(1 - b) \cdot ab(1 - c) > \left(\tfrac{4}{27}\right)^3.$$

? **Ask yourself** How does the first result say anything about these mixed products?

Regroup the left-hand side so that each letter is collected with its own bracket:

$$bc(1 - a) \cdot ca(1 - b) \cdot ab(1 - c) = \left[a^2(1 - a)\right]\left[b^2(1 - b)\right]\left[c^2(1 - c)\right].$$

Each of a, b, c is non-negative, so by the first result each bracket lies between 0 (as $1 - a \geq 0$ and so on) and $4/27$. Multiplying these three inequalities between non-negative numbers,

$$\left[a^2(1 - a)\right]\left[b^2(1 - b)\right]\left[c^2(1 - c)\right] \leq \left(\tfrac{4}{27}\right)^3.$$

This contradicts the strict inequality above. So our assumption was false, and at least one of $bc(1 - a)$, $ca(1 - b)$, $ab(1 - c)$ is less than or equal to $4/27$.

Part (ii)

The same idea works with a quadratic. For any real t,

$$t(1-t) = \tfrac{1}{4} - \left(t - \tfrac{1}{2}\right)^2 \leq \tfrac{1}{4},$$

by completing the square. Suppose, for contradiction, that $p(1-q) > \frac{1}{4}$ and $q(1-p) > \frac{1}{4}$. Both sides are positive, so multiplying,

$$p(1-q)\cdot q(1-p) > \tfrac{1}{16}.$$

But the left-hand side regroups as $\big[p(1-p)\big]\big[q(1-q)\big]$, and since $0 < p, q < 1$ each bracket lies between 0 and $\frac{1}{4}$, so the product is at most $\frac{1}{16}$. This is a contradiction, so at least one of $p(1-q)$ and $q(1-p)$ is less than or equal to $\frac{1}{4}$.

Checks

- Equality cases: $a = b = c = \frac{2}{3}$ makes all three numbers in (i) equal to $\frac{4}{9}\cdot\frac{1}{3} = \frac{4}{27}$, and $p = q = \frac{1}{2}$ makes both numbers in (ii) equal to $\frac{1}{4}$. So the bounds $4/27$ and $1/4$ cannot be improved.
- Expanding $(3x-2)^2(3x+1) = (9x^2 - 12x + 4)(3x+1) = 27x^3 - 27x^2 + 4$ confirms the factorisation.
- Numerically, the largest value of $x^2(1-x)$ on $[0,3]$ is $0.148148\ldots = 4/27$. Over a million random triples (a,b,c) in $(0,1)$, the smallest of the three products never exceeded 0.1468, and over a million random pairs the smaller of $p(1-q)$, $q(1-p)$ never exceeded 0.24998.

Common slips

- Multiplying inequalities without saying that everything is positive. The step "$A > \alpha$, $B > \beta$ implies $AB > \alpha\beta$" fails for negative numbers, so the hypothesis $0 < a, b, c < 1$ really is needed.
- Using the bound $x^2(1-x) \leq 4/27$ for $x < 0$, where it is false (try $x = -1$).
- Proving only that one particular number, say $bc(1-a)$, is small. It need not be: with a close to 0 and $b = c$ close to 1 it is close to 1.

Beyond STEP

Taking the product of quantities so that a symmetric bound applies is a close cousin of the AM–GM inequality, which gives the first result directly: for $0 \leq x \leq 1$, applying AM–GM to $\frac{x}{2}, \frac{x}{2}, 1-x$ (which sum to 1) gives $\frac{x^2}{4}(1-x) \leq \left(\frac{1}{3}\right)^3$. The argument also has a pigeonhole flavour: if a product is small, not every factor can be large. These ideas of bounding and optimising are developed in first-year analysis; Abbott's *Understanding*

Analysis is a friendly place to start, and Siklos's *Advanced Problems in Mathematics* has more STEP questions in this style.

WORKED SOLUTION

STEP II 2008, Question 4

Pure • Implicit Differentiation

STEP II 2008, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q4

The key idea

The angle between two lines is easiest to handle with vectors: the normal to $F(x,y)=1$ points along ∇F, and for two vectors $\mathbf{p}$ and $\mathbf{n}$ in the plane, tan of the angle between them is |cross product| divided by the dot product. Here the dot product turns out to be exactly the left-hand side of the curve's equation, which is 1. After that, θ is stationary exactly when y^2-x^2 is, and the rest is algebra with x^2+y^2 and xy.

Solution

The gradient

Differentiating $x^2+y^2+2axy=1$ implicitly with respect to x,

$$2x+2y\frac{\mathrm{d}y}{\mathrm{d}x}+2a\left(y+x\frac{\mathrm{d}y}{\mathrm{d}x}\right)=0 \quad\Longrightarrow\quad (ax+y)\frac{\mathrm{d}y}{\mathrm{d}x}=-(x+ay).$$

So, provided $ax+y\neq 0$, the gradient is $-\dfrac{x+ay}{ax+y}$, as required.

The angle θ

The tangent has direction $(ax+y,\ -(x+ay))$, so the normal has direction

$$\mathbf{n}=(x+ay,\ ax+y),$$

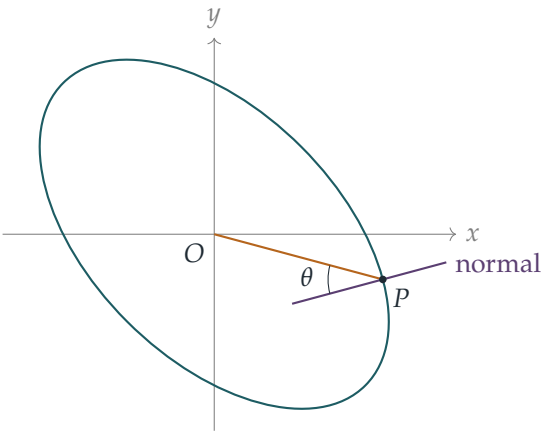
the vector perpendicular to it. (This is $\frac{1}{2}\nabla F$ for $F=x^2+y^2+2axy$, and it makes sense even where $ax+y=0$.) The vector $\overrightarrow{OP}$ is $\mathbf{p}=(x,y)$. Let ϕ be the angle between $\mathbf{p}$ and $\mathbf{n}$. Then

$$|\mathbf{p}||\mathbf{n}|\cos\phi=\mathbf{p}\cdot\mathbf{n}=x(x+ay)+y(ax+y)=x^2+y^2+2axy=1,$$
$$|\mathbf{p}||\mathbf{n}|\sin\phi=|x(ax+y)-y(x+ay)|=|ax^2-ay^2|=a|y^2-x^2|,$$

where the second line is the size of the two-dimensional cross product (the area of the parallelogram spanned by $\mathbf{p}$ and $\mathbf{n}$), and $a > 0$.

Because the dot product is $+1 > 0$, ϕ is already acute (or zero), so $\phi = \theta$. Dividing the two lines,

$$\tan\theta = a\,|y^2 - x^2|.$$



The sketch (with $a = \frac{1}{2}$) shows the curve: since $0 < a < 1$ it is an ellipse with its axes along $y = \pm x$. The point P drawn is one where θ is largest.

Part (i)

The function $t \mapsto \arctan(t)$ is strictly increasing, so θ is a smooth increasing function of $f = y^2 - x^2$ wherever $f \neq 0$.

Ask yourself What about points where $y^2 = x^2$, where the modulus has a corner?

There $\theta = 0$. If θ has zero derivative at such a point then $a|f|$ divided by the change in x tends to 0, which forces $\mathrm{d}f/\mathrm{d}x = 0$ as well. So in every case

$$\frac{\mathrm{d}\theta}{\mathrm{d}x} = 0 \implies \frac{\mathrm{d}}{\mathrm{d}x}(y^2 - x^2) = 0 \implies y\frac{\mathrm{d}y}{\mathrm{d}x} = x.$$

Here we are using the gradient formula, so $ax + y \neq 0$ (we'll confirm below that this holds automatically). Substituting and multiplying by $ax + y$:

$$-y(x + ay) = x(ax + y) \implies -xy - ay^2 = ax^2 + xy \implies a(x^2 + y^2) + 2xy = 0.$$

That's (i).

Incidentally, if $ax + y = 0$ then (i) would give $ax^2(1 + a^2) - 2ax^2 = -a(1 - a^2)x^2 = 0$, so $x = y = 0$, which is not on the curve. So such points never cause trouble.

Part (ii)

Write $S = x^2 + y^2$. By (i), $2xy = -aS$. Putting this into the curve's equation,

$$S + a(-aS) = 1 \quad \Longrightarrow \quad S = \frac{1}{1-a^2}.$$

Hence

$$x^2 + y^2 + 2xy = S - aS = \frac{1-a}{1-a^2} = \frac{1}{1+a},$$

that is, $(1+a)(x^2+y^2+2xy) = 1$, as required.

Part (iii)

We need $|y^2 - x^2|$. Using $(y^2-x^2)^2 = (x^2+y^2)^2 - 4x^2y^2 = S^2 - (2xy)^2$,

$$(y^2-x^2)^2 = S^2 - a^2S^2 = (1-a^2)\cdot\frac{1}{(1-a^2)^2} = \frac{1}{1-a^2}.$$

Taking the positive square root (valid as $0 < a < 1$), $|y^2 - x^2| = 1/\sqrt{1-a^2}$, and so

$$\tan\theta = \frac{a}{\sqrt{1-a^2}}.$$

Checks

- Rotate axes: with $u = (x+y)/\sqrt{2}$ and $v = (x-y)/\sqrt{2}$ the curve becomes $(1+a)u^2 + (1-a)v^2 = 1$ and $y^2 - x^2 = -2uv$. By AM–GM, $1 \geq 2\sqrt{(1-a^2)}\,|uv|$, so $\tan\theta = 2a|uv| \leq a/\sqrt{1-a^2}$, with equality when $(1+a)u^2 = \frac{1}{2}$. That equality condition is exactly (ii), since $x^2+y^2+2xy = 2u^2$. So the stationary value is the maximum of θ.
- With $a = \frac{1}{2}$: $P \approx (1.1154,\ -0.2989)$ satisfies the curve equation and (i), and $\tan\theta = \frac{1}{2}|y^2 - x^2| \approx 0.5774 = 1/\sqrt{3}$, so $\theta = 30°$. A numerical scan of $\tan\theta$ round the ellipse for several values of a finds its maxima at exactly these values.
- As $a \to 0$ the curve becomes the unit circle, whose normal passes through O, so $\theta \to 0$, consistent with $\tan\theta \to 0$.

Common slips

- Using $\tan(\alpha - \beta)$ with the gradients of OP and the normal: it works, but it needs care with $x = 0$ and with signs, and the vector route avoids both.
- Forgetting why the angle found is the *acute* one: the dot product being $+1$ is what settles it.
- Squaring to find $|y^2 - x^2|$ and then forgetting to take the positive root.

Beyond STEP

Writing $x^2 + y^2 + 2axy$ as a quadratic form $\mathbf{x}^\mathsf{T} A\mathbf{x}$ with $A = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$, the axes of the ellipse are the eigenvectors of A, $(1,1)$ and $(1,-1)$, and the eigenvalues $1 \pm a$ give the semi-axes. This diagonalisation of symmetric matrices is a central idea of first-year linear algebra; see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP II 2008, Question 5

Pure • Integration by Substitution

STEP II 2008, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q5

The key idea

Both integrals yield to spotting a derivative. In the first, the numerator $\sin 2x = 2\sin x \cos x$ is exactly the derivative of the denominator $1 + \sin^2 x$, so the integral is a logarithm on sight. In the second, the lone $\sin x$ on top suggests $c = \cos x$, after writing $1 + \sin^2 x = 2 - \cos^2 x$, which leaves a standard partial-fractions integral. Comparing the two answers then turns into comparing two numbers involving $\sqrt{2}$, and the inequalities in the final part are built to settle exactly that.

Solution

The two integrals

Call the integrals I_1 (numerator $\sin 2x$) and I_2 (numerator $\sin x$).

For I_1, since $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(1 + \sin^2 x\right) = 2\sin x \cos x = \sin 2x$, the integrand has the form $f'(x)/f(x)$ with $f(x) = 1 + \sin^2 x > 0$. Hence

$$I_1 = \Big[\ln\left(1 + \sin^2 x\right)\Big]_0^{\pi/2} = \ln 2 - \ln 1 = \ln 2.$$

For I_2, write $1 + \sin^2 x = 2 - \cos^2 x$ and substitute $c = \cos x$, so $\mathrm{d}c = -\sin x\,\mathrm{d}x$; as x runs from 0 to $\frac{1}{2}\pi$, c runs from 1 down to 0. The minus sign reverses the limits:

$$I_2 = \int_0^1 \frac{\mathrm{d}c}{2 - c^2}.$$

Using partial fractions,

$$\frac{1}{2 - c^2} = \frac{1}{(\sqrt{2} - c)(\sqrt{2} + c)} = \frac{1}{2\sqrt{2}}\left(\frac{1}{\sqrt{2} - c} + \frac{1}{\sqrt{2} + c}\right),$$

and both denominators stay positive for $0 \le c \le 1$, so

$$I_2 = \frac{1}{2\sqrt{2}}\Big[\ln(\sqrt{2}+c) - \ln(\sqrt{2}-c)\Big]_0^1 = \frac{1}{2\sqrt{2}}\ln\frac{\sqrt{2}+1}{\sqrt{2}-1}.$$

This tidies up nicely: $(\sqrt{2}+1)(\sqrt{2}-1) = 1$, so $\dfrac{\sqrt{2}+1}{\sqrt{2}-1} = (\sqrt{2}+1)^2$, and therefore

$$I_2 = \frac{1}{2\sqrt{2}}\cdot 2\ln(1+\sqrt{2}) = \frac{\ln(1+\sqrt{2})}{\sqrt{2}}.$$

The inequalities

By the binomial theorem, using $(\sqrt{2})^2 = 2$, $(\sqrt{2})^3 = 2\sqrt{2}$, $(\sqrt{2})^4 = 4$ and $(\sqrt{2})^5 = 4\sqrt{2}$,

$$\begin{aligned}(1+\sqrt{2})^5 &= 1 + 5\sqrt{2} + 10\cdot 2 + 10\cdot 2\sqrt{2} + 5\cdot 4 + 4\sqrt{2}\\ &= 41 + 29\sqrt{2}.\end{aligned}$$

Now $\sqrt{2} < 2$ (as $2 < 4$), so $29\sqrt{2} < 58$ and $(1+\sqrt{2})^5 < 41 + 58 = 99$, as required.

Next, $1.4^2 = 1.96 < 2$, and since both 1.4 and $\sqrt{2}$ are positive, squaring preserves order, so $\sqrt{2} > 1.4$.

Ask yourself How can $(1+\sqrt{2})^5 < 99$ and $\sqrt{2} > 1.4$ possibly tell us anything about $2^{\sqrt{2}}$?

Write $1.4 = \frac{7}{5}$. Then $2^{7/5}$ is the fifth root of $2^7 = 128$, and the fifth power of $1+\sqrt{2}$ is what we have just bounded. Precisely: since $t \mapsto 2^t$ is increasing and $\sqrt{2} > \frac{7}{5}$,

$$2^{\sqrt{2}} > 2^{7/5}.$$

Also $(1+\sqrt{2})^5 < 99 < 128 = \left(2^{7/5}\right)^5$, and since $y \mapsto y^5$ is increasing for $y > 0$, taking positive fifth roots gives $1+\sqrt{2} < 2^{7/5}$. Combining,

$$2^{\sqrt{2}} > 2^{7/5} > 1 + \sqrt{2}.$$

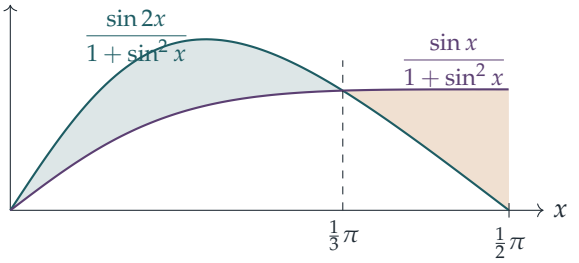
Which integral is greater?

The function ln is increasing, so the last inequality gives $\ln\left(2^{\sqrt{2}}\right) > \ln(1+\sqrt{2})$, that is, $\sqrt{2}\ln 2 > \ln(1+\sqrt{2})$. Dividing by $\sqrt{2} > 0$,

$$I_1 = \ln 2 > \frac{\ln(1+\sqrt{2})}{\sqrt{2}} = I_2.$$

So the integral with $\sin 2x$ in the numerator is the greater.

The comparison is not something we could have read off the integrands directly. Since $\sin 2x - \sin x = \sin x\,(2\cos x - 1)$, the first integrand is the larger for $0 < x < \frac{1}{3}\pi$ and the smaller for $\frac{1}{3}\pi < x < \frac{1}{2}\pi$:



The question is whether the teal region outweighs the amber one, and it does, but only by about 0.07. That is why some careful arithmetic with $\sqrt{2}$ was needed.

Checks

- Simpson's rule on the original integrands gives $I_1 \approx 0.693147$ and $I_2 \approx 0.623225$, matching $\ln 2$ and $\ln(1+\sqrt{2})/\sqrt{2}$ to ten decimal places.
- $(1+\sqrt{2})^5 \approx 82.01$ and $41 + 29\sqrt{2} \approx 82.01$; well below 99.
- $2^{\sqrt{2}} \approx 2.665$, $2^{1.4} \approx 2.639$ and $1+\sqrt{2} \approx 2.414$, in the claimed order.
- Sanity check on I_2: the integrand lies between $\frac{1}{2}\sin x$ and $\sin x$, so $\frac{1}{2} < I_2 < 1$, as 0.62 does.

Common slips

- Forgetting that $\mathrm{d}c = -\sin x\,\mathrm{d}x$ flips the limits, and ending with a negative value for a positive integrand.
- Leaving I_2 as $\frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2}+1}{\sqrt{2}-1}$: without spotting that this is $(\sqrt{2}+1)^2$, the link to $1+\sqrt{2}$ in the final part is hidden.
- Asserting $2^{\sqrt{2}} > 1+\sqrt{2}$ from a calculator. The question wants the chain through $2^{7/5}$, with the monotonicity of 2^t and y^5 stated.

Beyond STEP

What does $2^{\sqrt{2}}$ even mean? At university it is *defined* as $e^{\sqrt{2}\ln 2}$, and the fact we used, that 2^t is increasing in t for irrational as well as rational t, follows from properties of the exponential function that a first analysis course proves carefully. (A famous theorem of Gelfond and Schneider shows that $2^{\sqrt{2}}$ is in fact transcendental.) For the careful construction of e^x and $\ln x$, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2008, Question 6

Pure • Sum-to-Product Formulae

STEP II 2008, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q6

The key idea

A sum of a cosine and a sine with different arguments is awkward, but a sum of two cosines factorises into a product by the sum-to-product formula. Rewrite the sine as a cosine first; then the zeros of f are the zeros of two simple factors, and a point where *both* factors vanish is where the curve touches the axis. For the last part, remember that a cosine and a sine can each be at most 1.

Solution

Throughout, write

$$A = 2x + \frac{\pi}{3}, \qquad \sin\left(\frac{3x}{2} - \frac{\pi}{4}\right) = \cos\left(\frac{\pi}{2} - \frac{3x}{2} + \frac{\pi}{4}\right) = \cos B, \quad \text{where } B = \frac{3\pi}{4} - \frac{3x}{2},$$

so that $f(x) = \cos A + \cos B$.

Part (i)

The term $\cos(2x + \frac{\pi}{3})$ has period π, and $\sin(\frac{3x}{2} - \frac{\pi}{4})$ has period $\frac{2\pi}{3/2} = \frac{4\pi}{3}$. A common period must be a whole number of each: $p\pi = q \cdot \frac{4\pi}{3}$ means $3p = 4q$, and the smallest positive solution is $p = 4$, $q = 3$. So $f(x + 4\pi) = f(x)$ for all x, and the period of f is

$$4\pi.$$

Nothing smaller works: we shall see in part (iii) that $f(x) = 2$ exactly when $x = \frac{11\pi}{6} + 4t\pi$ for integer t, so the points where f reaches its maximum are spaced 4π apart, and any period must be a multiple of 4π.

Part (ii)

Using $\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$, with

$$\frac{A+B}{2} = \frac{1}{2}\left(\frac{x}{2} + \frac{13\pi}{12}\right) = \frac{x}{4} + \frac{13\pi}{24},$$
$$\frac{A-B}{2} = \frac{1}{2}\left(\frac{7x}{2} - \frac{5\pi}{12}\right) = \frac{7x}{4} - \frac{5\pi}{24},$$

we get

$$f(x) = 2\cos\left(\frac{x}{4} + \frac{13\pi}{24}\right)\cos\left(\frac{7x}{4} - \frac{5\pi}{24}\right).$$

So $f(x) = 0$ exactly when one of the two cosines is zero.

First factor. $\frac{x}{4} + \frac{13\pi}{24} = \frac{\pi}{2} + k\pi$ gives $x = -\frac{\pi}{6} + 4k\pi$. The only one in $[-\pi, \pi]$ is $x = -\frac{\pi}{6}$.

Second factor. $\frac{7x}{4} - \frac{5\pi}{24} = \frac{\pi}{2} + k\pi$ gives $\frac{7x}{4} = \frac{17\pi}{24} + k\pi$, so

$$x = \frac{(17 + 24k)\pi}{42}.$$

We need $-42 \le 17 + 24k \le 42$, so $k = -2, -1, 0, 1$, giving $x = -\frac{31\pi}{42}$, $-\frac{7\pi}{42} = -\frac{\pi}{6}$, $\frac{17\pi}{42}$, $\frac{41\pi}{42}$.

Combining, the zeros of f in $-\pi \le x \le \pi$ are

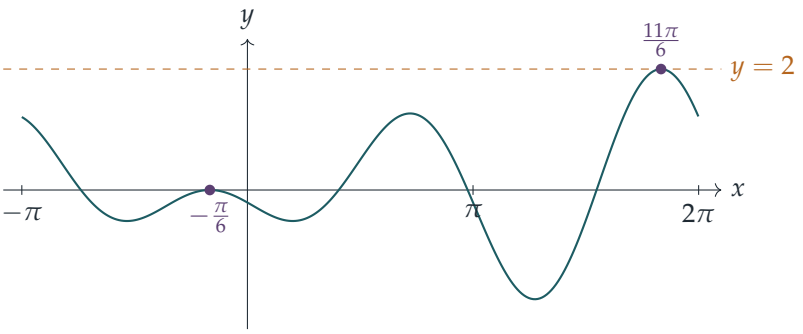
$$x = -\frac{31\pi}{42}, \quad -\frac{\pi}{6}, \quad \frac{17\pi}{42}, \quad \frac{41\pi}{42}.$$

Ask yourself Why should a value that appears in *both* lists be special?

At $x = -\frac{\pi}{6}$ both cosine factors are zero. Each is a cosine passing through a zero, so each changes sign there (its derivative is non-zero, since sin is ± 1 where cos is 0). Near $x = -\frac{\pi}{6}$, then, f behaves like a constant times $(x + \frac{\pi}{6})^2$: the product of two sign changes is no sign change. In fact, as x increases through $-\frac{\pi}{6}$, the first factor goes from positive to negative (its argument passes up through $\frac{\pi}{2}$) and the second from negative to positive (its argument passes up through $-\frac{\pi}{2}$), so f is negative on either side. Hence the curve touches the x-axis without crossing it at

$$x = -\frac{\pi}{6}.$$

At the other three zeros only one factor vanishes, so f changes sign and the curve crosses.



Part (iii)

Since $\cos A \leq 1$ and $\cos B \leq 1$, we have $f(x) \leq 2$, with equality if and only if $\cos A = 1$ *and* $\cos B = 1$. In terms of the original functions:

$$\cos\left(2x + \frac{\pi}{3}\right) = 1 \iff 2x + \frac{\pi}{3} = 2m\pi \iff x = -\frac{\pi}{6} + m\pi,$$
$$\sin\left(\frac{3x}{2} - \frac{\pi}{4}\right) = 1 \iff \frac{3x}{2} - \frac{\pi}{4} = \frac{\pi}{2} + 2n\pi \iff x = \frac{\pi}{2} + \frac{4n\pi}{3},$$

for integers m, n. Both hold when $-\frac{\pi}{6} + m\pi = \frac{\pi}{2} + \frac{4n\pi}{3}$; multiplying by $\frac{6}{\pi}$, this is $6m - 1 = 3 + 8n$, that is,

$$3m - 4n = 2.$$

One solution is $m = 2$, $n = 1$, and the general solution is $m = 2 + 4t$, $n = 1 + 3t$ (since 3 and 4 are coprime). So $f(x) = 2$ exactly when

$$x = -\frac{\pi}{6} + (2 + 4t)\pi = \frac{11\pi}{6} + 4t\pi, \qquad t \in \mathbb{Z}.$$

The only such value in $0 \leq x \leq 2\pi$ is

$$x = \frac{11\pi}{6}.$$

Check: $2 \cdot \frac{11\pi}{6} + \frac{\pi}{3} = 4\pi$ and $\frac{3}{2} \cdot \frac{11\pi}{6} - \frac{\pi}{4} = \frac{5\pi}{2}$, so $f = \cos 4\pi + \sin\frac{5\pi}{2} = 1 + 1 = 2$.

Checks

- The factorised form agrees with the original f at 10^4 random points (to 10^{-12}).
- Scanning $[-\pi, \pi]$ numerically finds sign changes only near $-\frac{31\pi}{42}$, $\frac{17\pi}{42}$ and $\frac{41\pi}{42}$; at $-\frac{\pi}{6}$ we find $f(-\frac{\pi}{6} \pm 10^{-3}) \approx -8.7 \times 10^{-7}$ on *both* sides, confirming a touching zero with no sign change.
- The maximum of f on $[0, 2\pi]$ is 2, attained at $x \approx 1.8333\pi = \frac{11\pi}{6}$.

Common slips

- Listing $-\frac{\pi}{6}$ twice, or missing that it comes from both factors, which is exactly the clue to the touching point.
- Solving $\cos A = 1$ and $\cos B = 1$ separately and stopping: both conditions must hold at the same x, which needs the integer equation $3m - 4n = 2$.

Beyond STEP

Part (iii) is really a question about when two periodic motions line up: the condition $3m - 4n = 2$ is a linear Diophantine equation, solvable because $\gcd(3, 4)$ divides 2. The same idea, applied to many frequencies at once, leads to Fourier series and to the question of when a sum of periodic functions is itself periodic (only when the periods are rationally related). For the number theory behind it, try Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 2008, Question 7

Pure • First Order Differential Equations

STEP II 2008, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q7

The key idea

The left-hand side $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}$ is the derivative of $\ln y$, so the substitution is best handled by taking logarithms: a product $y = u\,(1+x^2)^{1/2}$ turns into a sum, and the term $\frac{x}{1+x^2}$ on the right is exactly the derivative of $\ln(1+x^2)^{1/2}$. It cancels, leaving a separable equation for u. Part (ii) is the same idea with the cube root, and once the two answers sit side by side the pattern for (iii) is plain.

Solution

Part (i)

We have $y = 1 \neq 0$ at $x = 0$, and the equation involves $1/y$, so we work on an interval around $x = 0$ on which $y \neq 0$. Since $(1+x^2)^{1/2} > 0$, the substitution gives $u \neq 0$ there too, and $u(0) = 1$.

Taking logarithms of $|y| = |u|\,(1+x^2)^{1/2}$ and differentiating,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{x}{1+x^2}.$$

Comparing with the differential equation, the $\frac{x}{1+x^2}$ terms cancel and we are left with

$$\frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} = xy = x\,u\,(1+x^2)^{1/2}, \qquad \text{so} \qquad \frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}x} = x(1+x^2)^{1/2},$$

dividing by u, which we may do since $u \neq 0$.

Ask yourself How do we integrate $x(1+x^2)^{1/2}$ without a fuss?

The factor x is half the derivative of $1+x^2$, so by the reverse chain rule $\int x(1+x^2)^{1/2}\,\mathrm{d}x = \frac{1}{2}\cdot\frac{2}{3}(1+x^2)^{3/2} = \frac{1}{3}(1+x^2)^{3/2}$, plus a constant. (Differentiate to check.)

Integrating both sides with respect to x,

$$-\frac{1}{u} = \tfrac{1}{3}(1+x^2)^{3/2} + C.$$

At $x = 0$, $u = 1$, so $-1 = \frac{1}{3} + C$ and $C = -\frac{4}{3}$. Hence

$$\frac{1}{u} = \frac{4-(1+x^2)^{3/2}}{3}, \qquad u = \frac{3}{4-(1+x^2)^{3/2}},$$

and therefore

$$\boxed{y = \frac{3\,(1+x^2)^{1/2}}{4-(1+x^2)^{3/2}}}$$

This is valid on the interval containing 0 where the denominator is positive, namely $(1+x^2)^{3/2} < 4$, i.e. $|x| < \sqrt{2^{4/3}-1} \approx 1.233$; as $|x|$ approaches that value, $y \to \infty$.

Part (ii)

Now the extra term is $\frac{x^2}{1+x^3}$, which is the derivative of $\frac{1}{3}\ln(1+x^3) = \ln(1+x^3)^{1/3}$. That tells us to write $y = u\,(1+x^3)^{1/3}$, working near $x = 0$ (where $1+x^3 > 0$). Exactly as before,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{x^2}{1+x^3},$$

so the equation becomes

$$\frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} = x^2 y = x^2 u\,(1+x^3)^{1/3}, \qquad \frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}x} = x^2(1+x^3)^{1/3}.$$

This time x^2 is a third of the derivative of $1+x^3$, so

$$-\frac{1}{u} = \tfrac{1}{3}\cdot\tfrac{3}{4}(1+x^3)^{4/3} + C = \tfrac{1}{4}(1+x^3)^{4/3} + C.$$

With $u = 1$ at $x = 0$, $-1 = \frac{1}{4} + C$, so $C = -\frac{5}{4}$. Thus $\dfrac{1}{u} = \dfrac{5-(1+x^3)^{4/3}}{4}$ and

$$\boxed{y = \frac{4\,(1+x^3)^{1/3}}{5-(1+x^3)^{4/3}}}$$

valid near 0 for $-1 < x < \left(5^{3/4}-1\right)^{1/3} \approx 1.33$.

Part (iii)

Comparing the two answers, the root $(1+x^n)^{1/n}$ appears on top, the power $(1+x^n)^{(n+1)/n}$ appears below, and the constants are $n+1$ and $n+2$. So we conjecture

$$\boxed{y = \frac{(n+1)\,(1+x^n)^{1/n}}{n+2-(1+x^n)^{(n+1)/n}}}$$

which gives the answers to (i) and (ii) when $n = 2$ and $n = 3$. (No proof is asked for, but the same substitution $y = u(1+x^n)^{1/n}$ proves it: $\int x^{n-1}(1+x^n)^{1/n}\,\mathrm{d}x = \frac{1}{n+1}(1+x^n)^{(n+1)/n}$, and the condition at $x = 0$ gives the constant $-\frac{n+2}{n+1}$.)

Checks

- At $x = 0$ each formula gives $\frac{n+1}{n+2-1} = 1$, as required.
- Near $x = 0$ in (i), the equation gives $y'(0) = 0$, and indeed both $(1+x^2)^{1/2}$ and $(1+x^2)^{3/2}$ have zero gradient at 0.
- Solving the equations numerically (a Runge–Kutta step from $x = 0$) matches the formulas to better than 10^{-8} for $n = 2, 3, 4, 5, 7$ at several points. For example in (i) at $x = 1$, $y = \frac{3\sqrt{2}}{4-2\sqrt{2}} \approx 3.6213$, and in (ii) at $x = 1$, $y = \frac{4\cdot 2^{1/3}}{5-2^{4/3}} \approx 2.0320$.

Common slips

- Substituting $y = u(1+x^2)^{1/2}$ and differentiating with the product rule directly: it works, but it is messy. Dividing through by y (equivalently, taking logs) makes the cancellation obvious.
- Forgetting the minus sign in $\int u^{-2}\,\mathrm{d}u = -u^{-1}$, which spoils the constant.
- Fixing the constant at the wrong stage: apply $u(0) = 1$ (not $y(0) = 1$ in the u-equation) before rearranging.

Beyond STEP

The equation is a *Bernoulli equation*: $y' = P(x)y + Q(x)y^2$, and the general trick is the substitution $v = 1/y$, which makes it linear. Here the substitution in the question does that job in two steps: removing the P-term with an integrating-factor-like $(1+x^n)^{1/n}$ and then separating. Notice also that the solution blows up at a finite x even though the equation looks harmless; this "finite-time blow-up" is typical of equations with a y^2 term and is studied in a first course on ODEs. Boyce and DiPrima, *Elementary Differential Equations*, covers Bernoulli equations and the existence of solutions.

WORKED SOLUTION

STEP II 2008, Question 8

Pure • Vectors and Angle Bisectors

STEP II 2008, Question 8

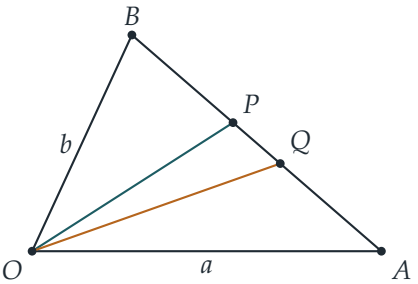
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q8

The key idea

A vector along the bisector of $\angle AOB$ is the sum of the *unit* vectors along OA and OB: adding two vectors of equal length gives the diagonal of a rhombus, and a rhombus diagonal bisects its angle. Matching this with the position vector of P fixes λ. After that, Q is just P with the roles of A and B swapped, and the final identity is a direct calculation with scalar products in which the awkward term $\mathbf{a}\cdot\mathbf{b}$ cancels.

Solution



In the picture $a > b$, so P (on the bisector, in teal) is nearer B, and Q is the point the same distance from B as P is from A.

The position vector of P

Since $AP : PB = (1-\lambda) : \lambda$ and the two parts add up to AB, we have $AP = (1-\lambda)AB$, so $\overrightarrow{AP} = (1-\lambda)(\mathbf{b}-\mathbf{a})$ and

$$\overrightarrow{OP} = \mathbf{a} + (1-\lambda)(\mathbf{b}-\mathbf{a}) = \lambda\mathbf{a} + (1-\lambda)\mathbf{b}.$$

(Check: $\lambda = 1$ gives A and $\lambda = 0$ gives B, as the ratio says.)

Finding λ

The vectors $\mathbf{a}/a$ and $\mathbf{b}/b$ are unit vectors along OA and OB. Their sum is the diagonal of a rhombus with sides along OA and OB, so it points along the bisector of $\angle AOB$. Since P lies on this bisector,

$$\lambda\mathbf{a} + (1-\lambda)\mathbf{b} = k\left(\frac{\mathbf{a}}{a} + \frac{\mathbf{b}}{b}\right)$$

for some scalar k.

Ask yourself Why may we compare the coefficients of $\mathbf{a}$ and of $\mathbf{b}$ separately?

Because O, A, B are not collinear, $\mathbf{a}$ and $\mathbf{b}$ are non-zero and not parallel, so $\mathbf{a}$ and $\mathbf{b}$ are linearly independent: if $\alpha\mathbf{a} + \beta\mathbf{b} = \mathbf{0}$ then $\alpha = \beta = 0$. Hence

$$\lambda = \frac{k}{a}, \qquad 1-\lambda = \frac{k}{b}.$$

Adding, $1 = k\left(\frac{1}{a} + \frac{1}{b}\right)$, so $k = \dfrac{ab}{a+b}$ and

$$\boxed{\lambda = \frac{b}{a+b}}, \qquad 1-\lambda = \frac{a}{a+b}.$$

This is the angle bisector theorem in disguise: $AP : PB = (1-\lambda) : \lambda = a : b = OA : OB$. So

$$\overrightarrow{OP} = \frac{b\,\mathbf{a} + a\,\mathbf{b}}{a+b}.$$

The point Q and the identity

Q is on AB with $BQ = AP = (1-\lambda)AB$, so by the same reasoning as for P, measured from B instead of from A,

$$\overrightarrow{OQ} = \mathbf{b} + (1-\lambda)(\mathbf{a}-\mathbf{b}) = (1-\lambda)\mathbf{a} + \lambda\mathbf{b} = \frac{a\,\mathbf{a} + b\,\mathbf{b}}{a+b}.$$

Write $c = \mathbf{a}\cdot\mathbf{b}$, and use $\mathbf{a}\cdot\mathbf{a} = a^2$, $\mathbf{b}\cdot\mathbf{b} = b^2$. Then

$$(a+b)^2\,OQ^2 = (a\mathbf{a} + b\mathbf{b})\cdot(a\mathbf{a} + b\mathbf{b}) = a^2\cdot a^2 + 2ab\,c + b^2\cdot b^2 = a^4 + b^4 + 2abc,$$
$$(a+b)^2\,OP^2 = (b\mathbf{a} + a\mathbf{b})\cdot(b\mathbf{a} + a\mathbf{b}) = b^2a^2 + 2ab\,c + a^2b^2 = 2a^2b^2 + 2abc.$$

Subtracting, the c terms cancel:

$$(a+b)^2\left(OQ^2 - OP^2\right) = a^4 - 2a^2b^2 + b^4 = (a^2-b^2)^2 = (a-b)^2(a+b)^2.$$

Dividing by $(a+b)^2 \neq 0$,

$$OQ^2 - OP^2 = (a-b)^2 = (b-a)^2,$$

as required.

Checks

Take $\mathbf{a} = (3,0)$ and $\mathbf{b} = (0,4)$, so $a = 3$, $b = 4$ and $\angle AOB$ is a right angle. Then $\lambda = \frac{4}{7}$ and $\overrightarrow{OP} = \left(\frac{12}{7}, \frac{12}{7}\right)$, which lies on $y = x$, the bisector of the right angle. Also $\overrightarrow{OQ} = \left(\frac{9}{7}, \frac{16}{7}\right)$, so $OQ^2 - OP^2 = \frac{337}{49} - \frac{288}{49} = 1 = (4-3)^2$. If $a = b$ the triangle is isosceles, $P = Q$ is the midpoint of AB, and the difference is 0, as it should be. A numerical test on 20,000 random triangles in two and three dimensions agreed with every result to within 10^{-13}.

Common slips

- Writing $\overrightarrow{OP} = (1-\lambda)\mathbf{a} + \lambda\mathbf{b}$: the larger share of the ratio, $1-\lambda$, is the distance from A, so it goes with $\mathbf{b}$. Test with $\lambda = 1$.
- Using $\mathbf{a} + \mathbf{b}$ as the bisector direction. That only works when $a = b$; you need the unit vectors.
- Expanding $|\mathbf{a} + \mathbf{b}|^2$ as $a^2 + b^2$, forgetting the cross term $2\,\mathbf{a} \cdot \mathbf{b}$. Here it cancels, but only if it is there in both expansions.

Beyond STEP

The final calculation is a special case of how lengths transform under a linear map: OP^2 and OQ^2 are values of the quadratic form $\mathbf{x} \mapsto \mathbf{x} \cdot \mathbf{x}$, and the cross terms are controlled by the Gram matrix of $\mathbf{a}$ and $\mathbf{b}$. The fact that unit vectors add to give a bisector also generalises: in any inner-product space, normalising and adding is how we find the direction "halfway" between two vectors. Strang's *Introduction to Linear Algebra* develops scalar products, lengths and angles in this spirit.

WORKED SOLUTION

STEP II 2008, Question 9

Mechanics • Projectile Motion

STEP II 2008, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

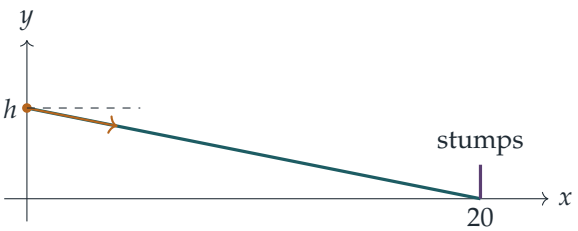
step.maths.org/questions/08-s2-q9

The key idea

The horizontal motion fixes the time of flight in terms of α, and putting that time into the vertical motion gives one equation linking α and h. Writing $\sec^2\alpha = 1 + \tan^2\alpha$ turns it into a quadratic in $\tan\alpha$, the standard trick for "which angle hits the target?" problems. After that, common sense about cricket picks the right root, and small-angle approximations finish the job.

Solution

We model the ball as a particle moving freely under gravity, with no air resistance and no spin. Take the origin on the ground directly below the point of projection, with x horizontal towards the stumps and y vertically up, so the ball starts at $(0, h)$ with speed $40\,\mathrm{m\,s^{-1}}$ at angle α above the horizontal (a negative α means it is aimed downwards).



Part (i)

Horizontally there is no force, so $x = 40t\cos\alpha$. The ball reaches $x = 20$ when

$$t = \frac{20}{40\cos\alpha} = \frac{1}{2\cos\alpha}.$$

Vertically the acceleration is $-10\,\mathrm{m\,s^{-2}}$, so

$$y = h + 40t\sin\alpha - 5t^2.$$

The ball is at ground level, $y = 0$, at this time, so

$$0 = h + 40\sin\alpha \cdot \frac{1}{2\cos\alpha} - \frac{5}{4\cos^2\alpha} = h + 20\tan\alpha - \tfrac{5}{4}\sec^2\alpha.$$

Now we use $\sec^2\alpha = 1 + \tan^2\alpha$ and write $T = \tan\alpha$:

$$h + 20T - \tfrac{5}{4}(1 + T^2) = 0 \quad \Longleftrightarrow \quad 5T^2 - 80T + 5 - 4h = 0.$$

By the quadratic formula,

$$T = \frac{80 \pm \sqrt{6400 - 20(5 - 4h)}}{10} = \frac{80 \pm \sqrt{6300 + 80h}}{10},$$

so the two possible values are

$$\tan\alpha = 8 \pm \sqrt{63 + \tfrac{4}{5}h}.$$

Both are real, since $h \geq 0$.

Ask yourself Which root would a fast bowler actually use?

Since $\sqrt{63} \approx 7.9$, the larger root gives $\tan\alpha \approx 16$, so $\alpha \approx 86°$: a towering lob that would rise about 80 m and take around 8 s to come down on the stumps. That is no fast bowler's delivery. The smaller root,

$$\tan\alpha = 8 - \sqrt{63 + \tfrac{4}{5}h},$$

is the appropriate one. For a realistic release height (a couple of metres), $63 + \frac{4}{5}h$ is close to 64, so $\tan\alpha$ is close to 0. For example, any h between 0 and 3 gives $|\tan\alpha| < 0.09$. So α is a small angle and $\cos\alpha \approx 1$, and the time to reach the stumps is

$$t = \frac{1}{2\cos\alpha} \approx \tfrac{1}{2}\text{ s},$$

as required. (In fact $t = \frac{1}{2}\sqrt{1 + \tan^2\alpha}$, which is within 0.5% of $\frac{1}{2}$ whenever $|\tan\alpha| < 0.1$.)

Part (ii)

The ball is projected below the horizontal exactly when $\tan\alpha < 0$. The larger root is always positive, so we use the smaller one:

$$8 - \sqrt{63 + \tfrac{4}{5}h} < 0 \iff \sqrt{63 + \tfrac{4}{5}h} > 8 \iff 63 + \tfrac{4}{5}h > 64 \iff h > \tfrac{5}{4},$$

where squaring is reversible because both sides are positive. So the bowler projects the ball below the horizontal when $h > 1.25$. When $h = 1.25$ exactly, the ball is projected horizontally.

Part (iii)

When $h = 2.5$ we know from part (ii) that α is small and negative. With $\cos\alpha \approx 1$ and $\sin\alpha \approx \alpha$ (in radians), we have $\tan\alpha \approx \alpha$ and $\sec^2\alpha \approx 1$, so the equation $0 = h + 20\tan\alpha - \frac{5}{4}\sec^2\alpha$ from part (i) becomes

$$0 \approx 2.5 + 20\alpha - 1.25, \qquad \text{so} \qquad \alpha \approx -\frac{1.25}{20} = -0.0625 \text{ rad}.$$

Converting to degrees,

$$\alpha \approx -0.0625 \times \frac{180}{\pi} \approx -3.58^\circ,$$

so $\alpha \approx -3.6^\circ$. The ball is released about 3.6° below the horizontal.

Another way to see it: $\tan\alpha = 8 - \sqrt{65} = 8 - 8\sqrt{1 + \frac{1}{64}} \approx 8 - 8\left(1 + \frac{1}{128}\right) = -\frac{1}{16}$, using the binomial approximation. This gives the same answer.

Checks

- *Exact value.* $\tan\alpha = 8 - \sqrt{65} = -0.06226$, so $\alpha = -3.563^\circ$, which is -3.6° to two significant figures, in agreement with the approximation.
- *Simulation.* Stepping the motion numerically from $(0, 2.5)$ at $40\,\text{m}\,\text{s}^{-1}$ and -3.563°, the ball reaches $x = 20$ at height $0.0000\,\text{m}$ after $0.501\,\text{s}$.
- *Boundary.* At $h = 1.25$ the smaller root is $8 - \sqrt{64} = 0$: the ball is projected horizontally, falls $5(0.5)^2 = 1.25\,\text{m}$ in the half-second flight, and lands exactly at the stumps.

Common slips

- Forgetting $\sec^2\alpha = 1 + \tan^2\alpha$ and trying to solve in $\sin\alpha$ and $\cos\alpha$ separately, which gets messy.
- Giving the angle in radians, or using the small-angle approximation in degrees: $\sin\theta \approx \theta$ only holds with θ in radians.
- Dropping the minus sign in part (iii). The angle is below the horizontal, as part (ii) predicts.

Beyond STEP

The two-root structure is the general fact that a projectile with a fixed speed reaches any point inside its "envelope" (a parabola of safety) by two trajectories. On the envelope the two merge into one, and points outside it cannot be reached at all. Real cricket balls are also affected by drag and by the Magnus effect from spin, which leads to nonlinear differential equations that usually have to be solved numerically. Kleppner

and Kolenkow, *An Introduction to Mechanics*, treats projectiles and resisted motion at first-year level.

WORKED SOLUTION

STEP II 2008, Question 10

Mechanics • Oblique Impacts with Walls

STEP II 2008, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q10

The key idea

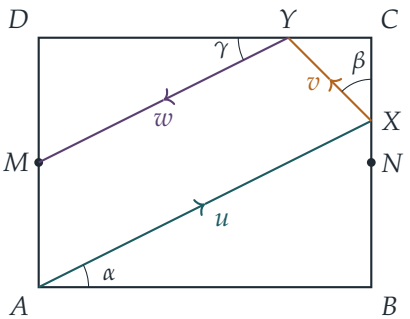
At a smooth wall the impulse is perpendicular to the wall, so the velocity component along the wall is unchanged and only the component into the wall is multiplied by e. Splitting the velocity into components parallel to AB and to AD makes everything transparent: each bounce scales just one of the two components. After both bounces, both components have been scaled by e, which settles the final direction and the kinetic energy at once. The geometry of part (ii) is then a matter of adding up horizontal distances.

Solution

We model the ball as a particle and the sides as smooth walls. Take axes with origin A, the x-axis along AB and the y-axis along AD, so that

$$B = (a, 0), \quad C = (a, 2b), \quad D = (0, 2b), \quad M = (0, b), \quad N = (a, b).$$

Write $t = \tan\alpha$ for short.



Part (i)

Initially the velocity is $(u\cos\alpha,\ u\sin\alpha)$.

Impact at X. The side BC is parallel to the y-axis. Since it is smooth, the impulse on the ball is along the x-direction, so the y-component $u\sin\alpha$ is unchanged. By Newton's law of restitution the x-component is reversed and multiplied by e. The velocity after the impact is

$$(-eu\cos\alpha,\ u\sin\alpha).$$

The angle $\beta = \angle CXY$ is measured from XC, which points in the positive y-direction. So

$$\tan\beta = \frac{eu\cos\alpha}{u\sin\alpha} = \frac{e}{\tan\alpha}, \qquad \text{that is,} \qquad \tan\alpha\tan\beta = e,$$

as required.

Impact at Y. The side DC is parallel to the x-axis, so now the x-component is unchanged and the y-component is reversed and multiplied by e. The velocity after the impact is

$$(-eu\cos\alpha,\ -eu\sin\alpha) = -e\,(u\cos\alpha,\ u\sin\alpha).$$

This is a negative multiple of the initial velocity, so the final stage of the path is parallel to AX and travels in the opposite direction. The angle $\gamma = \angle DYM$ is measured from YD, which points in the negative x-direction, and the direction $(-\cos\alpha, -\sin\alpha)$ makes angle α with it. Hence

$$\gamma = \alpha.$$

Part (ii)

We follow the x-coordinate along the path.

From A to X. The ball rises $a\tan\alpha$ while crossing the width a, so $X = (a,\ at)$.

From X to Y. The ball rises a further $2b - at$, and on this stage it moves $\tan\beta$ units in x for each unit it rises (because β is measured from the vertical side). With $\tan\beta = e/t$ from part (i), the ball moves left a distance $(2b - at)\,e/t$, so

$$x_Y = a - \frac{e(2b - at)}{t}.$$

From Y to M. The ball drops from height $2b$ to height b, a drop of b, at angle $\gamma = \alpha$ below the horizontal. So the horizontal distance covered is $b/\tan\alpha$, and this must take it exactly to $x = 0$:

$$x_Y = \frac{b}{t}.$$

Ask yourself Why can we get away with tracking only the x-coordinate?

Each straight stage starts where the last one ended, and its direction is known from part (i). Once we know how far the ball rises or drops on a stage, the angle fixes how far it moves sideways. The heights at the ends of the stages are forced (0, then $2b$ at DC, then b at M), so the only condition left is that the sideways distances add up.

Equating the two expressions for x_Y and multiplying by t:

$$\begin{aligned} b &= at - e(2b - at) \\ &= at(1+e) - 2be, \end{aligned}$$

so $at(1+e) = b(1+2e)$, which gives

$$\tan\alpha = \frac{(1+2e)b}{(1+e)a},$$

as required.

The shot is always possible. This value of $\tan\alpha$ is positive for every $e > 0$, so it defines an angle α in $(0, \frac{\pi}{2})$. We still need the bounce points to land on the sides they are meant to.

- X is at height $at = \dfrac{1+2e}{1+e}\,b$. Since $1+e < 1+2e < 2+2e$ for $e > 0$, this lies strictly between b and $2b$. So X is genuinely on BC, below C, and above the pocket N, so the ball doesn't drop into N on the way.
- Y is at $x_Y = b/t = \dfrac{1+e}{1+2e}\,a$, which lies strictly between 0 and a for $e > 0$. So Y is genuinely on DC, between D and C.

The final stage then runs from Y to M in a straight line inside the table. Every stage of the path is therefore possible, whatever the value of e.

Part (iii)

From part (i), the final velocity is $-e$ times the initial velocity, so $w = eu$. (We can also see this directly: each component has been multiplied by e exactly once.) The kinetic energy falls from $\frac{1}{2}mu^2$ to $\frac{1}{2}me^2u^2$, where m is the mass of the ball. The fraction lost is therefore

$$\frac{\frac{1}{2}mu^2 - \frac{1}{2}me^2u^2}{\frac{1}{2}mu^2} = 1 - e^2.$$

This doesn't depend on a, b or the angles. (For $e > 1$ the "loss" is negative: the walls would be adding energy. That isn't physical, but the algebra holds.)

Checks

- With $a = 4$, $b = 1.5$ and $e = \frac{1}{2}$ (the diagram), the formula gives $\tan\alpha = \frac{1}{2}$. Then $X = (4, 2)$, $Y = (3, 3)$, and the line from Y with gradient $\frac{1}{2}$ drops 1.5 over a horizontal run of 3, landing exactly on $M = (0, 1.5)$. The kinetic energy lost is $1 - \frac{1}{4} = \frac{3}{4}$.
- When $e = 1$ (perfectly elastic), $\tan\alpha = 3b/(2a)$. Unfold the table by reflecting it in BC and then in DC: the path becomes a straight line from A to the image of M, which is $2a$ across and $3b$ up. That gives the same gradient, $3b/(2a)$. No energy is lost, and $1 - e^2 = 0$ agrees.
- A simulation of the bounces for 20,000 random choices of a, b, e and u confirmed $\tan\alpha\tan\beta = e$, $\gamma = \alpha$, arrival at M and the fraction $1 - e^2$, all to within 10^{-13}.

Common slips

- Applying e to the whole speed at an impact. Only the component perpendicular to the wall changes. The component along a smooth wall is untouched.
- Measuring β from BC's normal instead of from XC, which turns $\tan\beta$ into its reciprocal.
- Forgetting to check that X is above N and Y is between D and C. That check is the content of "the shot is possible".

Beyond STEP

For $e = 1$, "unfolding" a billiard path by reflecting the table turns it into a straight line. This is the starting point of mathematical billiards, which links to dynamical systems and, for rational tables, to geometry on surfaces. With $e < 1$ the reflection is no longer mirror-like, but splitting the velocity into components still makes it easy to handle. For the mechanics of impulses and restitution, see Kleppner and Kolenkow, *An Introduction to Mechanics*. For more problems of this flavour, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2008, Question 11

Mechanics • Particle on Moving Wedge

STEP II 2008, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

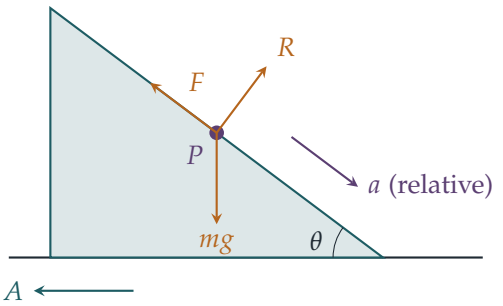
step.maths.org/questions/08-s2-q11

The key idea

Nothing outside the system pushes horizontally (the floor is smooth), so the horizontal momentum of particle plus wedge stays at zero; that gives the wedge's acceleration almost for free. Everything else comes from writing the particle's *true* acceleration as "its acceleration relative to the wedge plus the wedge's acceleration" and then resolving forces on the particle along and perpendicular to the slope. A path seen by the stationary observer is a straight line in the direction of the true acceleration, since everything starts from rest and the accelerations are constant.

Solution

We model P as a particle and the wedge as a rigid body. The floor is smooth; the inclined face is rough with coefficient μ. Put the vertical face on the left, so the slope runs down to the right. When P slides down to the right, the wedge is pushed to the left. Let the wedge have acceleration A to the *left*, and let P have acceleration a down the slope *relative to the wedge*. Let R be the normal reaction and F the friction on P, which acts up the slope while P slides down.



Take x to the right and y upwards. The unit vector down the slope is $(\cos\theta, -\sin\theta)$, so

P's true acceleration is

$$\big(a\cos\theta - A,\ -a\sin\theta\big).$$

Part (i): the wedge's acceleration

The only external forces on the system "P plus wedge" are gravity and the normal reaction of the smooth floor, and both are vertical. So the total horizontal momentum is constant, and it is zero since everything starts at rest. Hence the horizontal components of $m\times$(acceleration) add to zero:

$$m(a\cos\theta - A) + km(-A) = 0 \quad\Longrightarrow\quad A = \frac{a\cos\theta}{k+1},$$

which is the required result. (The forces between P and the wedge, R and F, are internal and cancel by Newton's third law.)

Part (i): the 45° path



Ask yourself Why must the observed path be a straight line in the direction of the true acceleration?

The equations of motion below fix a and A as constants, so P's true acceleration is a constant vector. Starting from rest, its velocity is always t times that vector and its displacement is $\frac{1}{2}t^2$ times it, so it moves along a fixed straight line in that direction.

The horizontal component of P's acceleration is

$$a\cos\theta - \frac{a\cos\theta}{k+1} = \frac{k}{k+1}\,a\cos\theta,$$

and the vertical component has size $a\sin\theta$. A 45° line means these are equal:

$$a\sin\theta = \frac{k}{k+1}\,a\cos\theta \quad\Longrightarrow\quad \tan\theta = \frac{k}{k+1},$$

dividing by $a\cos\theta \neq 0$ (as P does slide, $a > 0$). This is the required result.

Part (i): the case $k = 3$

Now $\tan\theta = \frac{3}{4}$, so $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, and $A = \frac{a\cdot 4/5}{4} = \frac{a}{5}$.

We resolve for P. Its true acceleration has component $a - A\cos\theta$ down the slope and $-A\sin\theta$ along the outward normal (the wedge's leftward acceleration has these components; the relative acceleration is purely along the slope).

Perpendicular to the slope:

$$R - mg\cos\theta = -mA\sin\theta \quad\Longrightarrow\quad R = m\Big(\tfrac{4}{5}g - \tfrac{3}{25}a\Big).$$

Notice that R is *less* than $mg\cos\theta$: the wedge is falling away from P.

Down the slope, with $F = \mu R$ because P is sliding:

$$mg\sin\theta - \mu R = m(a - A\cos\theta) = m\left(a - \tfrac{4}{25}a\right) = \tfrac{21}{25}ma.$$

Substituting for R and multiplying by $25/m$:

$$15g - 20\mu g + 3\mu a = 21a \quad\Longrightarrow\quad a = \frac{5g(3-4\mu)}{3(7-\mu)}.$$

This is positive exactly when $\mu < \frac{3}{4} = \tan\theta$, as it must be for P to slide, which leads straight into part (ii).

Part (ii)

We claim that if $\tan\theta \le \mu$, then P stays where it is and the wedge does not move.

Rest is possible. If neither body moves, then for P we need $R = mg\cos\theta$ and a friction force $F = mg\sin\theta$ up the slope. Rough contact can supply this provided $F \le \mu R$, that is $\sin\theta \le \mu\cos\theta$, or $\tan\theta \le \mu$, which is exactly our condition. The wedge then feels R and F reversed; their horizontal components are $R\sin\theta = mg\sin\theta\cos\theta$ to the left and $F\cos\theta = mg\sin\theta\cos\theta$ to the right, which cancel. So the wedge is in equilibrium too, and the floor supplies only a vertical force.

> **Ask yourself** Could P slide anyway, even though rest is possible?

Sliding is impossible. Suppose P did slide down, with $F = \mu R$, for general k and θ. Let $s = \sin\theta - \mu\cos\theta$. The horizontal equation for the wedge is $kmA = R\sin\theta - \mu R\cos\theta = Rs$. Combining this with the normal equation $R = m(g\cos\theta - A\sin\theta)$ gives

$$R = \frac{kmg\cos\theta}{k + s\sin\theta}, \qquad a = \frac{(k+1)\,gs}{k + s\sin\theta}.$$

For contact we need $R > 0$, which forces $k + s\sin\theta > 0$; then a has the same sign as s. But $\tan\theta \le \mu$ means $s \le 0$, so $a \le 0$, which contradicts sliding down. Nothing could make P slide *up* the slope either, since only gravity, which has a down-slope component, could start relative motion. So when $\tan\theta \le \mu$, P remains at rest on the wedge and the wedge remains at rest on the floor. (When $\tan\theta = \mu$, friction is limiting but P still doesn't move.)

Checks

- With $\mu = 0$ and $k = 3$ we get $a = \frac{5g}{7}$ and $A = \frac{g}{7}$. Then P's true acceleration has horizontal component $\frac{4}{5}\cdot\frac{5g}{7} - \frac{g}{7} = \frac{3g}{7}$ and vertical component $\frac{3}{5}\cdot\frac{5g}{7} = \frac{3g}{7}$, which is indeed a 45° direction.
- As $k \to \infty$ (an immovable wedge), $A \to 0$ and the general formula gives $a \to g(\sin\theta - \mu\cos\theta)$, the familiar fixed-slope result.

- Numerically, with $g = 9.8$, $k = 3$ and $\mu = 0.2$, solving the three equations of motion directly gives $a \approx 5.284\ \mathrm{m\,s^{-2}}$, matching $\frac{5g(3-0.8)}{3(6.8)}$. Stepping the motion forward in time keeps the horizontal momentum at zero and gives a path at exactly $45°$.

Common slips

- Taking $R = mg\cos\theta$. That is only true when the wedge is fixed; here the wedge accelerates away, so R is smaller.
- Resolving with the relative acceleration a alone, forgetting the wedge's acceleration. Newton's second law needs the true acceleration.
- In part (ii), assuming the friction is μR even though P is not sliding. Friction is *at most* μR and takes whatever value is needed.

Beyond STEP

The neat step in part (i) is a conservation law: because the external forces have no horizontal component, one component of momentum is conserved. In first-year mechanics this becomes a general principle linking symmetries to conservation laws: in Lagrangian mechanics, a coordinate that the Lagrangian doesn't depend on (here the horizontal position of the whole system) gives a conserved momentum automatically. The smooth wedge and block is a standard first example there. Kleppner and Kolenkow, *An Introduction to Mechanics*, works through problems of exactly this kind.

WORKED SOLUTION

STEP II 2008, Question 12

Probability and Statistics • Independent Events

STEP II 2008, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q12

The key idea

The expectation of a count is the sum of the probabilities of the individual events that make it up, so the first part is one quadratic in p. For the second part, the judges toss their mental coins without looking at the defendant, so the King's pronouncement is independent of the truth. Writing the probability of a correct pronouncement in terms of t and then rearranging it around $\frac{1}{2}$ makes the answer fall out in one line.

Solution

Finding p

Let A, B and C be the events that the ass, the beaver and the centaur respectively give a guilty verdict, and let I_A, I_B, I_C be their indicator variables (equal to 1 if the event happens and 0 otherwise). Then $X = I_A + I_B + I_C$, and since $\mathrm{E}(I_A) = \mathrm{P}(A)$ and so on, linearity of expectation gives

$$\mathrm{E}(X) = p + \frac{p}{3} + p^2.$$

(Linearity doesn't need independence; we use independence only later.) Setting this equal to $\frac{4}{3}$ and multiplying by 3:

$$3p^2 + 4p - 4 = 0 \quad \Longrightarrow \quad (3p-2)(p+2) = 0.$$

A probability can't be negative, so $p + 2 \neq 0$, and therefore

$$p = \frac{2}{3}.$$

So the three judges say "guilty" with probabilities

$$\mathrm{P}(A) = \frac{2}{3}, \qquad \mathrm{P}(B) = \frac{2}{9}, \qquad \mathrm{P}(C) = \frac{4}{9},$$

all of which lie in $[0, 1]$, as they must.

The probability that the King says "guilty"

Let K be the event that at least two judges give a guilty verdict, so that K is the event that the King pronounces the defendant guilty. Write a, b, c for the three probabilities above. Exactly two guilty verdicts can happen in three ways and all three in one way; using the independence of the judges,

$$\begin{aligned} \mathrm{P}(K) &= ab(1-c) + ac(1-b) + bc(1-a) + abc \\ &= ab + ac + bc - 2abc. \end{aligned}$$

Here $ab = \frac{4}{27}$, $ac = \frac{8}{27}$, $bc = \frac{8}{81}$ and $abc = \frac{16}{243}$, so

$$\mathrm{P}(K) = \frac{36}{243} + \frac{72}{243} + \frac{24}{243} - \frac{32}{243} = \frac{100}{243}.$$

Call this g for short.

Finding t

Let D be the event that the defendant really is guilty, so $\mathrm{P}(D) = t$. The King is correct when D and K both happen, or when neither does.

? **Ask yourself** Why can we multiply $\mathrm{P}(D)$ by $\mathrm{P}(K)$?

The judges decide "entirely at random": their verdicts do not depend on the evidence, so K is independent of D. Hence

$$\mathrm{P}(\text{correct}) = \mathrm{P}(D)\,\mathrm{P}(K) + \mathrm{P}(D')\,\mathrm{P}(K') = tg + (1-t)(1-g).$$

Now subtract $\frac{1}{2}$ and factorise:

$$\begin{aligned} tg + (1-t)(1-g) - \tfrac{1}{2} &= 2tg - t - g + \tfrac{1}{2} \\ &= (2t-1)\left(g - \tfrac{1}{2}\right). \end{aligned}$$

So the King is correct with probability $\frac{1}{2}$ exactly when $(2t-1)(g-\frac{1}{2}) = 0$. Since $g = \frac{100}{243} \neq \frac{1}{2}$, the second factor is non-zero, and we must have

$$t = \frac{1}{2}.$$

Equivalently, without the factorisation: $\frac{100}{243}t + \frac{143}{243}(1-t) = \frac{1}{2}$ gives $143 - 43t = \frac{243}{2}$, so $43t = \frac{43}{2}$ and $t = \frac{1}{2}$.

Notice that the value of p hardly matters here: whatever the judges' probabilities, provided $\mathrm{P}(K) \neq \frac{1}{2}$, a court that ignores the evidence is right half the time exactly when guilty and innocent defendants are equally common.

Checks

- With $p = \frac{2}{3}$: $\frac{2}{3} + \frac{2}{9} + \frac{4}{9} = \frac{12}{9} = \frac{4}{3}$, as required.

- Enumerating all eight verdict patterns exactly with fractions gives $\mathrm{P}(K) = \frac{100}{243} \approx 0.412$ and $\mathrm{E}(X) = \frac{4}{3}$.
- A simulation of 200 000 trials with $t = \frac{1}{2}$ gave the King correct in a proportion 0.5007 of cases.
- Sanity check of the formula: if $t = 1$ (everyone is guilty) the King is right with probability g, and if $t = 0$ with probability $1 - g$; the probability is linear in t, so it passes through $\frac{1}{2}$ at the midpoint.

Common slips

- Taking "at least two" as $ab + ac + bc$. This counts the all-three case three times; the correct expression subtracts $2abc$ (or add up the four disjoint cases directly).
- Forgetting to reject $p = -2$, or not checking that p^2 and $p/3$ are valid probabilities.
- Treating "the King pronounces correctly" as $\mathrm{P}(K)$ alone, forgetting the innocent defendants who are correctly acquitted.

Beyond STEP

The first part is a small instance of a powerful trick: writing a count as a sum of indicator variables, so that its expectation is just a sum of probabilities, even when the events are dependent. Undergraduate probability uses this constantly, for example to find the expected number of fixed points of a random permutation. The second part is about a classifier whose output is independent of the truth, which is the starting point for thinking about sensitivity, specificity and Bayes' theorem in testing. Blitzstein and Hwang's *Introduction to Probability* makes heavy use of indicator variables and is a good next step.

WORKED SOLUTION

STEP II 2008, Question 13

Probability and Statistics • Drawing Without Replacement

STEP II 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q13

The key idea

Don't track every counter: track only the black ones. Each black counter's fate is decided by two draws, and at each draw the chance that a particular counter is taken is simply (number drawn)/(number in the bag). Splitting into cases according to what happens at the first draw turns both parts into short sums, and both simplify far more than you might expect.

Solution

Throughout, "drawn at random" means every set of k counters in the bag is equally likely to be the one drawn. The fact we use repeatedly is this: if k counters are drawn at random from a bag of m, a *particular* counter is among them with probability k/m, by symmetry (each of the m counters is equally likely to be included, and on average k of them are).

After the first transfer, bag P holds $n-k$ counters and bag Q holds $n+k$. At the second transfer, k of those $n+k$ are drawn and moved to P.

Part (i)

Let B be the black counter, starting in P. Split according to whether B moves at the first transfer.

- B is not drawn from P: probability $\dfrac{n-k}{n}$. Then B stays in P throughout, since the second transfer only takes counters out of Q.
- B is drawn from P: probability $\dfrac{k}{n}$. Now B is one of the $n+k$ counters in Q, and it returns to P exactly when it is among the k drawn back, which has probability $\dfrac{k}{n+k}$.

These cases are disjoint and cover everything, so

$$\begin{aligned}\mathrm{P}(B \text{ ends in } P) &= \frac{n-k}{n} + \frac{k}{n}\cdot\frac{k}{n+k} = \frac{(n-k)(n+k)+k^2}{n(n+k)} \\ &= \frac{n^2}{n(n+k)} = \frac{n}{n+k}.\end{aligned}$$

This is strictly decreasing in k, so over $0 \le k \le n$ it is maximised at $k = 0$ only, where it equals 1. (Not surprising: if nothing is moved, the black counter certainly stays put.)

Part (ii)

Now let B_1 be the black counter starting in P and B_2 the one starting in Q. Let S be the event that they end in the same bag. Again split on whether B_1 moves at the first transfer.

Case 1: B_1 stays in P (probability $\frac{n-k}{n}$). Then B_1 ends in P whatever happens, so S happens exactly when B_2 is among the k drawn from the $n+k$ counters in Q:

$$\mathrm{P}(S \mid \text{case 1}) = \frac{k}{n+k}.$$

Case 2: B_1 moves to Q (probability $\frac{k}{n}$). Now both black counters are among the $n+k$ in Q, and S happens when the second draw takes *both* of them or *neither*.

> **Ask yourself** What is the probability that a random k-set from $n+k$ counters contains both, or neither, of two given counters?

Count subsets. There are $\binom{n+k}{k}$ equally likely choices. Those containing both blacks choose the other $k-2$ from the remaining $n+k-2$; those containing neither choose all k from the $n+k-2$ whites. Dividing,

$$\begin{aligned}\mathrm{P}(\text{both}) &= \binom{n+k-2}{k-2}\Big/\binom{n+k}{k} = \frac{k(k-1)}{(n+k)(n+k-1)}, \\ \mathrm{P}(\text{neither}) &= \binom{n+k-2}{k}\Big/\binom{n+k}{k} = \frac{n(n-1)}{(n+k)(n+k-1)}.\end{aligned}$$

(These formulas also hold for $k = 0, 1$, where $\mathrm{P}(\text{both}) = 0$.) So

$$\mathrm{P}(S \mid \text{case 2}) = \frac{k(k-1)+n(n-1)}{(n+k)(n+k-1)}.$$

Putting the cases together. By the law of total probability,

$$\begin{aligned}\mathrm{P}(S) &= \frac{n-k}{n}\cdot\frac{k}{n+k} + \frac{k}{n}\cdot\frac{k(k-1)+n(n-1)}{(n+k)(n+k-1)} \\ &= \frac{k\big[(n-k)(n+k-1)+k^2-k+n^2-n\big]}{n(n+k)(n+k-1)}.\end{aligned}$$

Inside the square bracket, $(n-k)(n+k-1) = n^2 - k^2 - n + k$, so the bracket is $2n^2 - 2n = 2n(n-1)$. Hence

$$\boxed{\mathrm{P}(S) = \frac{2k(n-1)}{(n+k)(n+k-1)}}$$

Maximising. At $k = 0$ the probability is 0, so take $1 \le k \le n$ and minimise the reciprocal. Up to the positive constant $2(n-1)$,

$$\frac{(n+k)(n+k-1)}{k} = k + (2n-1) + \frac{n(n-1)}{k},$$

so we need to minimise $g(k) = k + \dfrac{n(n-1)}{k}$ over integers. Compare neighbours:

$$g(k+1) - g(k) = 1 - \frac{n(n-1)}{k(k+1)}.$$

This is negative when $k(k+1) < n(n-1)$, that is for $k < n-1$, and it is exactly zero when $k = n-1$, because then $k(k+1) = (n-1)n$. So g strictly decreases from $k = 1$ to $k = n-1$, and $g(n) = g(n-1)$. Therefore the probability is maximised at

$$k = n-1 \quad \text{and} \quad k = n,$$

both of which lie in the allowed range since $n \ge 2$. At $k = n-1$ the maximum value is

$$\frac{2(n-1)^2}{(2n-1)(2n-2)} = \frac{n-1}{2n-1},$$

and $k = n$ gives the same, $\dfrac{2n(n-1)}{2n(2n-1)} = \dfrac{n-1}{2n-1}$. (The continuous minimiser of g is $\sqrt{n(n-1)}$, which lies strictly between $n-1$ and n; the tie between the two integers is exact, not approximate.)

Checks

- Part (i) at $k = n$: $\frac{1}{2}$. Then all $2n$ counters pass through Q and n of them return, so the black one is equally likely to end in either bag. Good.
- Part (ii) at $k = 0$ gives 0, correct since the blacks never meet. At $n = 2$, $k = 1$ it gives $\frac{2}{3 \cdot 2} = \frac{1}{3}$, which you can confirm by hand.
- We enumerated every possible pair of draws exactly (with fractions) for $2 \le n \le 7$ and every k; both formulas matched in every case, and the maximisers in (ii) were always $\{n-1, n\}$, e.g. maximum $\frac{3}{7} \approx 0.4286$ for $n = 4$.

Common slips

- Using k/n instead of $k/(n+k)$ at the second draw: bag Q has grown to $n+k$ counters.

- In part (ii), forgetting the "neither drawn" possibility in case 2: if both blacks stay in Q they are still together.
- Maximising by calculus and rounding $\sqrt{n(n-1)}$ to a single integer. The discrete comparison shows two equal maximisers.

Beyond STEP

The tool behind both parts is the *hypergeometric* distribution: the number of marked items in a random sample drawn without replacement. Conditioning on the outcome of the first stage and then using it for the second is the law of total probability, and the same two-stage reasoning underlies models of mixing and diffusion between containers, such as the Ehrenfest urn. Blitzstein and Hwang, *Introduction to Probability*, treats the hypergeometric distribution and conditioning in exactly this style.

WORKED SOLUTION

STEP III 2008, Question 1

Pure • Non-linear Simultaneous Equations

STEP III 2008, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q1

The key idea

Write $s_k = ax^k + by^k$, so the equations give s_0, s_1, s_2, s_3. Since x and y are the roots of $t^2 - pt + q = 0$, where $p = x + y$ and $q = xy$, the power sums obey a recurrence: each s_k is p times the previous one minus q times the one before that. That is exactly what the hint's "multiply by $x + y$" produces. The four non-linear equations become two *linear* equations in p and q. After that, a quadratic gives x and y, and the first two equations give a and b.

Solution

Let $s_k = ax^k + by^k$ for $k = 0, 1, 2, 3$, so that $s_0 = 1$, $s_1 = \frac{1}{3}$, $s_2 = \frac{1}{5}$, $s_3 = \frac{1}{7}$. Put $p = x + y$ and $q = xy$.

A recurrence for the power sums

For any a, b, x, y and any $k \geq 1$,

$$(x + y)(ax^k + by^k) = ax^{k+1} + by^{k+1} + xy\,(ax^{k-1} + by^{k-1}),$$

because the cross terms are $axyx^{k-1}$ and $bxy\,y^{k-1}$. In our notation,

$$s_{k+1} = p\,s_k - q\,s_{k-1}.$$

This is an identity, so it holds for *every* solution of the system.

Finding $x + y$ and xy

Taking $k = 1$ and $k = 2$ and substituting the known values:

$$\tfrac{1}{5} = \tfrac{1}{3}p - q,$$
$$\tfrac{1}{7} = \tfrac{1}{5}p - \tfrac{1}{3}q.$$

From the first, $q = \frac{1}{3}p - \frac{1}{5}$. Substituting into the second,

$$\tfrac{1}{7} = \tfrac{1}{5}p - \tfrac{1}{9}p + \tfrac{1}{15} \quad\Longrightarrow\quad \tfrac{4}{45}p = \tfrac{1}{7} - \tfrac{1}{15} = \tfrac{8}{105} \quad\Longrightarrow\quad p = \tfrac{6}{7},$$

and then $q = \frac{2}{7} - \frac{1}{5} = \frac{3}{35}$. The linear system has non-zero determinant, so these values are forced: *every* solution has $x + y = \frac{6}{7}$ and $xy = \frac{3}{35}$.

Finding x and y

So x and y are the two roots of

$$t^2 - \tfrac{6}{7}t + \tfrac{3}{35} = 0, \qquad \text{that is,} \qquad 35t^2 - 30t + 3 = 0,$$

which gives

$$t = \frac{30 \pm \sqrt{900 - 420}}{70} = \frac{30 \pm 4\sqrt{30}}{70} = \frac{15 \pm 2\sqrt{30}}{35}.$$

The discriminant is positive, so $x \neq y$, and $\{x, y\}$ is this pair in one order or the other.

Ask yourself Can we be sure a and b are determined once we know x and y?

Yes. With x and y known, the first two equations, $a + b = 1$ and $ax + by = \frac{1}{3}$, are linear in a and b with determinant $y - x$. That is non-zero because $x \neq y$, so there is exactly one pair (a, b).

Finding a and b

Take $x = \dfrac{15 + 2\sqrt{30}}{35}$ and $y = \dfrac{15 - 2\sqrt{30}}{35}$, so $x - y = \dfrac{4\sqrt{30}}{35}$. Eliminating b from $a + b = 1$ and $ax + by = \frac{1}{3}$ gives $a(x - y) = \frac{1}{3} - y$. Now

$$\tfrac{1}{3} - y = \frac{35 - 3(15 - 2\sqrt{30})}{105} = \frac{6\sqrt{30} - 10}{105},$$

so

$$a = \frac{6\sqrt{30} - 10}{105} \cdot \frac{35}{4\sqrt{30}} = \frac{6\sqrt{30} - 10}{12\sqrt{30}} = \frac{1}{2} - \frac{5}{6\sqrt{30}} = \frac{1}{2} - \frac{\sqrt{30}}{36},$$

and $b = 1 - a = \dfrac{1}{2} + \dfrac{\sqrt{30}}{36}$.

These really are solutions

We have only shown that any solution must look like this, so we still have to check that it works. By construction $s_0 = 1$ and $s_1 = \frac{1}{3}$. Since p and q satisfy the two linear equations above, the recurrence gives $s_2 = ps_1 - qs_0 = \frac{1}{5}$ and then $s_3 = ps_2 - qs_1 = \frac{1}{7}$. So all four equations hold.

Swapping the roles of x and y swaps a and b. So there are exactly two solutions:

$$(a, b, x, y) = \left(\frac{18 - \sqrt{30}}{36}, \frac{18 + \sqrt{30}}{36}, \frac{15 + 2\sqrt{30}}{35}, \frac{15 - 2\sqrt{30}}{35} \right)$$

and the same with $a \leftrightarrow b$ and $x \leftrightarrow y$ swapped together. In both, the larger of x, y is paired with the smaller coefficient.

Checks

Numerically, $x \approx 0.741556$, $y \approx 0.115587$, $a \approx 0.347855$ and $b \approx 0.652145$. Then $ax + by \approx 0.333333$, $ax^2 + by^2 \approx 0.200000$ and $ax^3 + by^3 \approx 0.142857$, which agree with $\frac{1}{3}$, $\frac{1}{5}$ and $\frac{1}{7}$ to machine precision. A quick sanity check: x and y both lie in $(0, 1)$, and a and b are both positive. That fits the right-hand sides being averages of t^{2k} over $[0, 1]$ (see below).

Common slips

- Solving for x and y but forgetting that the pairing matters: a goes with x, so you can't swap x and y without also swapping a and b.
- Stopping after finding necessary conditions. The question says "find all", so you need both that every solution has this form and that these values do satisfy all four equations.

Beyond STEP

The numbers $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}$ are $\frac{1}{2}\int_{-1}^{1} t^{2k}\,\mathrm{d}t$. So the system asks for a rule $\frac{1}{2}\int_{-1}^{1} f \approx \frac{a}{2}\big(f(\sqrt{x}) + f(-\sqrt{x})\big) + \frac{b}{2}\big(f(\sqrt{y}) + f(-\sqrt{y})\big)$ that is exact for polynomials up to degree 7. This is four-point *Gauss–Legendre quadrature*: its nodes are $\pm\sqrt{x}$ and $\pm\sqrt{y}$, and $a \approx 0.3479$ and $b \approx 0.6521$ are its famous weights. Finding a recurrence from power sums, as we did here, is also the idea behind Prony's method in signal processing. For more, try Süli and Mayers, *An Introduction to Numerical Analysis*.

WORKED SOLUTION

STEP III 2008, Question 2

Pure • Sums of Powers

STEP III 2008, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q2

The key idea

The sum of $(r+1)^k - r^k$ telescopes, so it collapses to a single term. But we can also expand each bracket by the binomial theorem, and then the same sum becomes a combination of the lower power sums $S_j(n)$. Equating the two gives a recurrence that builds each S_{k-1} from the ones before it, and that recurrence is also exactly what we need to see the polynomial structure in part (ii).

Solution

Part (i)

First, the telescoping. Writing the sum out,

$$\sum_{r=0}^{n}\left[(r+1)^k - r^k\right] = (1^k - 0^k) + (2^k - 1^k) + \cdots + \left((n+1)^k - n^k\right) = (n+1)^k,$$

since every intermediate term cancels and $0^k = 0$ (as $k \geq 1$).

Second, the binomial theorem. For each r,

$$(r+1)^k - r^k = \sum_{j=0}^{k-1}\binom{k}{j}r^j = 1 + \binom{k}{1}r + \binom{k}{2}r^2 + \cdots + \binom{k}{k-1}r^{k-1},$$

because the r^k terms cancel. Summing over $r = 0, 1, \ldots, n$, the constant 1 contributes $n+1$ (there are $n+1$ values of r), and each r^j with $j \geq 1$ contributes $S_j(n)$. So

$$(n+1)^k = (n+1) + \binom{k}{1}S_1(n) + \binom{k}{2}S_2(n) + \cdots + \binom{k}{k-2}S_{k-2}(n) + \binom{k}{k-1}S_{k-1}(n).$$

The last coefficient is $\binom{k}{k-1} = k$. For the others we use the symmetry $\binom{k}{j} = \binom{k}{k-j}$, so that the coefficient of $S_{k-j}(n)$ is $\binom{k}{k-j} = \binom{k}{j}$; in particular the coefficient of $S_1(n)$ is

$\binom{k}{1} = \binom{k}{k-1}$. Keeping $kS_{k-1}(n)$ on one side and moving everything else across gives

$$kS_{k-1}(n) = (n+1)^k - (n+1) - \binom{k}{2}S_{k-2}(n) - \binom{k}{3}S_{k-3}(n) - \cdots - \binom{k}{k-1}S_1(n),$$

which is $(*)$, as required. (We need $k \geq 2$ here so that S_{k-1} is one of our sums.)

Ask yourself Why did we treat the $j = 0$ term separately instead of writing it as $S_0(n)$?

S_k is only defined for positive k, and "0^0" at $r = 0$ is ambiguous. Pulling the constant out and counting it directly as $n+1$ avoids both problems, and it explains the lone $-(n+1)$ in $(*)$.

$S_3(n)$. Take $k = 4$ in $(*)$, so that $\binom{4}{2} = 6$ and $\binom{4}{3} = 4$:

$$\begin{aligned} 4S_3(n) &= (n+1)^4 - (n+1) - 6S_2(n) - 4S_1(n) \\ &= (n+1)^4 - (n+1) - n(n+1)(2n+1) - 2n(n+1). \end{aligned}$$

Every term has a factor $n+1$, so

$$\begin{aligned} 4S_3(n) &= (n+1)\left[(n+1)^3 - 1 - n(2n+1) - 2n\right] \\ &= (n+1)\left[n^3 + 3n^2 + 3n - 2n^2 - n - 2n\right] \\ &= (n+1)(n^3 + n^2) = n^2(n+1)^2. \end{aligned}$$

Hence

$$S_3(n) = \tfrac{1}{4}n^2(n+1)^2 = \big(S_1(n)\big)^2.$$

$S_4(n)$. Take $k = 5$, so that $\binom{5}{2} = \binom{5}{3} = 10$ and $\binom{5}{4} = 5$:

$$5S_4(n) = (n+1)^5 - (n+1) - 10S_3(n) - 10S_2(n) - 5S_1(n).$$

Again we take out the factor $n+1$, using our expressions for S_3, S_2 and S_1:

$$\begin{aligned} 5S_4(n) &= (n+1)\left[(n+1)^4 - 1 - \tfrac{5}{2}n^2(n+1) - \tfrac{5}{3}n(2n+1) - \tfrac{5}{2}n\right] \\ &= (n+1)\Big[n^4 + 4n^3 + 6n^2 + 4n - \tfrac{5}{2}n^3 - \tfrac{5}{2}n^2 \\ &\qquad - \tfrac{10}{3}n^2 - \tfrac{5}{3}n - \tfrac{5}{2}n\Big] \\ &= (n+1)\left[n^4 + \tfrac{3}{2}n^3 + \tfrac{1}{6}n^2 - \tfrac{1}{6}n\right] \\ &= \tfrac{1}{6}n(n+1)\left(6n^3 + 9n^2 + n - 1\right). \end{aligned}$$

Since S_4 ought to contain the factor $2n+1$ like S_2 does, we try $n = -\frac{1}{2}$ in the cubic: $-\frac{3}{4} + \frac{9}{4} - \frac{1}{2} - 1 = 0$. So $2n+1$ is a factor, and $6n^3 + 9n^2 + n - 1 = (2n+1)(3n^2 + 3n - 1)$. Therefore

$$S_4(n) = \tfrac{1}{30}\,n(n+1)(2n+1)(3n^2 + 3n - 1).$$

Part (ii)

We use strong induction on k. The claim is true for $k = 1$, since $S_1(n) = \frac{1}{2}n^2 + \frac{1}{2}n$ has degree 2 (and also for $k = 2$).

Suppose that, for some $k \geq 2$, each of $S_1(n), \ldots, S_{k-1}(n)$ is a polynomial in n with S_j of degree $j+1$. Apply $(*)$ with $k+1$ in place of k:

$$(k+1)\,S_k(n) = (n+1)^{k+1} - (n+1) - \binom{k+1}{2} S_{k-1}(n) - \cdots - \binom{k+1}{k} S_1(n).$$

On the right:

- $(n+1)^{k+1}$ is a polynomial of degree $k+1$ with leading coefficient 1;
- $n+1$ has degree 1;
- each $S_j(n)$ with $1 \leq j \leq k-1$ is, by hypothesis, a polynomial of degree $j+1 \leq k$.

So the right-hand side is a polynomial whose n^{k+1} term comes only from $(n+1)^{k+1}$ and cannot be cancelled. Dividing by $k+1 \neq 0$, $S_k(n)$ is a polynomial of degree exactly $k+1$ (with leading coefficient $\frac{1}{k+1}$). This completes the induction.

> **Ask yourself** What does "the constant term" and "the sum of the coefficients" of a polynomial $P(n)$ correspond to?

They are $P(0)$ and $P(1)$. Write $S_k(n) = P(n) = a_{k+1}n^{k+1} + \cdots + a_1 n + a_0$; this agrees with the sum $\sum_{r=0}^{n} r^k$ for every whole number $n \geq 0$, so we may evaluate at $n = 0$ and $n = 1$.

- Constant term: $a_0 = P(0) = S_k(0) = 0^k = 0$, since $k \geq 1$.
- Sum of coefficients: $a_{k+1} + \cdots + a_1 + a_0 = P(1) = S_k(1) = 0^k + 1^k = 1$.

Both results are shown.

Checks

- Small values: $S_3(3) = 1 + 8 + 27 = 36 = \frac{1}{4} \cdot 9 \cdot 16$, and $S_4(3) = 1 + 16 + 81 = 98 = \frac{1}{30} \cdot 3 \cdot 4 \cdot 7 \cdot 35$.
- Part (ii) on our answers: $S_4(n) = \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n$ has constant term 0 and coefficients summing to $\frac{6+15+10-1}{30} = 1$; its leading coefficient is $\frac{1}{5} = \frac{1}{k+1}$, as the induction predicts.
- We checked $(*)$ exactly for $2 \leq k \leq 9$ and $0 \leq n \leq 30$, and both closed forms for $0 \leq n < 200$, by direct summation.

Common slips

- Forgetting the $r = 0$ term of $\sum 1$: it contributes $n + 1$, not n.
- Expanding everything instead of taking out the common factor $n + 1$ first, which makes the algebra for S_4 much heavier.
- In part (ii), saying only that the degree is "at most $k + 1$". The question wants exactly $k + 1$, so note that the n^{k+1} term cannot cancel.

Beyond STEP

The coefficients of these polynomials are governed by the Bernoulli numbers, and Faulhaber's formula writes $S_k(n)$ as a combination of powers of n whose coefficients are binomial coefficients times Bernoulli numbers. The fact that $S_k(n)$ is a polynomial of degree $k + 1$ with leading term $\frac{n^{k+1}}{k+1}$ is the discrete analogue of $\int_0^x t^k \, \mathrm{d}t = \frac{x^{k+1}}{k+1}$, an idea made precise in the calculus of finite differences. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2008, Question 3

Pure • Ellipse Tangents and Foci

STEP III 2008, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

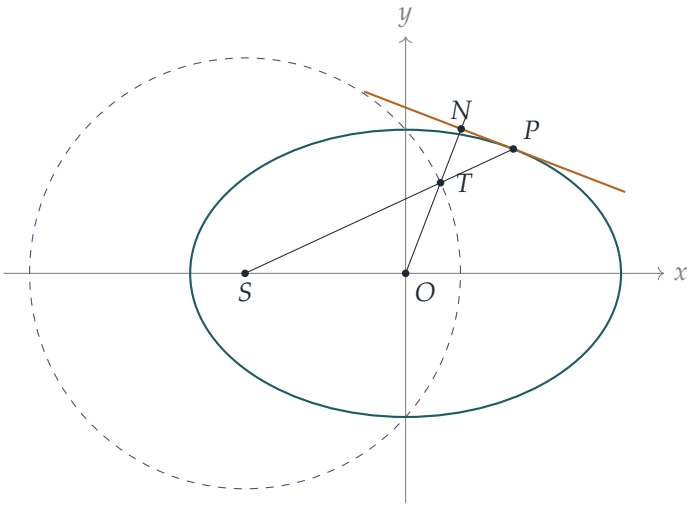
step.maths.org/questions/08-s3-q3

The key idea

Everything here is about two straight lines, so the job is to write down their equations cleanly and intersect them. The line ON is perpendicular to the tangent, so its direction is the tangent's normal vector, which we can read straight off the tangent's equation. Parametrising SP from S makes the intersection a one-line calculation, and the parameter we find turns out to be exactly what is needed for the circle: it tells us what fraction of the way along SP the point T sits.

Solution

Throughout, $0 < e < 1$: we have $e^2 = 1 - b^2/a^2$, which lies strictly between 0 and 1 because $a > b > 0$. We also take $\sin\theta \neq 0$. If $\sin\theta = 0$ then $P = (\pm a, 0)$, the tangent is $x = \pm a$, $N = P$, and both ON and SP are the x-axis, so T isn't defined.



(The picture has $a = 3$, $b = 2$, $\theta = 60^\circ$. The ellipse is teal, the tangent at P amber, and the dashed circle has centre S and radius a.)

The y-coordinate of T

Differentiating $x^2/a^2 + y^2/b^2 = 1$ implicitly, or using the standard result, the tangent at (x_1, y_1) is $xx_1/a^2 + yy_1/b^2 = 1$. At P this is

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1.$$

A line $px + qy = 1$ has normal vector (p, q), so the tangent has normal $\left(\frac{\cos\theta}{a}, \frac{\sin\theta}{b}\right)$, or, multiplying by ab, $(b\cos\theta,\, a\sin\theta)$. The line ON passes through O in this direction (it is perpendicular to the tangent), so its points are $\mu(b\cos\theta, a\sin\theta)$, and its equation is

$$a\sin\theta\, x = b\cos\theta\, y. \tag{1}$$

Now parametrise SP from S: its points are

$$\big(-ea + \lambda a(\cos\theta + e),\ \lambda b\sin\theta\big),$$

with $\lambda = 0$ at S and $\lambda = 1$ at P. Substituting into (1) and cancelling the factor $\sin\theta \neq 0$,

$$\begin{aligned} a\big(-ea + \lambda a(\cos\theta + e)\big) &= \lambda b^2\cos\theta \\ \lambda\big(a^2\cos\theta + a^2 e - b^2\cos\theta\big) &= ea^2. \end{aligned}$$

Here is where the focus condition earns its keep: $a^2 - b^2 = a^2e^2$, so the bracket is $a^2e^2\cos\theta + a^2e = a^2e(1 + e\cos\theta)$. Dividing by $a^2e \neq 0$,

$$\lambda = \frac{1}{1 + e\cos\theta},$$

which is legitimate since $1 + e\cos\theta \geq 1 - e > 0$. The y-coordinate of T is therefore

$$y_T = \lambda b\sin\theta = \frac{b\sin\theta}{1 + e\cos\theta},$$

as required. (For later reference, (1) then gives $x_T = \dfrac{b\cos\theta}{a\sin\theta}\, y_T = \dfrac{a(1 - e^2)\cos\theta}{1 + e\cos\theta}$.)

T lies on the circle centre S, radius a

Ask yourself We know what fraction of SP the segment ST is. What else would we need?

Since $T = S + \lambda(P - S)$ with $\lambda > 0$, we have $ST = \lambda\, SP$. So we only need the length SP, the focal distance of P:

$$\begin{aligned} SP^2 &= (a\cos\theta + ea)^2 + b^2\sin^2\theta \\ &= a^2\big[\cos^2\theta + 2e\cos\theta + e^2 + (1 - e^2)\sin^2\theta\big] \\ &= a^2\big[(\cos^2\theta + \sin^2\theta) + 2e\cos\theta + e^2(1 - \sin^2\theta)\big] \\ &= a^2\big[1 + 2e\cos\theta + e^2\cos^2\theta\big] = a^2(1 + e\cos\theta)^2, \end{aligned}$$

using $b^2 = a^2(1-e^2)$ in the second line. As $a > 0$ and $1 + e\cos\theta > 0$, this gives $SP = a(1 + e\cos\theta)$. Hence

$$ST = \lambda\, SP = \frac{a(1+e\cos\theta)}{1+e\cos\theta} = a,$$

so T lies on the circle with centre S and radius a.

(If you'd rather work directly from coordinates: $x_T + ea = \dfrac{a(\cos\theta + e)}{1 + e\cos\theta}$, and $(x_T + ea)^2 + y_T^2$ reduces to a^2 by exactly the same algebra as SP^2 above.)

Checks

- At $\theta = \frac{\pi}{2}$, $P = (0, b)$, the tangent is $y = b$, so ON is the y-axis and $T = (0, y_T)$ with $y_T = b$ from the formula. Then $ST^2 = e^2a^2 + b^2 = a^2$, as it should be.
- With $a = 3$, $b = 2$, $\theta = 60^\circ$ (the picture), intersecting the two lines numerically gives $T \approx (0.4857, 1.2618)$; the formula gives $y_T = \sqrt{3}/(1 + \sqrt{5}/6) \approx 1.2618$, and $ST \approx 3.0000$.
- In the circle limit $e \to 0$, $S \to O$ and $T \to P$, which does lie on the circle of radius a about O.

Common slips

- Taking ON parallel to the tangent rather than perpendicular to it: the direction of ON is the normal $(b\cos\theta, a\sin\theta)$, not $(-a\sin\theta, b\cos\theta)$.
- Forgetting to replace $a^2 - b^2$ by a^2e^2, which leaves an expression for λ that doesn't simplify.
- Dividing by $\sin\theta$ without comment; the case $\sin\theta = 0$ genuinely fails.

Beyond STEP

The result $SP = a(1 + e\cos\theta) = a + ex_P$ is the focus–directrix property in disguise: distance to the focus equals e times distance to the directrix $x = -a/e$. Writing a conic in polar coordinates about a focus, $r = \ell/(1 + e\cos\phi)$, is how Kepler orbits are described in first-year mechanics, and the same denominator $1 + e\cos\theta$ appeared here. For the geometry of conics, and many more STEP-style problems, see Siklos, *Advanced Problems in Mathematics*; for orbits, Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2008, Question 4

Pure • Inverse Hyperbolic Inequalities

STEP III 2008, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q4

The key idea

Put $x = \cosh y$. The half-angle formula turns $\tanh(y/2)$ into an algebraic function of x, so a trivial inequality about tanh becomes a non-trivial one about arcosh. After that the question runs a little machine: integrate an inequality from 1 to x, rearrange, and get a sharper inequality. Running the machine twice gives (ii) and then (iii).

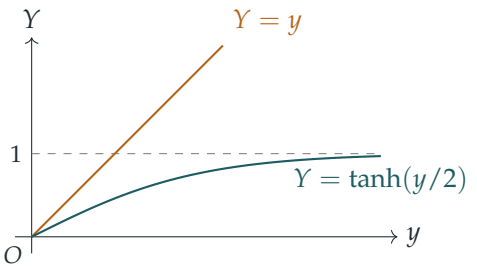
Solution

Throughout, write $A(x) = \operatorname{arcosh} x$ and $S(x) = \sqrt{x^2 - 1}$ for $x \geq 1$. Two derivatives we shall need are

$$\frac{\mathrm{d}}{\mathrm{d}x} \operatorname{arcosh} x = \frac{1}{\sqrt{x^2 - 1}}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \sqrt{x^2 - 1} = \frac{x}{\sqrt{x^2 - 1}} \qquad (x > 1).$$

Part (i)

The curve $Y = \tanh(y/2)$ passes through the origin with gradient $\frac{1}{2}$, is concave for $y > 0$ and never reaches 1, whereas the line $Y = y$ has gradient 1:



The line lies above the curve for every $y > 0$. To make this precise, let $g(t) = t - \tanh t$. Then $g(0) = 0$ and

$$g'(t) = 1 - \operatorname{sech}^2 t = \tanh^2 t > 0 \quad \text{for } t > 0,$$

so g is strictly increasing and $\tanh t < t$ for $t > 0$. With $t = y/2$,

$$\tanh(y/2) < \tfrac{1}{2}y < y \qquad (y > 0).$$

Now let $x > 1$ and put $y = \operatorname{arcosh} x$, so $y > 0$, $\cosh y = x$ and $\sinh y = +\sqrt{x^2 - 1}$ (positive because $y > 0$). Using $\sinh y = 2\sinh \frac{y}{2}\cosh \frac{y}{2}$ and $1 + \cosh y = 2\cosh^2 \frac{y}{2}$,

$$\tanh \frac{y}{2} = \frac{\sinh y}{1 + \cosh y} = \frac{\sqrt{x^2 - 1}}{x + 1} = \frac{x^2 - 1}{(x+1)\sqrt{x^2 - 1}} = \frac{x - 1}{\sqrt{x^2 - 1}}.$$

So $y > \tanh(y/2)$ becomes

$$\operatorname{arcosh} x > \frac{x - 1}{\sqrt{x^2 - 1}} \qquad (x > 1),$$

which is $(*)$.

Part (ii)

Fix $x > 1$ and integrate $(*)$, with t as the variable, over $[1, x]$.

Ask yourself The right-hand side of $(*)$ has $\sqrt{t^2 - 1}$ in the denominator. Is it safe to integrate from $t = 1$, and does integrating keep the inequality *strict*?

Yes on both counts. For $t > 1$, $\dfrac{t-1}{\sqrt{t^2-1}} = \sqrt{\dfrac{t-1}{t+1}}$, which tends to 0 as $t \to 1$, so both sides are continuous on $[1, x]$. Their difference is continuous, and strictly positive on $(1, x]$, so its integral over $[1, x]$ is strictly positive.

The left-hand side, integrating by parts with $u = \operatorname{arcosh} t$:

$$\int_1^x \operatorname{arcosh} t \, \mathrm{d}t = \Big[t \operatorname{arcosh} t\Big]_1^x - \int_1^x \frac{t}{\sqrt{t^2 - 1}} \, \mathrm{d}t = x\,A(x) - S(x),$$

since $\operatorname{arcosh} 1 = 0$ and $\sqrt{1^2 - 1} = 0$.

The right-hand side splits into two standard pieces:

$$\int_1^x \frac{t-1}{\sqrt{t^2-1}} \, \mathrm{d}t = \int_1^x \frac{t}{\sqrt{t^2-1}} \, \mathrm{d}t - \int_1^x \frac{1}{\sqrt{t^2-1}} \, \mathrm{d}t = S(x) - A(x).$$

Therefore

$$x\,A(x) - S(x) > S(x) - A(x) \quad \Longrightarrow \quad (x+1)A(x) > 2S(x).$$

Dividing by $x + 1 > 0$ and simplifying as in part (i),

$$\operatorname{arcosh} x > \frac{2\sqrt{x^2 - 1}}{x + 1} = 2\,\frac{x - 1}{\sqrt{x^2 - 1}} \qquad (x > 1),$$

as required. This is exactly twice the bound in $(*)$.

Part (iii)

Part (ii) says $\operatorname{arcosh} t > 2 \cdot \dfrac{t-1}{\sqrt{t^2-1}}$ for $t > 1$: the same right-hand side as $(*)$, doubled. So we run the same machine again. Integrating over $[1, x]$, with the same justification as before, and using the two integrals already found,

$$x\,A(x) - S(x) > 2\big(S(x) - A(x)\big) \quad \Longrightarrow \quad (x+2)A(x) > 3S(x).$$

Since $x + 2 > 0$,

$$\operatorname{arcosh} x > 3\,\frac{\sqrt{x^2-1}}{x+2} \qquad (x > 1),$$

which is the required result.

Ask yourself Why stop here? Can we integrate once more?

The machine needs the right-hand side to be a multiple of $\dfrac{t-1}{\sqrt{t^2-1}}$, whose integral we know. The new bound $3\sqrt{t^2-1}/(t+2)$ isn't of that form, so the trick runs out.

As an independent check (an "otherwise" route), let $f(x) = A(x) - 3S(x)/(x+2)$. Then $f(1) = 0$ and, by the quotient rule,

$$f'(x) = \frac{(x+2)^2 - 3x(x+2) + 3(x^2-1)}{(x+2)^2\sqrt{x^2-1}} = \frac{(x-1)^2}{(x+2)^2\sqrt{x^2-1}} > 0 \quad (x > 1),$$

so f is strictly increasing and $f(x) > 0$ for $x > 1$, as before.

Checks

At $x = 2$: $\operatorname{arcosh} 2 = \ln(2+\sqrt{3}) \approx 1.3170$, while the three lower bounds are $1/\sqrt{3} \approx 0.577$, $2/\sqrt{3} \approx 1.155$ and $3\sqrt{3}/4 \approx 1.299$. Each bound beats the last and all lie below 1.3170. Near $x = 1$ the bounds become very good: at $x = 1.001$, $\operatorname{arcosh} x \approx 0.04471763$, the bound in (ii) is short by about 7×10^{-6} and the bound in (iii) by only about 10^{-9}, because each matches $\operatorname{arcosh} x$ to higher order in $x - 1$. A numerical check of the two integrals (by midpoint quadrature at $x = 1.5$, 2, 5) and of all three inequalities at 10 000 random $x > 1$ found no discrepancy.

Common slips

- Taking $\sinh(\operatorname{arcosh} x) = \pm\sqrt{x^2-1}$ without fixing the sign. It is the positive root because $\operatorname{arcosh} x > 0$.
- Integrating from 0 instead of 1: arcosh is only defined for $x \geq 1$, and the lower limit 1 is what makes the boundary terms vanish.
- Forgetting to say why the integrated inequality is still strict.

Beyond STEP

The bounds get better because each integration makes the error vanish to one higher order at $x = 1$; this is the start of a sequence of increasingly accurate rational-type approximations, close cousins of Padé approximants. The principle used throughout (if $f(a) = 0$ and $f' > 0$, or the integrated form of that, then $f > 0$) is the mean value theorem in disguise, and it underlies a great deal of analysis. Abbott's *Understanding Analysis* treats the mean value theorem and integration rigorously.

WORKED SOLUTION

STEP III 2008, Question 5

Pure • Recurrence Relations

STEP III 2008, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q5

The key idea

The quantity $\mathrm{T}_n^2 - \mathrm{T}_{n-1}\mathrm{T}_{n+1}$ is an *invariant* of the recurrence: when we step n up by one, the difference collapses because the recurrence lets us replace $\mathrm{T}_{n+1} + \mathrm{T}_{n-1}$ by $2x\mathrm{T}_n$. When the invariant is zero, consecutive ratios are all equal, so the sequence is geometric. Substituting a geometric sequence into the recurrence then pins down the common ratio as a root of a quadratic.

Solution

Throughout, x is fixed and we write T_n for $\mathrm{T}_n(x)$ and f for $\mathrm{f}(x)$. The recurrence $(*)$ says

$$\mathrm{T}_{n+1} + \mathrm{T}_{n-1} = 2x\,\mathrm{T}_n \qquad (n \geq 1).$$

The invariant

Let $D_n = \mathrm{T}_n^2 - \mathrm{T}_{n-1}\mathrm{T}_{n+1}$ for $n \geq 1$. We prove $D_n = f$ for all $n \geq 1$ by induction.

Base case. $D_1 = \mathrm{T}_1^2 - \mathrm{T}_0\mathrm{T}_2$, which is f by definition.

Inductive step. Suppose $D_k = f$ for some $k \geq 1$. Then

$$\begin{aligned} D_{k+1} - D_k &= \mathrm{T}_{k+1}^2 - \mathrm{T}_k\mathrm{T}_{k+2} - \mathrm{T}_k^2 + \mathrm{T}_{k-1}\mathrm{T}_{k+1} \\ &= \mathrm{T}_{k+1}\big(\mathrm{T}_{k+1} + \mathrm{T}_{k-1}\big) - \mathrm{T}_k\big(\mathrm{T}_{k+2} + \mathrm{T}_k\big) \\ &= \mathrm{T}_{k+1} \cdot 2x\mathrm{T}_k - \mathrm{T}_k \cdot 2x\mathrm{T}_{k+1} \\ &= 0, \end{aligned}$$

where in the third line we used $(*)$ with $n = k$ for the first bracket and with $n = k+1$ for the second (both allowed, as $k \geq 1$). Hence $D_{k+1} = D_k = f$, and by induction

$$\mathrm{T}_n^2 - \mathrm{T}_{n-1}\mathrm{T}_{n+1} = f \quad \text{for all } n \geq 1,$$

as required.

The case $f \equiv 0$

First let us see what $f = 0$ says about $r = \mathrm{T}_1/\mathrm{T}_0$. By $(*)$ with $n = 1$, $\mathrm{T}_2 = 2x\mathrm{T}_1 - \mathrm{T}_0$, so

$$f = \mathrm{T}_1^2 - \mathrm{T}_0(2x\mathrm{T}_1 - \mathrm{T}_0) = \mathrm{T}_0^2\big(r^2 - 2xr + 1\big),$$

on writing $\mathrm{T}_1 = r\mathrm{T}_0$. So, as $\mathrm{T}_0 \neq 0$,

$$f = 0 \iff r^2 = 2xr - 1. \qquad (\dagger)$$

Ask yourself The invariant says $\mathrm{T}_n^2 = \mathrm{T}_{n-1}\mathrm{T}_{n+1}$, so the ratios are equal. Why not just divide?

Dividing by T_{n-1} needs it to be nonzero, which takes extra argument. It is cleaner to guess the answer from the "equal ratios" idea and then prove it with the recurrence itself, using $(\dagger)$.

Claim. $\mathrm{T}_n = \mathrm{T}_0\, r^n$ for all $n \geq 0$.

Proof. We use strong induction (two consecutive cases at a time). The claim holds for $n = 0$ trivially and for $n = 1$ because $\mathrm{T}_1 = r\mathrm{T}_0$ by the definition of r. Suppose it holds for $n = k - 1$ and $n = k$, where $k \geq 1$. Then by $(*)$,

$$\begin{aligned}\mathrm{T}_{k+1} &= 2x\mathrm{T}_k - \mathrm{T}_{k-1} = 2x\mathrm{T}_0 r^k - \mathrm{T}_0 r^{k-1} \\ &= \mathrm{T}_0 r^{k-1}\big(2xr - 1\big) = \mathrm{T}_0 r^{k-1} \cdot r^2 = \mathrm{T}_0 r^{k+1},\end{aligned}$$

using $(\dagger)$ in the fourth step. So the claim holds for $n = k + 1$, and by induction for all $n \geq 0$:

$$\boxed{\mathrm{T}_n(x) = \mathrm{T}_0(x)\,\big(\mathrm{r}(x)\big)^n}$$

The possible forms of $\mathrm{r}(x)$

By $(\dagger)$, r satisfies $r^2 - 2xr + 1 = 0$, so by the quadratic formula

$$\mathrm{r}(x) = x + \sqrt{x^2 - 1} \qquad \text{or} \qquad \mathrm{r}(x) = x - \sqrt{x^2 - 1}.$$

Conversely, either choice satisfies $(\dagger)$, so $f \equiv 0$ and the sequence $\mathrm{T}_0 r^n$ really does obey $(*)$. The two roots have product 1, so they are reciprocals of each other. For $|x| < 1$ they are complex, $x \pm \mathrm{i}\sqrt{1 - x^2}$; writing $x = \cos\theta$, they become $\mathrm{e}^{\pm \mathrm{i}\theta}$, and $\mathrm{T}_n = \mathrm{T}_0 \mathrm{e}^{\pm \mathrm{i}n\theta}$.

Checks

- Chebyshev polynomials $\mathrm{T}_0 = 1$, $\mathrm{T}_1 = x$, $\mathrm{T}_2 = 2x^2 - 1$ give $f = x^2 - (2x^2 - 1) = 1 - x^2$, and indeed $\cos^2 n\theta - \cos(n-1)\theta\cos(n+1)\theta = \sin^2\theta$ for every n. With $x = \frac{3}{7}$ the first fourteen values of D_n all equal $\frac{40}{49}$.
- Take $r = 2$, so $x = \frac{1}{2}(2 + \frac{1}{2}) = \frac{5}{4}$, and $\mathrm{T}_0 = 1$, $\mathrm{T}_1 = 2$. The recurrence gives $\mathrm{T}_2 = \frac{5}{2} \cdot 2 - 1 = 4$ and $\mathrm{T}_3 = \frac{5}{2} \cdot 4 - 2 = 8$: powers of 2, as claimed. Also $x - \sqrt{x^2 - 1} = \frac{5}{4} - \frac{3}{4} = \frac{1}{2}$ is the other root.

Common slips

- Applying $(*)$ with $n = 0$: the recurrence starts at $n = 1$, so the base case must be D_1, not D_0 (which is not even defined).
- Dividing $\mathrm{T}_n^2 = \mathrm{T}_{n-1}\mathrm{T}_{n+1}$ by T_{n-1} without checking it is nonzero.

Beyond STEP

A linear recurrence with constant coefficients has solutions $\alpha r_1^n + \beta r_2^n$, where r_1, r_2 are the roots of its characteristic equation, here $r^2 - 2xr + 1 = 0$. The condition $f = 0$ picks out exactly the cases where only one geometric term survives; in general f is a multiple of $\alpha\beta$. The invariant D_n is a discrete cousin of the Wronskian for second-order differential equations, which is constant for $y'' + q(x)y = 0$. Boyce and DiPrima, *Elementary Differential Equations*, develops the Wronskian, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style recurrence work.

WORKED SOLUTION

STEP III 2008, Question 6

Pure • First-Order Differential Equations

STEP III 2008, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q6

The key idea

Each equation gives y directly in terms of x and the slope p, but it is not solved for p, so we can't simply integrate. Differentiating with respect to x removes y (because $\mathrm{d}y/\mathrm{d}x = p$) and leaves a relation between x, p and $\mathrm{d}p/\mathrm{d}x$. Turned upside down, that relation is a *linear* first-order equation for x as a function of p, which an integrating factor solves. Then we put $x(p)$ back into the original equation to get y.

Solution

Part (i)

Differentiate $y = p^2 + 2xp$ with respect to x, using the chain rule on p^2 and the product rule on $2xp$, and replace $\mathrm{d}y/\mathrm{d}x$ by p:

$$p = 2p\frac{\mathrm{d}p}{\mathrm{d}x} + 2p + 2x\frac{\mathrm{d}p}{\mathrm{d}x} \quad \Longrightarrow \quad -p = (2p + 2x)\frac{\mathrm{d}p}{\mathrm{d}x}.$$

? **Ask yourself** Are we allowed to flip $\mathrm{d}p/\mathrm{d}x$ over to get $\mathrm{d}x/\mathrm{d}p$?

We need $\mathrm{d}p/\mathrm{d}x \neq 0$. The equation above shows that wherever $p \neq 0$, both $\mathrm{d}p/\mathrm{d}x$ and $2p + 2x$ are non-zero. (We divide by p anyway, so $p \neq 0$ is needed throughout.) So we may write

$$\frac{\mathrm{d}x}{\mathrm{d}p} = -\frac{2p + 2x}{p} = -2 - \frac{2x}{p},$$

as required.

Rearranged, this is linear in x:

$$\frac{\mathrm{d}x}{\mathrm{d}p} + \frac{2}{p}x = -2.$$

The integrating factor is $e^{\int 2/p\,dp} = p^2$, and multiplying through gives

$$\frac{d}{dp}(p^2x) = -2p^2 \quad \Longrightarrow \quad p^2x = -\tfrac{2}{3}p^3 + A \quad \Longrightarrow \quad x = -\tfrac{2}{3}p + Ap^{-2},$$

with A an arbitrary constant. That is the stated result.

Now use $p = -3$ when $x = 2$:

$$2 = -\tfrac{2}{3}(-3) + \frac{A}{9} = 2 + \frac{A}{9},$$

so $A = 0$ and $x = -\frac{2}{3}p$, that is, $p = -\frac{3}{2}x$. Substituting into the original equation,

$$y = p^2 + 2xp = \tfrac{9}{4}x^2 + 2x\left(-\tfrac{3}{2}x\right) = \tfrac{9}{4}x^2 - 3x^2,$$

so

$$y = -\tfrac{3}{4}x^2.$$

As a check, $dy/dx = -\frac{3}{2}x$, which really is p, and it equals -3 at $x = 2$.

Part (ii)

Here $\ln p$ appears, so $p > 0$. Differentiate $y = 2xp + p\ln p$ with respect to x, using $\frac{d}{dp}(p\ln p) = \ln p + 1$:

$$p = 2p + 2x\frac{dp}{dx} + (\ln p + 1)\frac{dp}{dx} \quad \Longrightarrow \quad -p = (2x + \ln p + 1)\frac{dp}{dx}.$$

As before, $p \neq 0$ forces $dp/dx \neq 0$, so

$$\frac{dx}{dp} = -\frac{2x + \ln p + 1}{p} \quad \Longrightarrow \quad \frac{dx}{dp} + \frac{2}{p}x = -\frac{\ln p + 1}{p}.$$

The integrating factor is again p^2:

$$\frac{d}{dp}(p^2x) = -p\ln p - p.$$

Integrating by parts with $u = \ln p$, $dv = p\,dp$,

$$\int p\ln p\,dp = \tfrac{1}{2}p^2\ln p - \int \tfrac{1}{2}p\,dp = \tfrac{1}{2}p^2\ln p - \tfrac{1}{4}p^2,$$

so

$$\begin{aligned} p^2x &= -\tfrac{1}{2}p^2\ln p + \tfrac{1}{4}p^2 - \tfrac{1}{2}p^2 + B \\ &= -\tfrac{1}{2}p^2\ln p - \tfrac{1}{4}p^2 + B, \end{aligned}$$

and

$$x = -\tfrac{1}{2}\ln p - \tfrac{1}{4} + Bp^{-2}.$$

Using $p = 1$ when $x = -\frac{1}{4}$: $-\frac{1}{4} = 0 - \frac{1}{4} + B$, so $B = 0$ and

$$x = -\tfrac{1}{2}\ln p - \tfrac{1}{4},$$

which is the result we were asked to show.

To find y, notice that this says $\ln p = -2x - \frac{1}{2}$. Substituting this into the original equation is quicker than substituting for p everywhere:

$$y = 2xp + p\left(-2x - \tfrac{1}{2}\right) = -\tfrac{1}{2}p,$$

and $p = \mathrm{e}^{-2x-1/2}$, so

$$y = -\tfrac{1}{2}\,\mathrm{e}^{-2x-\frac{1}{2}}.$$

Check: $\mathrm{d}y/\mathrm{d}x = \mathrm{e}^{-2x-\frac{1}{2}} = p$, as it must be, and at $x = -\frac{1}{4}$ this is $\mathrm{e}^0 = 1$.

Checks

In each part, differentiate your final y and confirm that the derivative really is the p you used: $y = -\frac{3}{4}x^2$ gives $p = -\frac{3}{2}x$, and $p^2 + 2xp = \frac{9}{4}x^2 - 3x^2 = y$. For part (ii) at $x = 0$, $p = \mathrm{e}^{-1/2} \approx 0.6065$ and $2xp + p\ln p = 0.6065 \times (-0.5) \approx -0.3033 = y(0)$. A numerical check also confirmed that, for random values of A and B, the general parametric curves $(x(p), y(p))$ have slope $\mathrm{d}y/\mathrm{d}x$ equal to p.

Common slips

- Forgetting that $\mathrm{d}y/\mathrm{d}x$ on the left-hand side is p, and so failing to see the cancellation that leaves $-p$.
- Dropping the constant of integration before using the initial condition. Here it happens to vanish, but you only know that after finding it.
- In part (ii), differentiating $p\ln p$ with respect to x as $\ln p$ alone, forgetting the factor $\mathrm{d}p/\mathrm{d}x$ and the extra $+1$.

Beyond STEP

Equations of the form $y = xf(p) + g(p)$ are called d'Alembert (or Lagrange) equations; the special case $y = xp + g(p)$ is Clairaut's equation. The trick of treating p as the independent variable works for all of them. They can also have *singular solutions*, which arise where we divided by $\mathrm{d}p/\mathrm{d}x$ and which are envelopes of the family of general solutions. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and their methods of solution.

WORKED SOLUTION

STEP III 2008, Question 7

Pure • Complex Numbers and Geometry

STEP III 2008, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q7

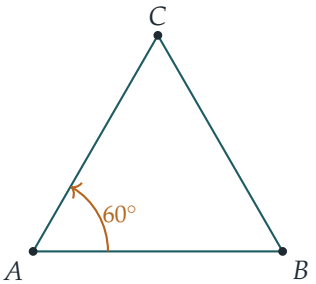
The key idea

In the Argand diagram, multiplying by $e^{i\pi/3}$ rotates through 60° anticlockwise. An equilateral triangle is exactly one side rotated through 60° about a vertex, so the opening relation turns "equilateral" into a linear equation in a, b, c. After that, both parts are bookkeeping: write every new point in terms of the old ones and check the condition we want, whether that is $p + r = q + s$ for a parallelogram or the opening relation itself for an equilateral triangle.

Solution

The opening relation

Write $\omega = e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$. Since ABC is equilateral, AC has the same length as AB and makes an angle of 60° with it. Because the triangle is described anticlockwise, we turn from $\overrightarrow{AB}$ to $\overrightarrow{AC}$ *anticlockwise*. So $\overrightarrow{AC}$ is $\overrightarrow{AB}$ rotated through $+60^\circ$:



$$c - a = \omega(b - a) = \tfrac{1}{2}(b - a) + \tfrac{\sqrt{3}}{2}i(b - a),$$
$$2c = 2a + (b - a) + i\sqrt{3}(b - a) = (a + b) + i\sqrt{3}(b - a),$$

which is the required result.

If the triangle is described clockwise, $\overrightarrow{AC}$ is $\overrightarrow{AB}$ rotated through $-60°$, so $c-a=\bar{\omega}(b-a)$ with $\bar{\omega}=\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$. The same algebra gives

$$2c=(a+b)-\mathrm{i}\sqrt{3}(b-a).$$

Conversely, any c satisfying either relation makes ABC equilateral, because each relation is equivalent to $c-a=\omega^{\pm 1}(b-a)$. So to *show* that a triangle is equilateral, it is enough to verify one of these relations.

Part (i)

We get to choose the four apexes, and each one may go on either side of its edge. Using the opening results with the apex in the role of c, put

$$\begin{aligned}
2p&=(d+e)+\mathrm{i}\sqrt{3}(e-d) && (D,E,P \text{ anticlockwise}),\\
2q&=(e+f)-\mathrm{i}\sqrt{3}(f-e) && (E,F,Q \text{ clockwise}),\\
2r&=(f+g)+\mathrm{i}\sqrt{3}(g-f) && (F,G,R \text{ anticlockwise}),\\
2s&=(g+d)-\mathrm{i}\sqrt{3}(d-g) && (G,D,S \text{ clockwise}).
\end{aligned}$$

By the opening work, each of PDE, QEF, RFG, SGD is then equilateral.

Ask yourself Why alternate the orientations, rather than putting every triangle on the same side of its edge?

Try it and see. $PQRS$ is a parallelogram exactly when $\overrightarrow{PQ}=\overrightarrow{SR}$, that is, when $q-p=r-s$, or $p+r=q+s$. (This says that the diagonals PR and QS have the same midpoint.) With the choices above,

$$\begin{aligned}
2(p+r)&=(d+e+f+g)+\mathrm{i}\sqrt{3}\,(e-d+g-f),\\
2(q+s)&=(d+e+f+g)-\mathrm{i}\sqrt{3}\,(f-e+d-g)=(d+e+f+g)+\mathrm{i}\sqrt{3}\,(e-d+g-f).
\end{aligned}$$

These are equal, so $p+r=q+s$ and $PQRS$ is a parallelogram. If instead we had used the "+" sign throughout, the imaginary parts would have been $\mathrm{i}\sqrt{3}(e-d+g-f)$ and $\mathrm{i}\sqrt{3}(f-e+d-g)$. These agree only when $d+f=e+g$, that is, only when $DEFG$ is already a parallelogram. The alternating signs are what make the $\sqrt{3}$ terms match for every quadrilateral.

(For some quadrilaterals the four points could happen to be collinear, giving a degenerate parallelogram. The condition $\overrightarrow{PQ}=\overrightarrow{SR}$ still holds in that case.)

Part (ii)

If LMN is described clockwise, reflect the whole figure in the real axis (that is, take complex conjugates). Reflection preserves equilateral triangles, centroids, and whether a triangle on a side is drawn externally, so it is enough to treat the case where LMN is described *anticlockwise*.

Then N is to the left of the directed edge $L \to M$, and the external triangle LMX on LM has its apex X to the right, so L, M, X is described clockwise:

$$2x = (l+m) - \mathrm{i}\sqrt{3}(m-l).$$

The centroid of LMX is

$$u = \tfrac{1}{3}(l+m+x) = \tfrac{1}{3}(l+m) + \tfrac{1}{6}(l+m) - \tfrac{\mathrm{i}\sqrt{3}}{6}(m-l) = \frac{l+m}{2} + \frac{\mathrm{i}(l-m)}{2\sqrt{3}},$$

using $\frac{\sqrt{3}}{6} = \frac{1}{2\sqrt{3}}$. Cycling $l \to m \to n \to l$, which keeps the anticlockwise order, gives

$$v = \frac{m+n}{2} + \frac{\mathrm{i}(m-n)}{2\sqrt{3}}, \qquad w = \frac{n+l}{2} + \frac{\mathrm{i}(n-l)}{2\sqrt{3}}.$$

We show that U, V, W satisfy the anticlockwise relation $2w = (u+v) + \mathrm{i}\sqrt{3}(v-u)$. First,

$$u + v = \tfrac{1}{2}l + m + \tfrac{1}{2}n + \frac{\mathrm{i}(l-n)}{2\sqrt{3}},$$

$$v - u = \tfrac{1}{2}(n-l) + \frac{\mathrm{i}(2m-l-n)}{2\sqrt{3}},$$

$$\mathrm{i}\sqrt{3}(v-u) = \frac{\mathrm{i}\sqrt{3}}{2}(n-l) - \tfrac{1}{2}(2m-l-n).$$

? **Ask yourself** How does the m disappear?

Add the first and third lines. The real parts give $\frac{1}{2}l + m + \frac{1}{2}n - m + \frac{1}{2}l + \frac{1}{2}n = l + n$, so m cancels from the real part, and it never appeared in the imaginary parts. The imaginary parts combine as

$$(n-l)\Big(\frac{\sqrt{3}}{2} - \frac{1}{2\sqrt{3}}\Big)\mathrm{i} = (n-l)\,\frac{3-1}{2\sqrt{3}}\,\mathrm{i} = \frac{\mathrm{i}(n-l)}{\sqrt{3}}.$$

Hence

$$(u+v) + \mathrm{i}\sqrt{3}(v-u) = (n+l) + \frac{\mathrm{i}(n-l)}{\sqrt{3}} = 2w.$$

By the opening result (and its converse), UVW is an equilateral triangle, described anticlockwise. This is Napoleon's theorem.

Checks

- Opening relation: $a = 0$, $b = 2$ gives $c = 1 + \mathrm{i}\sqrt{3}$, the apex above the real axis, as it should be for an anticlockwise triangle.
- Part (ii) with $l = 0$, $m = 6$, $n = 6\mathrm{i}$: the formulas give $u = 3 - \mathrm{i}\sqrt{3}$, $v = (3+\sqrt{3})(1+\mathrm{i})$ and $w = -\sqrt{3} + 3\mathrm{i}$. All three sides of UVW then have squared length $24 + 12\sqrt{3}$.
- We tested the opening relations, the parallelogram condition and Napoleon's theorem numerically on 10^4 random configurations. We also confirmed that using the same orientation on every edge in part (i) generally fails.

Common slips

- Getting the external direction wrong in part (ii). For an anticlockwise LMN the external apex on LM is *clockwise* from L, M. With the wrong sign we get the inner Napoleon triangle instead. (That one is also equilateral, but it isn't what the question asks for.)
- In part (i), trying to make all four triangles point the same way. That works only when $DEFG$ is itself a parallelogram.
- Testing for a parallelogram with $p + q = r + s$. Opposite vertices pair up: the condition is $p + r = q + s$.

Beyond STEP

The underlying idea is that the maps $z \mapsto \alpha z + \beta$ are exactly the rotations, enlargements and translations of the plane, so geometric conditions become linear equations over $\mathbb{C}$. A neat way to restate the opening relation is that ABC is equilateral precisely when $a + \omega^2 b + \omega^4 c = 0$ or $a + \omega^4 b + \omega^2 c = 0$, and Napoleon's theorem then drops out of a few lines of algebra. Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully.

WORKED SOLUTION

STEP III 2008, Question 8

Pure • Series and Recurrence Relations

STEP III 2008, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q8

The key idea

A linear recurrence for the coefficients means that multiplying the series by the right polynomial makes almost every term cancel. The polynomial is read straight off the recurrence: $a_{r+1} + pa_r = 0$ suggests $1 + px$, and $a_{r+2} + pa_{r+1} + qa_r = 0$ suggests $1 + px + qx^2$. What survives is a short polynomial, so the sum is a rational function. For finite sums, the same trick leaves a few extra terms at the far end. Alternatively, partial fractions split the series into geometric series, which we know how to sum.

Solution

Part (i)

Here the series starts at the x term, so $a_1 = \frac{1}{3}$, $a_2 = \frac{1}{6}$, $a_3 = \frac{1}{12}$. Each coefficient is half the one before, so $a_{r+1} - \frac{1}{2}a_r = 0$ and

$$p = -\tfrac{1}{2}.$$

(There is no constant term. The recurrence links $a_1, a_2, a_3, \ldots$ and says nothing about $r = 0$.) Solving the recurrence gives $a_r = \frac{1}{3}\left(\frac{1}{2}\right)^{r-1}$.

Sum to infinity. Multiply by $1 - \frac{1}{2}x$:

$$\left(1 - \tfrac{1}{2}x\right) S = a_1 x + \sum_{r \geq 1} \left(a_{r+1} - \tfrac{1}{2}a_r\right) x^{r+1} = \tfrac{1}{3}x,$$

because every bracket vanishes by the recurrence. Hence

$$S = \frac{\frac{1}{3}x}{1 - \frac{1}{2}x} = \frac{2x}{3(2 - x)}.$$

This is a geometric series with ratio $\frac{1}{2}x$, so the sum to infinity exists exactly when $|x| < 2$.

The first $n+1$ terms. These are the terms in $x, x^2, \ldots, x^{n+1}$. Write $S_{n+1} = \sum_{r=1}^{n+1} a_r x^r$. Repeating the same multiplication, all the middle terms still cancel, but the top term $-\frac{1}{2}a_{n+1}x^{n+2}$ has no partner:

$$\left(1 - \tfrac{1}{2}x\right) S_{n+1} = \tfrac{1}{3}x - \tfrac{1}{2}a_{n+1}x^{n+2} = \tfrac{1}{3}x - \tfrac{1}{3}\left(\tfrac{1}{2}\right)^{n+1} x^{n+2} = \tfrac{1}{3}x\left(1 - \left(\tfrac{x}{2}\right)^{n+1}\right).$$

So, for $x \neq 2$,

$$S_{n+1} = \frac{2x}{3(2-x)}\left(1 - \left(\frac{x}{2}\right)^{n+1}\right).$$

When $x = 2$ every term equals $\frac{2}{3}$, so $S_{n+1} = \frac{2}{3}(n+1)$.

Part (ii)

Now $a_0 = 2$, $a_1 = 8$, $a_2 = 18$, $a_3 = 37$. Putting $r = 0$ and $r = 1$ into the recurrence gives two equations:

$$18 + 8p + 2q = 0, \qquad 37 + 18p + 8q = 0.$$

The first gives $q = -9 - 4p$. Substituting into the second gives $37 + 18p - 72 - 32p = 0$, so $p = -\frac{5}{2}$ and then $q = 1$. The recurrence is

$$a_{r+2} - \tfrac{5}{2}a_{r+1} + a_r = 0.$$

Sum to infinity. Multiply by $1 + px + qx^2 = 1 - \frac{5}{2}x + x^2$. The coefficient of x^{r+2} in the product is $a_{r+2} - \frac{5}{2}a_{r+1} + a_r = 0$ for every $r \geq 0$. Only the x^0 and x^1 coefficients survive:

$$\left(1 - \tfrac{5}{2}x + x^2\right) T = a_0 + \left(a_1 - \tfrac{5}{2}a_0\right) x = 2 + 3x.$$

Doubling top and bottom, and factorising $2 - 5x + 2x^2 = (1-2x)(2-x)$,

$$T = \frac{4 + 6x}{(1-2x)(2-x)}.$$

Partial fractions. Write $\dfrac{4+6x}{(1-2x)(2-x)} = \dfrac{A}{1-2x} + \dfrac{B}{2-x}$. Cover-up at $x = \frac{1}{2}$ gives $A = \dfrac{7}{3/2} = \frac{14}{3}$, and at $x = 2$ gives $B = \dfrac{16}{-3} = -\frac{16}{3}$. So

$$T = \frac{14}{3} \cdot \frac{1}{1-2x} - \frac{8}{3} \cdot \frac{1}{1 - \frac{1}{2}x}.$$

Expanding each term as a geometric series,

$$a_r = \tfrac{14}{3}\, 2^r - \tfrac{8}{3}\left(\tfrac{1}{2}\right)^r.$$

Check: $r = 0, 1, 2, 3$ give 2, $\frac{28}{3} - \frac{4}{3} = 8$, $\frac{56}{3} - \frac{2}{3} = 18$ and $\frac{112}{3} - \frac{1}{3} = 37$, as they should.

Ask yourself For which x does the sum to infinity actually exist?

We need both geometric series to converge, so $|2x| < 1$ and $|\frac{1}{2}x| < 1$. The first condition is the stricter one, so we need $|x| < \frac{1}{2}$.

The first $n+1$ terms. These are the terms in x^0 to x^n. Summing each geometric series to $n+1$ terms gives, for $x \neq \frac{1}{2}, 2$,

$$\sum_{r=0}^{n} a_r x^r = \frac{14}{3} \cdot \frac{1-(2x)^{n+1}}{1-2x} - \frac{16}{3} \cdot \frac{1-\left(\frac{x}{2}\right)^{n+1}}{2-x}.$$

(The second fraction is $\frac{8}{3} \cdot \frac{1-(x/2)^{n+1}}{1-x/2}$, with top and bottom multiplied by 2.) If $x = \frac{1}{2}$ or $x = 2$, the matching geometric sum is just $n+1$ copies of 1.

Checks

- Let $n \to \infty$ with $|x| < \frac{1}{2}$: the powers $(2x)^{n+1}$ and $(x/2)^{n+1}$ vanish and we recover the partial-fraction form of T. The same happens in part (i) with $|x| < 2$.
- With $n = 0$ the part (i) formula gives $\frac{2x}{3(2-x)} \cdot \frac{2-x}{2} = \frac{1}{3}x$. The part (ii) formula gives $\frac{14}{3} - \frac{16}{3} \cdot \frac{1}{2} = 2$. Both are the first term, as they should be.
- Numerically, at $x = 0.3$ we have $T = \dfrac{5.8}{0.4 \times 1.7} \approx 8.5294$, and the partial sums of $2 + 8x + 18x^2 + 37x^3 + \cdots$ approach this value.

Common slips

- In part (i), taking the first term to be $a_0 x^0$. The series starts at x^1, so the first $n+1$ terms run up to x^{n+1}.
- Forgetting the leftover end terms when multiplying a *finite* sum by $1 + px$. These terms are exactly what produce the $(\frac{x}{2})^{n+1}$ factor.
- Quoting $|x| < 2$ for T. The root nearer the origin, $x = \frac{1}{2}$, sets the radius of convergence.

Beyond STEP

The power series $\sum a_r x^r$ is the *generating function* of the sequence (a_r). A linear recurrence with constant coefficients always gives a rational generating function, and the poles of that function decide both the closed form for a_r and the radius of convergence. Generating functions are a standard tool in combinatorics and probability, and the radius of convergence of power series is developed carefully in first-year analysis. Abbott's *Understanding Analysis* covers the convergence side well.

WORKED SOLUTION

STEP III 2008, Question 9

Mechanics • Friction and Variable Forces

STEP III 2008, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

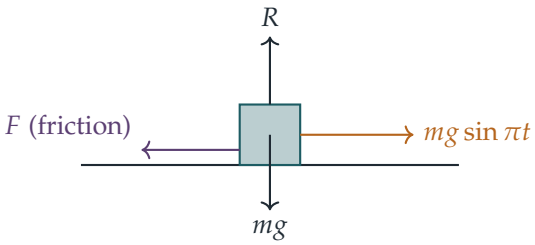
step.maths.org/questions/08-s3-q9

The key idea

Friction has two regimes. While the particle is at rest, static friction matches the applied force exactly, up to its limit μmg, so the acceleration is zero. Once the particle slides, friction has the fixed size μmg and opposes the velocity. So the acceleration graph is made of flat zero pieces joined to shifted sine curves. "Comes to rest" means the velocity, which is the area under the acceleration graph since the start of the motion, has returned to zero.

Solution

We model the body as a particle on a rough horizontal plane. The normal reaction is $R = mg$, so friction has size at most μmg at rest and exactly μmg when sliding. Take the positive direction to be the direction of the force when $\sin \pi t > 0$, and let a and v be the acceleration and velocity.



Starting to move

Resolving vertically, $R = mg$. While the particle is at rest, horizontal equilibrium needs a friction force of size $mg|\sin \pi t|$, and this is possible as long as $mg|\sin \pi t| \leq \mu mg$. The particle first moves when the applied force first exceeds the limiting friction, so (for

$\mu < 1$)

$$\sin \pi T_0 = \mu, \qquad \text{that is,} \qquad T_0 = \frac{1}{\pi} \arcsin \mu, \quad 0 \le T_0 < \tfrac{1}{2}.$$

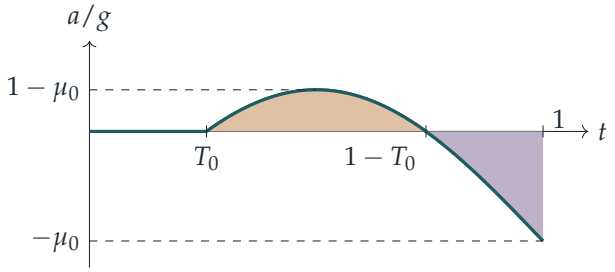
(If $\mu \ge 1$ the particle never moves.)

Part (i)

For $0 \le t \le T_0$ the particle is at rest, so $a = 0$. For $t > T_0$ it slides in the positive direction, friction $\mu_0 mg$ acts backwards, and Newton's second law gives

$$a = g(\sin \pi t - \mu_0) \qquad (\text{while } v > 0).$$

This starts at 0 when $t = T_0$, rises to $g(1 - \mu_0)$ at $t = \frac{1}{2}$, falls back to 0 at $t = 1 - T_0$ (by the symmetry $\sin \pi(1 - t) = \sin \pi t$), and is negative from then on, reaching $-\mu_0 g$ at $t = 1$.



The velocity is the area under this graph from T_0: it grows while the curve is above the axis (amber region) and shrinks after $1 - T_0$ (plum region). It stays positive until the two areas balance, and that is the moment the particle stops. So we need

$$v(1) = \int_{T_0}^{1} g(\sin \pi t - \mu_0)\, \mathrm{d}t = 0.$$

Now

$$\begin{aligned} \int_{T_0}^{1} (\sin \pi t - \mu_0)\, \mathrm{d}t &= \Big[-\frac{\cos \pi t}{\pi} \Big]_{T_0}^{1} - \mu_0 (1 - T_0) \\ &= \frac{1 + \cos \pi T_0}{\pi} - \mu_0 + \mu_0 T_0. \end{aligned}$$

Setting this to zero and multiplying by π:

$$1 + \cos \pi T_0 - \mu_0 \pi + \mu_0 \pi T_0 = 0.$$

Ask yourself Which sign does $\cos \pi T_0$ take when we write it in terms of μ_0?

Since $0 \le \pi T_0 < \frac{\pi}{2}$, the cosine is positive, so $\cos \pi T_0 = \sqrt{1 - \sin^2 \pi T_0} = (1 - \mu_0^2)^{1/2}$. Also $\pi T_0 = \arcsin \mu_0$. Substituting both,

$$1 + \left(1 - \mu_0^2\right)^{\frac{1}{2}} - \mu_0 \pi + \mu_0 \arcsin \mu_0 = 0,$$

as required. (Numerically $\mu_0 \approx 0.7246$ and $T_0 \approx 0.258$.)

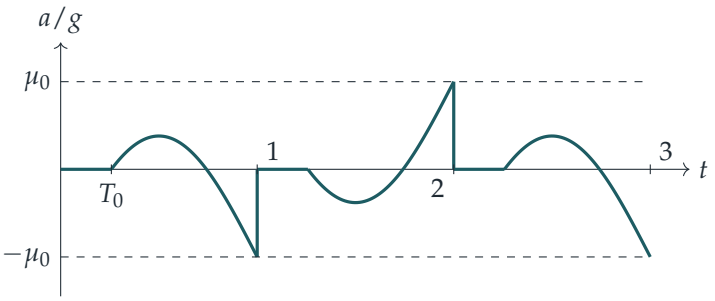
Part (ii)

At $t = 1$ the particle is at rest and the applied force is $mg \sin \pi = 0$. For $1 < t < 2$ the force points in the negative direction with size $mg|\sin \pi t|$. This is less than $\mu_0 mg$ until $|\sin \pi t| = \mu_0$, which first happens at $t = 1 + T_0$ (as $\sin \pi(1 + T_0) = -\sin \pi T_0$). So the particle stays at rest, with $a = 0$, for $1 \le t \le 1 + T_0$.

After that it slides in the negative direction, friction acts in the positive direction, and

$$a = g(\sin \pi t + \mu_0).$$

Since $\sin \pi(t + 1) = -\sin \pi t$, this is exactly the negative of the acceleration at time $t - 1$ in part (i). The motion on $[1, 2]$ is therefore the mirror image of the motion on $[0, 1]$: the velocity is the negative of the earlier one, so the particle comes to rest again at $t = 2$, having travelled back the same distance. It is then at rest with zero applied force, which is exactly the situation at $t = 0$, so on $[2, 3]$ the motion of $[0, 1]$ repeats.



The vertical segments at $t = 1$ and $t = 2$ mark jumps in the acceleration: at the instant the particle stops, kinetic friction (of size $\mu_0 mg$) is replaced by static friction, which only has to balance a zero applied force.

Motion for $\mu = \mu_0$. The particle is at rest until $t = T_0$, slides forwards and stops at $t = 1$, waits until $t = 1 + T_0$, slides back and stops at its starting point at $t = 2$, and so on. It shuttles back and forth over the same stretch of ground with period 2, pausing for time T_0 at each end. (Integrating once more gives the length of the stretch as $g\left[\mu_0/\pi^2 - (1 - T_0)(1 - \sqrt{1 - \mu_0^2})/(2\pi)\right] \approx 0.0367g$, though this is not asked for.)

Motion for $\mu = 0$. Now $T_0 = 0$ and $a = g \sin \pi t$ at all times, so

$$v = \frac{g}{\pi}(1 - \cos \pi t) \ge 0, \qquad x = \frac{g}{\pi}\left(t - \frac{\sin \pi t}{\pi}\right).$$

The velocity is never negative and vanishes only at $t = 0, 2, 4, \ldots$. So the particle never turns back: it moves forwards in surges, momentarily at rest at even times, and advances $2g/\pi$ in each period of 2, an average speed of g/π. Friction is what stops it at $t = 1$ in the rough case; without friction the backward half of the force only slows the particle down to zero velocity.

Checks

Solving the equation of part (i) by bisection gives $\mu_0 \approx 0.72461$, and so $T_0 \approx 0.25798$. A numerical simulation of the motion for $0 \le t \le 3$ with $g = 9.8$, switching between static and kinetic friction, shows the particle starting at $t \approx 0.258$, stopping at $t = 1.000$ after travelling 0.3598, starting back at $t \approx 1.258$ and stopping at $t = 2.000$ at the starting point, in agreement with $0.0367g$. For $\mu = 0$ the simulation matches $x = g(t - \sin \pi t/\pi)/\pi$. A quick sanity check on the equation: at $\mu_0 = 0$ the left side is $2 > 0$ and at $\mu_0 = 1$ it is $1 - \pi/2 < 0$, so a root in $(0,1)$ exists.

Common slips

- Taking $a = g(\sin \pi t - \mu)$ from $t = 0$: before T_0 the particle is at rest and friction is static, so $a = 0$.
- Keeping the friction in the same direction after the particle reverses in part (ii). Friction always opposes the velocity.
- Thinking the particle stops at $t = 1 - T_0$, where the acceleration (not the velocity) is zero.

Beyond STEP

Friction that switches between "stuck" and "sliding" makes the equation of motion piecewise and the acceleration discontinuous. This is the stick–slip behaviour behind squeaking doors and bowed violin strings, and a first example of a non-smooth dynamical system. Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses friction models and forced motion at the right level.

WORKED SOLUTION

STEP III 2008, Question 10

Mechanics • Elastic Strings and Energy

STEP III 2008, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q10

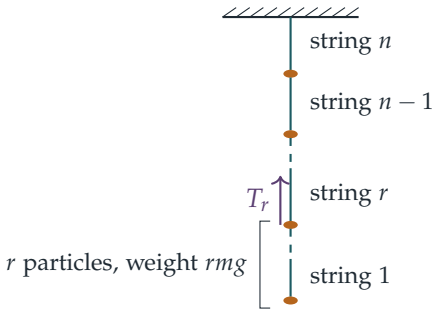
The key idea

Each short string holds up everything that hangs below it, so its tension is easy to write down; Hooke's law then gives its extension and its stored energy, and the totals are standard sums of r and r^2. The rope is the same picture with the mass spread out: chop it into n equal pieces, treat each piece's mass as a particle at its lower end, and let $n \to \infty$. The sums become integrals, and the factor $\frac{1}{2}$ in the length appears because, on average, a piece of rope supports half the rope's weight.

Solution

The string of particles

Model the short strings as light elastic strings obeying Hooke's law, and the masses as particles. Number the strings from the bottom, as in the question, so string r has r particles below it: its own particle and those of strings $1, \ldots, r-1$.



The whole system is at rest. Consider everything below the top of string r: string r, strings $1, \ldots, r-1$ and the r particles on them. The only external forces on it are its weight rmg downwards and the tension T_r in string r upwards (the strings are light). So

$$T_r = rmg.$$

By Hooke's law, the extension of string r is

$$x_r = \frac{T_r\,\ell}{\lambda} = \frac{rmg\ell}{\lambda}.$$

So the length of the long string is

$$\sum_{r=1}^{n}(\ell + x_r) = n\ell + \frac{mg\ell}{\lambda}\sum_{r=1}^{n} r = n\ell + \frac{n(n+1)mg\ell}{2\lambda}.$$

The elastic energy in a string of natural length ℓ stretched by x is $\lambda x^2/(2\ell)$. For string r this is

$$\frac{\lambda}{2\ell}\Big(\frac{rmg\ell}{\lambda}\Big)^2 = \frac{r^2m^2g^2\ell}{2\lambda},$$

and using $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$, the total elastic energy is

$$\frac{m^2g^2\ell}{2\lambda}\cdot\frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(2n+1)\,m^2g^2\ell}{12\lambda}.$$

The heavy rope

Ask yourself How does the first part help with a rope whose mass is spread continuously?

Cut the rope into n equal pieces. Each has natural length $\ell = L_0/n$ and mass $m = M/n$, and each piece has the same modulus λ as the rope (the modulus doesn't depend on the length of the piece). If we lump each piece's mass at its lower end, we get exactly the long string above. As $n \to \infty$ the pieces get shorter and the lumped model approaches the real rope.

Substituting into the length,

$$n\ell + \frac{n(n+1)mg\ell}{2\lambda} = L_0 + \frac{n(n+1)}{2\lambda}\cdot\frac{Mg}{n}\cdot\frac{L_0}{n} = L_0 + \frac{MgL_0}{2\lambda}\Big(1+\frac{1}{n}\Big) \;\longrightarrow\; L_0\Big(1+\frac{Mg}{2\lambda}\Big)$$

as $n \to \infty$. So

$$L = L_0\Big(1+\frac{Mg}{2\lambda}\Big),$$

as required.

You can also do this directly. Let s be the natural length of rope below a given point, so that the point supports weight Mgs/L_0. A small element of natural length δs there has tension $T = Mgs/L_0$, so it stretches by $T\,\delta s/\lambda$. Adding up,

$$L - L_0 = \int_0^{L_0}\frac{Mgs}{\lambda L_0}\,\mathrm{d}s = \frac{MgL_0}{2\lambda}.$$

For the energy, the same substitution gives

$$\frac{n(n+1)(2n+1)}{12\lambda}\cdot\frac{M^2g^2}{n^2}\cdot\frac{L_0}{n} = \frac{M^2g^2L_0}{12\lambda}\Big(1+\frac{1}{n}\Big)\Big(2+\frac{1}{n}\Big) \;\longrightarrow\; \frac{M^2g^2L_0}{6\lambda}.$$

(Directly: the element stores $T^2\,\delta s/(2\lambda)$, and $\int_0^{L_0} \frac{M^2g^2s^2}{2\lambda L_0^2}\,\mathrm{d}s = \frac{M^2g^2L_0}{6\lambda}$.)

To write this in terms of L, L_0 and λ, eliminate Mg using the length:

$$L - L_0 = \frac{MgL_0}{2\lambda} \quad\Longrightarrow\quad Mg = \frac{2\lambda(L-L_0)}{L_0}.$$

Hence the elastic energy stored in the rope is

$$\frac{L_0}{6\lambda}\cdot\frac{4\lambda^2(L-L_0)^2}{L_0^2} = \frac{2\lambda(L-L_0)^2}{3L_0}.$$

Checks

- With $n = 1$ the first part gives length $\ell + mg\ell/\lambda$ and energy $m^2g^2\ell/(2\lambda)$: a single string holding a single particle, as it should be.
- Compare a rope of the same total extension $e = L - L_0$ but under *uniform* tension: it would store $\lambda e^2/(2L_0)$. Our answer is $\frac{4}{3}$ times this, which is plausible because the stretching is concentrated near the top, and energy grows like the square of the local strain.
- Numerically, with $M = 3$, $L_0 = 2$, $\lambda = 50$, $g = 9.8$: the formula gives $L = 2.588$ and energy 5.7624; the discrete chain with $n = 100\,000$ gives 2.58801 and 5.76249, and $2\lambda(L-L_0)^2/(3L_0) = 5.7624$.

Common slips

- Taking the tension in string r as $(n - r + 1)mg$: the numbering puts string n at the top, so string r carries r particles.
- Giving each piece of rope modulus λ/n. The modulus is a property of the material and cross-section, not of the length; it is the stiffness λ/ℓ that changes.
- Leaving the energy in terms of M and g when it is asked for in terms of L, L_0 and λ.

Beyond STEP

Passing from n lumped masses to a continuous rope is the basic move of continuum mechanics: the tension becomes a function $T(s)$, Hooke's law becomes a relation between local stress and strain, and the energy becomes an integral of an energy density. The same chain of masses and springs, set vibrating, gives in the limit the wave equation for a string. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see these ideas developed.

WORKED SOLUTION

STEP III 2008, Question 11

Mechanics • Rotational Dynamics

STEP III 2008, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

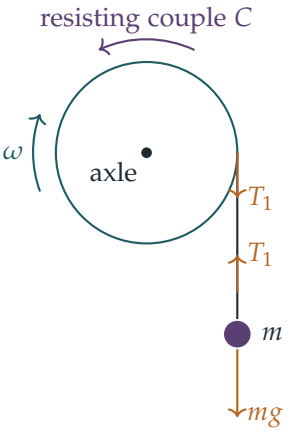
step.maths.org/questions/08-s3-q11

The key idea

A constant couple means constant angular deceleration, so once the string has gone the wheel obeys the rotational version of the "suvat" equations, and the first result is just "distance = average speed × time". That same stage tells us the size of the resisting couple in terms of I. For the descent, an energy equation is the natural tool: we know how far everything moves, and work done against a constant couple is simply couple × angle.

Solution

Model. The wheel is a rigid body turning about a fixed smooth axle, except for a resisting couple of constant magnitude C, which acts throughout the motion (while the particle descends as well as afterwards). The string is light and inextensible and does not slip, so while it is attached the particle's speed is $v = r\omega$, where ω is the wheel's angular speed.



The string leaves the right-hand side of the wheel, so the wheel turns clockwise as the particle falls, and the couple C acts anticlockwise, opposing the motion. (We won't actually need the tension T_1 in the string, because we use energy.)

After the string detaches

Now the only couple on the wheel is C, so taking moments about the axle,

$$I\dot{\omega} = -C,$$

and the angular deceleration C/I is constant. The angular speed therefore falls *linearly* from ω_0 to 0 in time T, so

$$\omega_0 = \frac{C}{I}\,T, \qquad \text{that is,} \qquad C = \frac{I\omega_0}{T}. \tag{1}$$

With a linear fall in angular speed, the average angular speed is $\frac{1}{2}(\omega_0 + 0)$, and the angle turned is the average angular speed times the time. That angle is n_2 revolutions, or $2\pi n_2$ radians, so

$$\tfrac{1}{2}\omega_0 T = 2\pi n_2,$$

which is the first result. (It is the rotational version of $s = \frac{1}{2}(u+v)t$.) It also gives

$$\omega_0 = \frac{4\pi n_2}{T}, \qquad \text{and so by (1)} \qquad C = \frac{4\pi n_2 I}{T^2}. \tag{2}$$

While the particle descends

While the wheel turns through n_1 revolutions, that is $2\pi n_1$ radians, the string unwinds a length $2\pi n_1 r$, so the particle falls a distance $2\pi n_1 r$. At the moment of detachment the wheel has angular speed ω_0 and, as the string has not slipped, the particle has speed $r\omega_0$.

Ask yourself Why can we use energy here, when there is a resistance?

Energy is not conserved, but the work–energy principle still holds: the loss of potential energy equals the gain in kinetic energy plus the work done against resistance. The work done against a constant couple C while turning through angle θ is $C\theta$, just as the work against a constant force is force $\times$ distance. The tension does no net work on the system, since the string is light and inextensible and does not slip, so the work it does on the wheel is exactly the work it takes from the particle.

So, from release to detachment,

$$mg\,(2\pi n_1 r) = \tfrac{1}{2}m(r\omega_0)^2 + \tfrac{1}{2}I\omega_0^2 + C\,(2\pi n_1).$$

Now substitute ω_0 and C from (2):

$$2\pi mgrn_1 = \tfrac{1}{2}\,(mr^2 + I)\,\frac{16\pi^2 n_2^2}{T^2} + 2\pi n_1 \cdot \frac{4\pi n_2 I}{T^2}.$$

Multiplying through by $T^2/(2\pi)$,

$$\begin{aligned} mgrn_1T^2 &= 4\pi n_2^2\,(mr^2 + I) + 4\pi n_1 n_2 I \\ &= 4\pi m r^2 n_2^2 + 4\pi n_2(n_1 + n_2)\,I. \end{aligned}$$

Rearranging, and dividing by $4\pi n_2(n_1 + n_2)$, which is positive,

$$I = \frac{mgrn_1T^2 - 4\pi m r^2 n_2^2}{4\pi n_2(n_1 + n_2)},$$

as required.

Checks

- *Dimensions.* Both terms in the numerator, $mgrT^2$ and mr^2, have the units of a moment of inertia ($\mathrm{kg\,m^2}$), and n_1, n_2 are pure numbers.
- *Numerical check.* We stepped the motion with $I = 0.5\,\mathrm{kg\,m^2}$, $m = 1\,\mathrm{kg}$, $r = 0.2\,\mathrm{m}$, $C = 0.5\,\mathrm{N\,m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $n_1 = 5$. This gives $\omega_0 \approx 13.03\,\mathrm{rad\,s^{-1}}$, $T \approx 13.03\,\mathrm{s}$ and $n_2 \approx 13.52$. Then $\frac{1}{2}\omega_0 T \approx 84.94 \approx 2\pi n_2$, and the formula returns $I = 0.5000$.
- *No resistance.* If $C \to 0$ the wheel never stops, so T and n_2 both become infinite; the formula behaves sensibly, since its numerator grows like T^2 and its denominator like $n_2^2 \propto T^2$.

Common slips

- Leaving out the work done against resistance during the descent. The question says the couple is constant, and it acts all the time, not just after the string detaches. Without it the formula comes out wrong: in the example above, it would give $I \approx 0.39$ instead of 0.5.
- Forgetting the particle's kinetic energy $\frac{1}{2}mr^2\omega_0^2$; this is where the $-4\pi m r^2 n_2^2$ term comes from.
- Mixing revolutions and radians: n revolutions is $2\pi n$ radians.

Beyond STEP

This is a neat experimental method: by timing the run-down of a wheel and counting revolutions, you can measure its moment of inertia *and* the friction in its bearing at the same time, even though you don't know either in advance. Undergraduate mechanics builds this into the full theory of rigid-body rotation, where angular momentum and torque play the roles of momentum and force, and the moment of inertia becomes a tensor for bodies spinning about arbitrary axes. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this carefully, with many problems of exactly this flavour.

WORKED SOLUTION

STEP III 2008, Question 12

Probability and Statistics • Moment Generating Functions

STEP III 2008, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.24, *The Central Limit Theorem*: https://beyondstep.keithmcnulty.org/ch12/clt.html

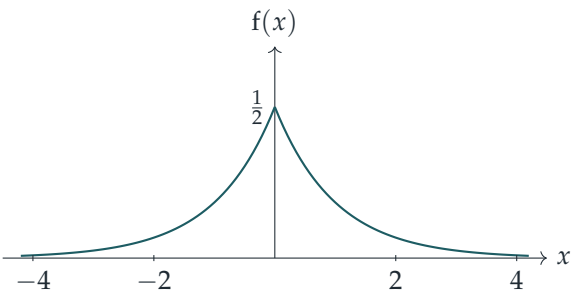
The key idea

The absolute value in the density splits every integral into two halves, $x < 0$ and $x > 0$, each of which is a plain exponential integral. After that, the whole question runs on two facts about moment generating functions: the m.g.f. of a sum of independent variables is the product of their m.g.f.s, and $M_{aY}(\theta) = M_Y(a\theta)$. Taking logarithms turns the nth power into n times something small, which is exactly what a Maclaurin series handles. This is the central limit theorem, seen in a case where you can do every step by hand.

Solution

The sketch

The density is even, has its maximum value $\frac{1}{2}$ at $x = 0$, where there is a sharp corner (the left and right derivatives there are $+\frac{1}{2}$ and $-\frac{1}{2}$), and decays exponentially to 0 on both sides.



The moment generating function

By definition $\mathrm{M}_X(\theta) = \mathrm{E}(\mathrm{e}^{\theta X})$. Splitting at 0,

$$\begin{aligned}\mathrm{M}_X(\theta) &= \tfrac{1}{2}\int_{-\infty}^{0} \mathrm{e}^{(1+\theta)x}\,\mathrm{d}x + \tfrac{1}{2}\int_{0}^{\infty} \mathrm{e}^{-(1-\theta)x}\,\mathrm{d}x \\ &= \frac{1}{2(1+\theta)} + \frac{1}{2(1-\theta)} = \frac{(1-\theta)+(1+\theta)}{2(1-\theta^2)} = \frac{1}{1-\theta^2}.\end{aligned}$$

The first integral converges only if $1+\theta > 0$ and the second only if $1-\theta > 0$, so this holds for $-1 < \theta < 1$, which is all we need near $\theta = 0$. This is the required result.

For the variance, expand as a geometric series for $|\theta| < 1$:

$$\mathrm{M}_X(\theta) = 1 + \theta^2 + \theta^4 + \cdots.$$

Comparing with $\mathrm{M}_X(\theta) = 1 + \mathrm{E}(X)\,\theta + \frac{1}{2}\mathrm{E}(X^2)\,\theta^2 + \cdots$, we get $\mathrm{E}(X) = 0$ and $\frac{1}{2}\mathrm{E}(X^2) = 1$. Hence

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \big(\mathrm{E}(X)\big)^2 = 2.$$

(Differentiating twice and setting $\theta = 0$ gives the same, but the series is quicker.)

The frog: m.g.f. of $Y/\sqrt{2n}$

Since the jumps $X_1, \ldots, X_n$ are independent, $\mathrm{e}^{\theta X_1}, \ldots, \mathrm{e}^{\theta X_n}$ are independent too, so the expectation of their product is the product of the expectations:

$$\mathrm{M}_Y(\theta) = \mathrm{E}\big(\mathrm{e}^{\theta X_1}\cdots\mathrm{e}^{\theta X_n}\big) = \prod_{i=1}^{n} \mathrm{M}_{X_i}(\theta) = (1-\theta^2)^{-n}.$$

Writing $Z = Y/\sqrt{2n}$, we have $\mathrm{E}(\mathrm{e}^{\theta Z}) = \mathrm{E}\big(\mathrm{e}^{(\theta/\sqrt{2n})Y}\big) = \mathrm{M}_Y\big(\theta/\sqrt{2n}\big)$, so

$$\mathrm{M}_Z(\theta) = \left(1 - \frac{\theta^2}{2n}\right)^{-n}, \qquad |\theta| < \sqrt{2n}.$$

The scaling by $\sqrt{2n}$ is natural: $\mathrm{Var}(Y) = 2n$, so Z has mean 0 and variance 1.

The limit

Fix θ. Once $n > \frac{1}{2}\theta^2$ we have $0 \le \theta^2/(2n) < 1$, so we may take logarithms and use $\ln(1-u) = -u - \frac{1}{2}u^2 - \frac{1}{3}u^3 - \cdots$ for $|u| < 1$:

$$\begin{aligned}\ln \mathrm{M}_Z(\theta) &= -n\ln\left(1 - \frac{\theta^2}{2n}\right) = n\left(\frac{\theta^2}{2n} + \frac{\theta^4}{8n^2} + \frac{\theta^6}{24n^3} + \cdots\right) \\ &= \frac{\theta^2}{2} + \frac{\theta^4}{8n} + \frac{\theta^6}{24n^2} + \cdots.\end{aligned}$$

Ask yourself How do we know the tail of that series really vanishes as $n \to \infty$?

With $u = \theta^2/(2n)$, everything after the first term is $n\left(\frac{1}{2}u^2 + \frac{1}{3}u^3 + \cdots\right) \le n \cdot \frac{1}{2}u^2(1 + u + u^2 + \cdots) = \dfrac{nu^2}{2(1-u)} = \dfrac{\theta^4}{8n(1-u)}$, which tends to 0 because $u \to 0$. So $\ln \mathrm{M}_Z(\theta) \to \frac{1}{2}\theta^2$, and since exp is continuous,

$$\mathrm{M}_{Y/\sqrt{2n}}(\theta) \to \exp\left(\tfrac{1}{2}\theta^2\right) \quad \text{as } n \to \infty,$$

as required.

The least number of jumps

Since the limit is the m.g.f. of the standard Normal, for large n we take $Z = Y/\sqrt{2n}$ to be approximately $\mathrm{N}(0,1)$. "Landing 25 cm or more from the start" means $|Y| \ge 25$, in either direction, so

$$\mathrm{P}(|Y| \ge 25) = \mathrm{P}\left(|Z| \ge \frac{25}{\sqrt{2n}}\right) \approx 2\left(1 - \Phi\left(\frac{25}{\sqrt{2n}}\right)\right).$$

This probability increases with n (the frog spreads out further), so we want the least n for which it reaches 5%. Since $\Phi(1.96) = 0.975$, we need

$$\frac{25}{\sqrt{2n}} \le 1.96 \iff 2n \ge \left(\frac{25}{1.96}\right)^2 \approx 162.7 \iff n \ge 81.3.$$

So the least number of jumps is $n = 82$. (With $n = 81$ the approximate probability is about 4.95%; with $n = 82$ it is about 5.09%.)

Checks

- $\mathrm{M}_X(0) = 1$, as every m.g.f. must, and M_X is even, matching the symmetry of the sketch.
- Numerical quadrature of $\int x^2 \mathrm{f}(x)\,\mathrm{d}x$ gives 2.0000, and of $\int \mathrm{e}^{0.3x}\mathrm{f}(x)\,\mathrm{d}x$ gives $1.0989 = 1/0.91$.
- With $\theta = 1$, $(1 - \frac{1}{2n})^{-n}$ is 1.6702 for $n = 10$ and 1.6508 for $n = 100$, closing in on $\mathrm{e}^{1/2} = 1.6487$.
- A simulation of 10^5 frogs making 82 Laplace jumps each had about 5.1% of them finish at least 25 cm from the start.

Common slips

- Using the one-tailed value 1.645: the frog can end up 25 cm away on either side, so both tails count.

- Writing $\mathrm{M}_{Y/\sqrt{2n}}(\theta) = \mathrm{M}_Y(\theta)/\sqrt{2n}$. Scaling a variable scales the *argument* of its m.g.f.
- Giving $n = 81$ by rounding 81.3 to the nearest integer: at $n = 81$ the chance is just under 5%.

Beyond STEP

What you have proved is the central limit theorem for Laplace steps, and the general proof follows the same route: expand $\ln \mathrm{M}(\theta/\sqrt{n})$ to second order and show the rest vanishes. In a first-year course you will meet the "continuity theorem", which justifies passing from convergence of m.g.f.s (or characteristic functions, which always exist) to convergence of probabilities, the step we took on trust here. Grimmett and Welsh, *Probability: An Introduction*, covers both.

WORKED SOLUTION

STEP III 2008, Question 13

Probability and Statistics • Sums of Indicator Variables

STEP III 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.10, *Tying Loose Ends*: https://beyondstep.keithmcnulty.org/ch12/loose-ends.html

The key idea

Whatever happens at a step, the number of pieces of string always drops by exactly one: either a ring is removed from play, or two pieces become one. So after every step we are back in the same situation with one fewer piece, whatever happened before. That lets us write the total number of rings as a sum of *independent* indicator variables, and then the mean and variance are simply sums.

Solution

The first step

With n pieces there are $2n$ free ends, so there are

$$\binom{2n}{2} = n(2n-1)$$

equally likely ways to choose the two ends. A ring is formed exactly when both ends belong to the same piece, and there are n such pairs (one per piece). So

$$\mathrm{P}(\text{ring at the first step}) = \frac{n}{n(2n-1)} = \frac{1}{2n-1}.$$

Equivalently: whichever end we pick first, exactly one of the remaining $2n-1$ ends is its partner. If X_n is 1 when the first step makes a ring and 0 otherwise, the expected number of rings at the first step is

$$\mathrm{E}(X_n) = \frac{1}{2n-1}.$$

Expected number of rings

After the first step there are $n-1$ pieces of string, each with two free ends. This is true in both cases: a ring takes one piece out of play, and joining two different pieces turns two pieces into one. The lengths of the pieces don't matter; only their ends do.

> **Ask yourself** Does it matter that some pieces are now longer than others, or that a ring was or wasn't made earlier?

No. The only thing that affects which pair of ends is chosen is how many pieces (and hence ends) remain. So when there are k pieces left, the step makes a ring with probability $\frac{1}{2k-1}$, regardless of all earlier steps.

Label the steps by the number of pieces present just before them, $k = n, n-1, \ldots, 1$, and let X_k be the indicator that a ring is made at that step. Then $\mathrm{P}(X_k = 1) = \frac{1}{2k-1}$, and the process stops after the step with $k = 1$ (which always makes a ring, as $X_1 = 1$). The total number of rings is

$$R = X_1 + X_2 + \cdots + X_n,$$

so by linearity of expectation

$$\mathrm{E}(R) = \sum_{k=1}^{n} \frac{1}{2k-1} = 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1}.$$

Variance

For the variance we need more than linearity: we need the X_k to be independent. They are. Given any outcomes of the earlier steps, the step with k pieces still makes a ring with probability $\frac{1}{2k-1}$. So the conditional distribution of X_k given all earlier X's is the same as its unconditional distribution, and so the X_k are independent.

Each X_k is Bernoulli with $p_k = \frac{1}{2k-1}$, so $\mathrm{Var}(X_k) = p_k(1-p_k)$, and for independent variables the variances add:

$$\begin{aligned}\mathrm{Var}(R) &= \sum_{k=1}^{n} \frac{1}{2k-1}\left(1 - \frac{1}{2k-1}\right) = \sum_{k=1}^{n} \frac{2k-2}{(2k-1)^2} \\ &= \sum_{k=1}^{n} \frac{1}{2k-1} \; - \; \sum_{k=1}^{n} \frac{1}{(2k-1)^2}.\end{aligned}$$

The case $n = 40\,000$

Write $H_m = 1 + \frac{1}{2} + \cdots + \frac{1}{m}$. The sum of reciprocals of odd numbers up to $2n-1$ is all reciprocals up to $2n$, minus the even ones:

$$\sum_{k=1}^{n} \frac{1}{2k-1} = H_{2n} - \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n}\right) = H_{2n} - \tfrac{1}{2}H_n.$$

Using $H_m \approx \ln m$ for large m,

$$\mathrm{E}(R) \approx \ln(2n) - \tfrac{1}{2}\ln n = \tfrac{1}{2}\left(2\ln 2n - \ln n\right) = \tfrac{1}{2}\ln(4n).$$

For $n = 40\,000$ we have $4n = 160\,000 = 20^4$, so

$$\mathrm{E}(R) \approx \tfrac{1}{2} \ln\left(20^4\right) = 2 \ln 20 \approx 6.$$

So from 40 000 pieces of string we expect only about six rings.

Checks

- $n = 1$: one piece, one ring for certain. The formulas give $\mathrm{E}(R) = 1$ and $\mathrm{Var}(R) = 0$, as they should.
- $n = 2$: the first step makes a ring with probability $\frac{1}{3}$, and the last step always does, so R is 2 with probability $\frac{1}{3}$ and 1 otherwise. That gives mean $\frac{4}{3}$ and variance $\frac{2}{9}$, matching $1 + \frac{1}{3}$ and $\frac{2}{9}$ from the formulas.
- Enumerating every sequence of choices exactly for $n \leq 5$, and simulating $n = 6$, agrees with both formulas. For $n = 40\,000$ the exact sum is $\mathrm{E}(R) \approx 6.28$. The extra 0.29 over our estimate comes from the crude approximation $H_m \approx \ln m$ (see below).

Common slips

- Taking the ring probability to be $\frac{1}{n}$ (choosing two *strings*) instead of $\frac{1}{2n-1}$ (choosing two *ends*).
- Adding the variances without first justifying that the X_k are independent. Linearity is enough for the mean, but not for the variance.
- Approximating $\sum \frac{1}{2k-1}$ by $\ln n$ or $\frac{1}{2} \ln n$ without splitting off the even terms.

Beyond STEP

A sharper estimate is $H_m = \ln m + \gamma + o(1)$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. This gives $\mathrm{E}(R) \approx \frac{1}{2} \ln(4n) + \frac{1}{2}\gamma$, and also $\mathrm{Var}(R) \approx \frac{1}{2} \ln(4n) + \frac{1}{2}\gamma - \frac{\pi^2}{8}$, since $\sum 1/(2k-1)^2 = \pi^2/8$. The same "sum of independent indicators with probabilities $1/k$" structure counts the cycles of a random permutation and the records in a random sequence; in both cases the count grows like $\ln n$. Grimmett and Welsh, *Probability: An Introduction*, is a good place to go further with indicator variables and these random structures.

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